

FlightOne AI Travel Operating System (AI-TOS)

Software Design Specification — Functional Design, Wireframes & Detailed
User Flows for all 17 Product Modules

DOCUMENT	VERSION	DATE	AUDIENCE
SDS / Functional Specification	1.0 — Derived from PRD v1.0	16 September 2026	Engineering • Product • QA • Management • Technology Partners

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About this Document

REVISION HISTORY

Version	Date	Description	Owner
1.0	16-Sep-2026	Initial SDS derived from PRD v1.0 — all 17 modules, wireframes and user flows	Product Management

1. Purpose

This Software Design Specification (SDS) converts the FlightOne AI-TOS Product Requirements Document into a build-ready functional design. For every module it defines the screens, low-fidelity web wireframes, step-by-step user flows (actor → action → system response), alternate and exception paths, business rules and core data fields. It is the reference for UI/UX design, engineering estimation, API contracts and QA test cases.

2. Scope

- All 17 PRD modules, covered one by one without omission, including the inline product notes captured in the PRD (short decision-led AI answers; OCR with manual verification and document expiry notifications; alternate departure/arrival city sales strategy; similar-option and cross-provider differentiation algorithm).
- A foundation section (Module 00) for Access, Authentication & Onboarding, since every module depends on it.
- Web application only (desktop/responsive web). Native mobile apps are out of scope for this version; the traveller portal is responsive.
- Phase 1 supplier integrations: Travelport Galileo GDS (flights) and RateHawk API (hotels). Future suppliers are listed for extensibility only.

3. How to Read a User Flow

Each flow is written in SDS style, e.g. “*User clicks Login → enters email and password → clicks Sign In → System validates credentials → System opens Home.*” Every step names the actor performing it:

UserSystemAIAgentApproverOpsSupplier

- User** — traveller, corporate employee, travel Booker or group/event organiser using the portal.
- System** — deterministic platform behaviour (validation, storage, notifications, integrations).
- AI** — the AI Conversation / Recommendation engine (LLM-driven reasoning, ranking, extraction).
- Agent** — FlightOne human travel consultant.
- Approver** — corporate line manager / travel approver.
- Ops** — FlightOne operations, finance or admin staff.
- Supplier** — Galileo, RateHawk, payment gateway, or other external API.

Wireframes are low-fidelity and indicate layout, content and controls — not final visual design. Final UI will apply the FlightOne brand system.

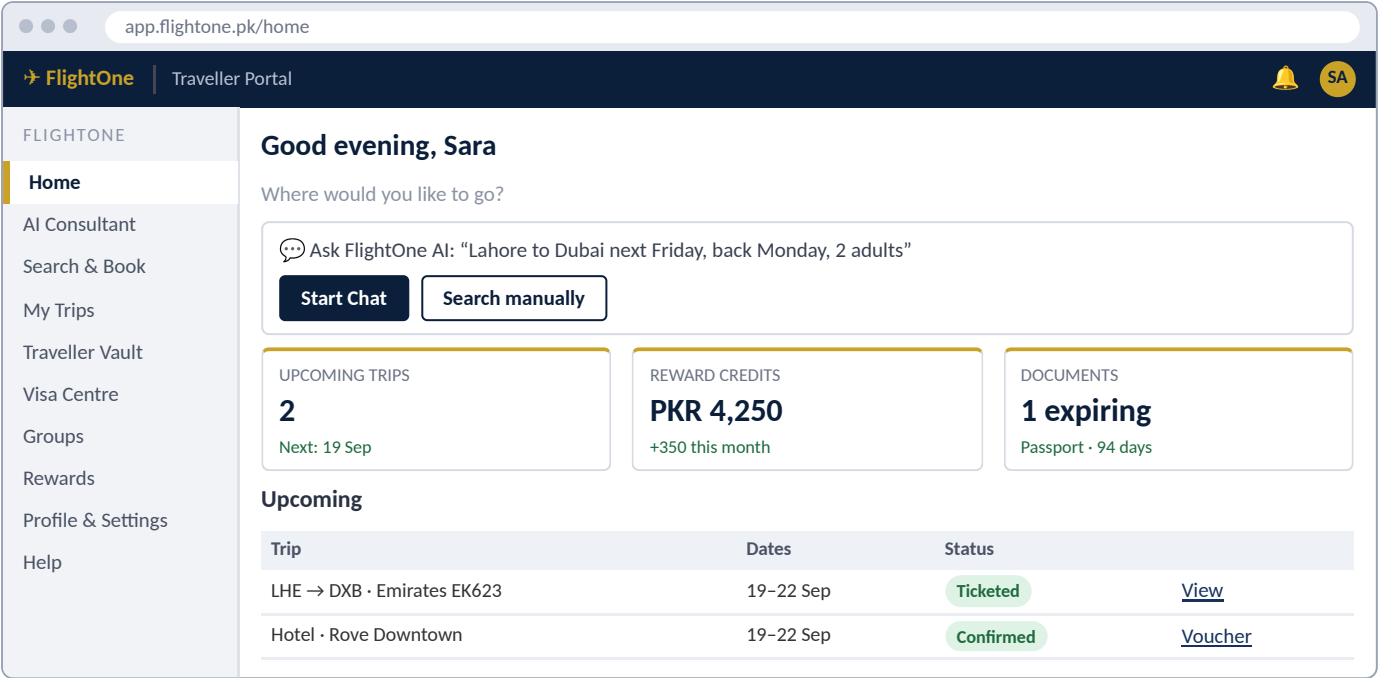
4. User Roles & Portals

Role	Portal	Summary of Access
Guest Visitor	Public Web	Browse, chat with AI (limited), must sign in to book
Traveller (Personal)	Traveller Portal	Search, book, pay, manage trips, vault, visas, rewards, groups
Corporate Traveller	Traveller Portal (Corporate profile)	Book within policy; approvals; company billing; switch to Personal
Travel Booker / Arranger	Traveller Portal	Book on behalf of colleagues in the same company

Role	Portal	Summary of Access
Corporate Approver	Traveller Portal + Email/WhatsApp	Approve / reject trip requests
Corporate Travel Admin	Corporate Admin Portal	Employees, policies, cost centres, budgets, invoices, reports
Group Organiser	Traveller Portal → Groups	Create groups, invite, announce, polls, attendance
Event (MICE) Manager	MICE Console	Events, delegates, travel & stay, agenda, badges, check-in, sponsors
FlightOne Consultant (Agent)	Operations Console	Escalations, manual booking, servicing, refunds
Operations / Finance	Operations Console	Queues, suppliers, reconciliation, invoicing, commission
Pricing Manager	Operations Console	Markups, discounts, AI discount limits
Knowledge Manager	Operations Console	SOPs, policies, contracts in AI Knowledge Base
Management	Management Dashboard	Real-time KPIs and analytics
Super Admin	Operations Console	Users, roles, permissions, system configuration, audit

5. Global Navigation — Traveller Portal

WIREFRAME — Traveller Portal — global layout (Home)



Left navigation is persistent. The top bar holds notifications, profile switcher (Personal/Corporate) and account menu. A floating AI chat launcher is available on every page.

6. Global Components

Component	Behaviour
Notification Centre (🔔)	Lists in-app alerts (flight changes, approvals, expiries, rewards). Unread count badge. Each alert deep-links to its record. Same events are mirrored to email and WhatsApp per user preference.
Profile Switcher	Shown only for users linked to a company. Toggles Corporate ↔ Personal context; changes payment methods, policies and booking history in view.
Floating AI Launcher	Opens the AI Consultant as a side drawer from any page, carrying page context (e.g. the open booking).
Global Search	Search bookings by PNR, ticket number, traveller name, or hotel confirmation number.

Component	Behaviour
Talk to a Consultant	Visible in chat and on every booking; starts the Human Escalation flow (Module 13).
Status Badges	Draft · On Hold · Pending Approval · Pending Payment · Confirmed · Ticketed · Changed · Cancelled · Refunded

Access, Authentication & Onboarding

Release: Phase 1 (MVP)

Foundation module used by every other module. Covers registration, sign-in, social login, OTP, password recovery, two-factor authentication, corporate invitation onboarding, session handling and role-based access.

SCREEN INVENTORY

ID	Screen	Purpose	Access
AUTH-01	Sign In	Email/phone + password, social sign-in, OTP option	Public
AUTH-02	Create Account	Self-registration for personal travellers	Public
AUTH-03	Verify OTP	Email / SMS / WhatsApp one-time code	Public
AUTH-04	Forgot / Reset Password	Recovery via email link or OTP	Public
AUTH-05	Two-Factor Setup	Authenticator app or SMS 2FA	Signed-in
AUTH-06	Accept Corporate Invitation	Employee activation from company invite link	Invite link
AUTH-07	Onboarding Wizard	Name, nationality, home airport, preferences, passport upload prompt	First sign-in
AUTH-08	Active Sessions & Devices	View and revoke logged-in devices	Signed-in

WIREFRAME — AUTH-01 Sign In

app.flightone.pk/login

FlightOnePublic Web

SA

Brand imagery / destination visual

Sign in to FlightOne

Your AI travel consultant

Email or mobile number

Password

☒ Keep me signed in [Forgot password?](#)

Sign In

— or —

[Continue with Google](#) [Continue with Apple](#) [Sign in with OTP](#)

New to FlightOne? [Create account](#)

UF-00.1 Sign In with Email & Password

PRIMARY ACTOR		PRE-CONDITIONS
Traveller / any user		User has a verified account.
#	Actor	Action / System Response
1	User	Opens app.flightone.pk and clicks Sign In .
2	System	Displays AUTH-01 Sign In screen.
3	User	Enters email (or mobile number) and password; optionally ticks <i>Keep me signed in</i> .
4	User	Clicks Sign In .

#	Actor	Action / System Response
5	System	Validates format (email/E.164 phone, password not empty); shows inline errors if invalid.
6	System	Verifies credentials against Identity Service; checks account status (active/locked).
7	System	If 2FA is enabled, opens AUTH-03 and sends a 6-digit code; user enters code and clicks Verify .
8	System	Creates session (JWT access token 15 min + refresh token 30 days if 'keep me signed in', else 12 h); records device and IP.
9	System	Resolves role(s) and profile context (Personal/Corporate) and redirects to Home, or to the originally requested page.
POST-CONDITIONS		
User is authenticated; login event is written to the audit log.		
ALTERNATE & EXCEPTION FLOWS		
• Wrong credentials — System shows "Email or password is incorrect" without revealing which; failed-attempt counter increments. • 5 failed attempts — Account is locked for 15 minutes; unlock link sent by email. • Unverified account — System resends verification OTP and opens AUTH-03. • New device — System sends a "New sign-in" alert by email/WhatsApp with a 'This wasn't me' link.		

WIREFRAME — AUTH-02 Create Account / AUTH-03 Verify OTP

app.flightone.pk/register

FlightOnePublic Web

Create your account

First name

Sara

Last name

Ahmed

Email

sara.ahmed@gmail.com

Country code

+92

Mobile (WhatsApp)

300 1234567

Password

••••••••

8+ chars, 1 uppercase, 1 number, 1 symbol

☒ I agree to the Terms & Privacy Policy

☐ Send me offers on WhatsApp

Create Account

Verify your mobile

Enter the 6-digit code sent to WhatsApp +92 300 •••4567

4

8

1

9

Resend code in 0:42

Verify

UF-00.2 Create Account (Self-Registration)		
PRIMARY ACTOR		PRE-CONDITIONS
Guest visitor		None.
#	Actor	Action / System Response
1	User	Clicks Create account on AUTH-01.
2	User	Enters first name, last name, email, country code, mobile number and password; accepts Terms; optionally opts in to WhatsApp offers.
3	User	Clicks Create Account .
4	System	Validates fields and checks email/mobile are not already registered.

#	Actor	Action / System Response
5	System	Creates account in <i>Pending Verification</i> state and sends OTP to WhatsApp (fallback SMS) and a verification link to email.
6	User	Enters the 6-digit OTP and clicks Verify .
7	System	Verifies OTP (valid 5 minutes, max 3 attempts); marks mobile verified; activates account.
8	System	Signs the user in and launches the Onboarding Wizard (AUTH-07).
POST-CONDITIONS Active account with verified mobile; email verification may complete later via link.		
ALTERNATE & EXCEPTION FLOWS Email already exists — System shows “An account already exists — Sign in or reset password”. OTP expired — User clicks Resend code (enabled after 60 s; max 5 per hour).		

UF-00.3 Sign In with Google / Apple

PRIMARY ACTOR Traveller		PRE-CONDITIONS User has a Google or Apple ID.
#	Actor	Action / System Response
1	User	Clicks Continue with Google (or Apple).
2	Supplier	Provider consent screen opens; user selects account and approves.
3	System	Receives OAuth token, reads verified email and name.
4	System	If email matches an existing account → links provider and signs in; otherwise creates a new account.
5	System	If mobile number is missing, asks user to add and verify mobile (required for WhatsApp alerts).
6	System	Redirects to Home (or Onboarding Wizard for new users).
POST-CONDITIONS User authenticated via SSO; provider linked to profile.		

UF-00.4 Sign In with OTP (Passwordless)

PRIMARY ACTOR Traveller		PRE-CONDITIONS Verified mobile or email.
#	Actor	Action / System Response
1	User	Clicks Sign in with OTP and enters mobile number or email.
2	User	Clicks Send Code .
3	System	Sends 6-digit code via WhatsApp/SMS or email.
4	User	Enters code and clicks Verify .
5	System	Validates code and signs user in.
POST-CONDITIONS User authenticated.		

UF-00.5 Forgot / Reset Password

PRIMARY ACTOR
Any user

PRE-CONDITIONS
Account exists.

#	Actor	Action / System Response
1	User	Clicks Forgot password? on AUTH-01.
2	User	Enters registered email or mobile and clicks Send reset link .
3	System	Always shows "If an account exists, we've sent instructions" (no account enumeration).
4	System	Sends reset link (valid 30 min, single use) by email or OTP by WhatsApp.
5	User	Opens link, enters new password twice, clicks Reset Password .
6	System	Validates strength and that it differs from last 3 passwords; updates password; revokes all other sessions.
7	System	Shows success and redirects to Sign In; sends "Password changed" alert.

POST-CONDITIONS

Password updated; other sessions logged out.

WIREFRAME — AUTH-06 Accept Corporate Invitation → AUTH-07 Onboarding Wizard

app.flightone.pk/onboarding

FlightOne
Traveller Portal
SA

1 Personal details
2 Travel preferences
3 Documents
4 Notifications

Tell us how you like to travel

Home airport

Preferred cabin

Seat preference

LHE — Allama Iqbal Intl.

Economy

Aisle

Meal preference

Preferred airlines

Muslim meal (MOML)

Emirates, Qatar Airways

Direct flights

Morning departures

4★+ hotels

Refundable fares

Near city centre

Corporate account detected: Acme Pharma (Pvt) Ltd · Cost centre CC-104 Sales · Approver: Imran Qureshi

Back

Skip for now

Continue

UF-00.6 Accept Corporate Invitation

PRIMARY ACTOR
Corporate employee

PRE-CONDITIONS
Corporate Admin has invited the employee (Module 6).

#	Actor	Action / System Response
1	System	Sends invitation email/WhatsApp with a unique link (valid 7 days).
2	User	Clicks the invitation link.
3	System	If the email already has a personal account, asks the user to sign in and Link to company ; otherwise opens a pre-filled registration form (name, email, employee ID, department locked).
4	User	Sets password (or company SSO if configured), verifies mobile OTP, clicks Activate .
5	System	Creates Corporate profile linked to company, department, cost centre, policy and approver; enables Profile Switcher.
6	System	Launches Onboarding Wizard.

POST-CONDITIONS

Employee is active under the company account.

ALTERNATE & EXCEPTION FLOWS

- **Link expired** — System shows “Invitation expired — request a new one” and notifies Corporate Admin.
- **Company SSO (SAML/OIDC)** — User is redirected to the company identity provider instead of setting a password.

UF-00.7 First-time Onboarding Wizard

PRIMARY ACTOR

New user

PRE-CONDITIONS

First successful sign-in.

#	Actor	Action / System Response
1	System	Opens AUTH-07 at step 1 <i>Personal details</i> (title, name as per passport, date of birth, gender, nationality).
2	User	Completes fields and clicks Continue .
3	User	Step 2: selects home airport, cabin, seat, meal, preferred airlines, and preference chips; clicks Continue .
4	User	Step 3: clicks Upload passport (launches OCR flow UF-02.2) or Skip for now .
5	User	Step 4: chooses notification channels (App, Email, WhatsApp) and quiet hours; clicks Finish .
6	System	Saves profile and preferences; AI Consultant loads them as context for future conversations; opens Home.

POST-CONDITIONS

Profile is at least partially complete; completion % is shown on Profile.

ALTERNATE & EXCEPTION FLOWS

- **User skips** — Wizard can be resumed from Profile; a banner shows “Complete your profile for 1-tap booking”.

UF-00.8 Enable Two-Factor Authentication & Manage Sessions

PRIMARY ACTOR

Any user

PRE-CONDITIONS

Signed in.

#	Actor	Action / System Response
1	User	Opens Profile & Settings → Security and clicks Enable 2FA .
2	User	Selects method: Authenticator app (QR) or WhatsApp/SMS OTP.
3	System	Displays QR code/secret or sends OTP; user enters code to confirm.
4	System	Enables 2FA and shows 10 backup codes to download.
5	User	Opens Active Sessions , reviews device, browser, location and last active time.
6	User	Clicks Sign out on a device (or Sign out all others).
7	System	Revokes tokens for the selected sessions immediately.

POST-CONDITIONS

2FA active; unwanted sessions revoked.

BUSINESS RULES

1. Role-based access control (RBAC) with permission sets per role; one user may hold several roles (e.g. Traveller + Approver).
2. 2FA is mandatory for Ops, Finance, Pricing, Admin and Corporate Admin roles.
3. Sessions expire after 30 min of inactivity on the Operations Console; 12 h for travellers unless ‘Keep me signed in’.
4. All authentication events (success, failure, lockout, password change, 2FA change) are written to the audit log (Module 15).
5. Name fields captured during onboarding must match passport spelling for ticketing; the system warns if they differ from OCR data.

DATA — USER ACCOUNT

Field	Type	Req.	Validation / Notes
user_id	UUID	Y	System generated
email	String	Y	Unique, RFC 5322
mobile	String	Y	E.164, unique, verified flag
password_hash	String	C	Argon2id; null for SSO-only
status	Enum	Y	pending, active, locked, suspended, closed
roles[]	Array	Y	RBAC role codes
company_id	UUID	N	Set when linked to a corporate account
mfa_method	Enum	N	none, totp, sms, whatsapp
last_login_at	Datetime	N	UTC
notification_prefs	JSON	Y	Channel × event matrix, quiet hours

AI Travel Consultant

Release: Phase 1 (MVP)

A conversational travel consultant that understands natural language, asks only the questions that matter, gives short decision-oriented answers, and moves the traveller from intent to confirmed booking.

1.1 Functional Overview

- Natural-language chat (English, Urdu and Roman Urdu) for planning, pricing, booking and servicing.
- Asks targeted follow-up questions only for missing essentials (origin, destination, dates, travellers, cabin, budget).
- **Answers are short, crisp and decision-led** — every reply ends with a clear next action (select, hold, book, pay) to drive conversion.
- Refines requirements, proposes practical alternatives (dates, airports, routings), recommends optimal itineraries and explains why.
- Learns and applies stored preferences (Module 2) and corporate policy (Module 6).
- Conversation history persists server-side and continues across browsers and devices.
- Can act: search, hold, book, pay, send documents, open refunds, escalate to a human (Module 13).

SCREEN INVENTORY

ID	Screen	Purpose	Access
AI-01	AI Consultant Workspace	Main chat with inline rich cards (options, fare summary, payment, documents)	Traveller
AI-02	Conversation History	List of past conversations with status and linked bookings	Traveller
AI-03	Trip Brief Panel	Live summary of captured requirements; editable fields	Traveller
AI-04	"Why this?" Explanation Drawer	Reasoning and trade-offs for a recommendation	Traveller
AI-05	What FlightOne AI Knows About Me	View / edit / delete learned preferences	Traveller

FLIGHTONE

Home

AI Consultant

Search & Book

My Trips

Traveller Vault

Visa Centre

Groups

Rewards

Profile & Settings

Help

Need Lahore to Dubai next Friday, back Monday. 2 of us.

Got it — LHE → DXB, Fri 19 Sep to Mon 22 Sep, 2 adults. Economy, and prefer direct?

Yes direct, morning if possible

Here are my top 3. **Best overall: Emirates 09:40, PKR 186,400 for both, refundable.**

BEST OVERALL

Emirates EK623 09:40 → 12:05

Direct · 3h25 · 30kg

PKR 186,400

Select

CHEAPEST

flydubai FZ360 04:15 → 06:30

Direct · 20kg

PKR 142,900

Select

FASTEST

PIA PK203 10:30 → 12:40

Direct · 3h10

PKR 171,200

Select

Shall I hold Emirates for 2 hours while you confirm? [Why this?](#)

Hold Emirates

Show evening flights

Add hotel

Talk to a consultant

Type a message... (Urdu / English)

Send

Trip Brief

From

Lahore (LHE)

To

Dubai (DXB)

Depart

19 Sep

Return

22 Sep

Travellers

2 Adults

Cabin

Economy

Direct

Morning

Profile: Personal · Pay by card

Edit brief

Replies are limited to ~2 short sentences plus actionable cards and quick-reply chips; the brief panel updates live as the AI extracts intent.

UF-01.1 Start a Conversation & Capture Travel Intent		
PRIMARY ACTOR		PRE-CONDITIONS
Traveller		Signed in (guests may chat but must sign in before holding/booking).
#	Actor	Action / System Response
1	User	Clicks AI Consultant in the side menu (or the floating chat launcher).
2	System	Opens AI-01 with a greeting and suggested prompts (“Plan a trip”, “Change my booking”, “Visa for UK?”).
3	User	Types a request, e.g. “Lahore to Dubai next Friday, back Monday, 2 of us”, and clicks Send (or presses Enter / uses voice input).
4	AI	Classifies intent (new trip / modify / cancel / info / visa / support) and extracts entities: origin, destination, dates (resolves “next Friday” in user time zone), travellers, cabin.
5	System	Populates the Trip Brief panel (AI-03) with extracted values.
6	AI	Identifies missing or ambiguous essentials and asks one concise follow-up covering them (e.g. “Economy, and prefer direct?”).
7	User	Answers the follow-up or taps a quick-reply chip.
8	AI	Confirms the brief in one line and triggers search (UF-03.1 / UF-03.3).
9	AI	Presents 3 curated options as cards (Module 4) with one-line summary and a recommended next action (Hold / Book).
POST-CONDITIONS		
Conversation saved with structured Trip Brief; search session ID linked.		
ALTERNATE & EXCEPTION FLOWS		
• Ambiguous city (e.g. “Jeddah or Madinah?”) — AI offers chips for each airport and a “both” option.		

- **Past date / invalid date** — AI corrects: “19 Sep has passed for this year — did you mean next Friday, 26 Sep?”
- **Out-of-scope request** — AI politely declines and steers back to travel; offers consultant.
- **Corporate profile active** — AI applies policy (Module 6) before searching and labels out-of-policy options.

UF-01.2 Refine Requirements & Get Practical Alternatives

PRIMARY ACTOR		PRE-CONDITIONS
Traveller		Options already shown.
#	Actor	Action / System Response
1	User	Types a refinement (“anything cheaper on Saturday?”, “evening only”, “under 150k”) or edits a field in the Trip Brief panel.
2	AI	Updates the brief (diff shown in panel) and re-runs search only for changed parameters (cached results reused within 10 minutes).
3	AI	If no options satisfy constraints, proposes practical alternatives: ±1–3 day flex, nearby airports, 1-stop routings, alternate cabin, or different hotel area (see UF-03.2).
4	AI	States the trade-off in one sentence (“Saturday saves PKR 22,000 but arrives 23:10”).
5	User	Selects an alternative chip or card.
POST-CONDITIONS		Brief and recommendations updated; refinement history kept.

WIREFRAME — AI-04 “Why this?” Explanation Drawer

app.flightone.pk/assistant/c/8812#why

FlightOne Traveller Portal

SA

...conversation...

Chat (dimmed)

Why Emirates EK623?

Price vs. best

72%

Journey time

95%

Arrival time fit

90%

Airline quality

92%

Refundability

100%

Your preferences

88%

✓ Direct, morning departure as requested

✓ Refundable with PKR 15,000 fee

✓ Earns ~2,480 Skywards miles on your saved number

✗ PKR 43,500 more than cheapest (flydubai 04:15, non-refundable)

Book this

Prefer cheapest

UF-01.3 Explain a Recommendation

PRIMARY ACTOR		PRE-CONDITIONS
Traveller		At least one recommendation displayed.
#	Actor	Action / System Response
1	User	Clicks Why this? on a card or asks “why Emirates?”.
2	System	Opens AI-04 drawer with factor scores from the Recommendation Engine (Module 4).
3	AI	Writes max 3 pros and 1–2 cons in plain language, including loyalty earning and refund terms.
4	User	Clicks Book this , or Prefer cheapest / Prefer fastest to re-weight.

#	Actor	Action / System Response
5	AI	Re-ranks and responds with the updated top pick.
POST-CONDITIONS User understands trade-offs; weight change recorded as preference signal.		

UF-01.4 Book Directly from Chat

PRIMARY ACTOR Traveller		PRE-CONDITIONS Option selected; profile has traveller details.
#	Actor	Action / System Response
1	User	Clicks Select on a card or types "book the Emirates one".
2	AI	Confirms passengers from saved profile/companions ("Sara Ahmed and Ali Ahmed — correct?") and shows a fare summary card (base, taxes, fees, total, baggage, refund rule).
3	User	Clicks Confirm travellers (or edits / adds a companion).
4	System	Runs price and availability re-check with supplier (UF-03.5).
5	AI	Offers Hold (if fare allows) or Pay now ; for corporate profile shows Send for approval .
6	User	Clicks Pay now .
7	System	Renders inline payment card (Module 3 payment flow UF-03.6).
8	System	On success, issues ticket, posts e-ticket and itinerary in chat, sends via email/WhatsApp.
9	AI	Asks one follow-up to upsell: "Need a hotel near Downtown for these dates?"
POST-CONDITIONS Booking created and linked to the conversation.		
ALTERNATE & EXCEPTION FLOWS • Missing passport data — AI asks for upload in chat; OCR runs inline (UF-02.2). • Price changed — AI states the new price and asks to proceed or see alternatives.		

UF-01.5 Learn & Apply Customer Preferences

PRIMARY ACTOR System / AI		PRE-CONDITIONS User has interacted at least once.
#	Actor	Action / System Response
1	AI	Detects explicit preferences ("I always fly aisle") and implicit signals (repeatedly chooses morning flights, 4★ hotels).
2	AI	Explicit preferences: asks "Save aisle seat as your default?" — user clicks Save .
3	System	Implicit signals are stored with a confidence score; applied as a ranking boost once confidence ≥ 0.7 .
4	User	Opens Profile → What FlightOne AI knows (AI-05) to view learned preferences.
5	User	Edits or deletes any preference, or turns Personalisation off.
6	System	Updates the preference store; AI uses updated context from the next message.
POST-CONDITIONS Preferences stored transparently and under user control.		

UF-01.6 Continue Conversation Across Devices

PRIMARY ACTOR
Traveller

PRE-CONDITIONS
Signed in on another device.

#	Actor	Action / System Response
---	-------	--------------------------

1	User	Signs in on a second device and opens AI Consultant .
---	------	--

2	System	Loads AI-02 history; the most recent active conversation is pinned on top with an “Active” badge.
---	--------	---

3	User	Clicks the conversation.
---	------	--------------------------

4	System	Restores full message history, Trip Brief, holds and pending payments with live timers.
---	--------	---

5	AI	Posts a context line: “Welcome back — your Emirates hold expires in 42 min. Pay now?”
---	----	---

POST-CONDITIONS

Same conversation state on all devices (real-time sync via WebSocket).

UF-01.7 Conversation Servicing Requests

PRIMARY ACTOR
Traveller

PRE-CONDITIONS
Existing booking.

#	Actor	Action / System Response
---	-------	--------------------------

1	User	Asks “change my return to Tuesday” or “cancel my hotel”.
---	------	--

2	AI	Identifies booking (asks to pick if several) and the service type.
---	----	--

3	System	Fetches fare rules / hotel policy and computes the cost via Module 14.
---	--------	--

4	AI	Presents the outcome in one card: new itinerary, fare difference, penalty, total payable/refundable.
---	----	--

5	User	Clicks Confirm change .
---	------	--------------------------------

6	System	Executes reissue/cancellation or routes to Ops queue if not automatable; posts updated documents.
---	--------	---

POST-CONDITIONS

Change completed or queued with ETA.

BUSINESS RULES

1. Response style: maximum 2 short sentences (≈40 words) before cards; always end with a call to action or quick-reply chips.
2. Never show more than 3 primary options in chat; “See more options” opens the full results page.
3. AI must re-confirm passenger names, dates and total price before any payment or ticketing action (human-in-the-loop confirmation).
4. AI cannot exceed discount limits configured in Module 5; cannot bypass corporate policy or approvals.
5. Sensitive data (passport numbers, card data) is masked in chat transcripts.
6. Escalation triggers (Module 13) are evaluated on every turn.
7. All AI replies are grounded in live supplier data or the Knowledge Base (Module 16); if uncertain, the AI says so and offers a consultant.
8. Supported languages: English, Urdu, Roman Urdu; reply in the language of the user’s last message.

DATA — CONVERSATION & TRIP BRIEF

Field	Type	Req.	Validation / Notes
conversation_id	UUID	Y	—
user_id / profile_context	UUID / Enum	Y	personal corporate
messages[]	Array	Y	role, content, cards, timestamp, channel

Field	Type	Req.	Validation / Notes
intent	Enum	Y	new_trip, modify, cancel, info, visa, support
trip_brief	JSON	Y	origin, destination, dates, pax, cabin, budget, constraints
linked_booking_ids[]	Array	N	—
status	Enum	Y	active, converted, abandoned, escalated, closed
csat	Int 1-5	N	Asked at close

Customer Identity & Profile

Release: Phase 1 (MVP)

A permanent, verified traveller profile — identity documents, loyalty, preferences, companions and history — powering one-tap booking and proactive expiry alerts.

2.1 Functional Overview

Every traveller has one permanent profile. Identity documents can be captured with OCR, must be verified by the user (or re-uploaded), and are monitored for expiry with advance notifications.

SCREEN INVENTORY

ID	Screen	Purpose	Access
PRF-01	Profile Overview	Completion %, personal info, quick links to all sections	Traveller
PRF-02	Identity Documents	Passport, CNIC/National ID, Visas, Residence Permits with expiry status	Traveller
PRF-03	Upload & OCR Capture	Upload/scan document; OCR extraction progress	Traveller
PRF-04	Verify Extracted Data	Side-by-side image and editable fields; confirm or re-upload	Traveller
PRF-05	Loyalty Programmes	Frequent flyer and hotel loyalty numbers	Traveller
PRF-06	Travel Preferences	Seat, meal, cabin, airlines, hotels, special assistance	Traveller
PRF-07	Companions & Family	Saved companions and family members with their own documents	Traveller
PRF-08	Emergency Contacts	Contact persons notified in emergencies	Traveller
PRF-09	Travel History	Past trips, spend, countries visited	Traveller

WIREFRAME — PRF-01 Profile Overview

app.flightone.pk/profile

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Photo

Sara Ahmed

Pakistani · DOB 14-Mar-1990 · +92 300 1234567 · sara.ahmed@gmail.com

Edit

Personal

Corporate: Acme Pharma

Profile completion 80%

80%

Documents

Loyalty

Preferences

Companions & Family

Emergency Contacts

Travel History

Document	Number	Country	Expiry	Status	
Passport	AB●●●4521	Pakistan	19-Dec-2026	Expires in 94 days	Renew / Update
CNIC	35202-●●●●●●-8	Pakistan	02-May-2031	Verified	View
UAE Visa	●●●7781	UAE	10-Nov-2026	Verified	View
UK Residence Permit	—	UK	—	Needs verification	Verify

+ Add Document

Scan with camera

app.flightone.pk/profile/documents/new

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1 Choose type2 Upload3 Verify data4 Saved

Passport image (bio page) — zoom / rotate

Re-uploadRotate

We read your passport. Please check each field — highlighted fields have low confidence.

Surname

AHMED

Given names

SARA

Passport no.

AB1234521

Nationality

Pakistani

Date of birth

14/03/1990

Gender

F

Issue date

20/12/2021

Expiry date

19/12/2026

Place of issue

LAHORE

MRZ check

Valid

I confirm these details match my document exactly

CancelConfirm & Save

UF-02.1 View & Edit Personal Details		
PRIMARY ACTOR Traveller		PRE-CONDITIONS Signed in.
#	Actor	Action / System Response
1	User	Clicks Profile & Settings in the side menu.
2	System	Displays PRF-01 with completion percentage and tabs.
3	User	Clicks Edit , updates title, name, DOB, gender, nationality, contact details or photo.
4	User	Clicks Save .
5	System	Validates fields; if name/DOB differs from a verified passport, warns “This must match your passport for ticketing”.
6	System	If email/mobile changed, sends OTP to the new value before applying; saves and logs change history.
POST-CONDITIONS Profile updated.		

UF-02.2 Upload Passport with OCR		
PRIMARY ACTOR Traveller		PRE-CONDITIONS Signed in.
#	Actor	Action / System Response
1	User	On Documents tab clicks + Add Document and selects Passport .
2	User	Clicks Upload (JPG/PNG/PDF ≤ 10 MB) or Scan with camera (webcam/phone camera).
3	System	Runs image quality checks (blur, glare, cropping, resolution); asks to retake if failed.
4	System	Sends image to OCR service; reads the MRZ and visual zone; validates MRZ check digits.
5	System	Opens PRF-04 with extracted fields pre-filled; fields below 85% confidence are highlighted amber.

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#	Actor	Action / System Response
6	User	Reviews each field against the image; corrects any value.
7	User	Ticks the confirmation checkbox and clicks Confirm & Save .
8	System	Encrypts and stores image in the Traveller Vault (Module 7); stores fields; sets status <i>Verified by user</i> .
9	System	Schedules expiry reminders (UF-02.6) and updates any upcoming bookings' APIS data.
POST-CONDITIONS Verified passport available for 1-tap booking.		
ALTERNATE & EXCEPTION FLOWS • OCR fails / unreadable — System shows “We couldn’t read this document” with tips; user clicks Re-upload or Enter manually. • MRZ check-digit mismatch — Fields are flagged red; save is blocked until user corrects or re-uploads. • Passport expires within 6 months — System warns that many countries require 6 months' validity. • Duplicate passport number — System asks whether to replace the existing record.		

UF-02.3 Manual Verification / Re-upload		
PRIMARY ACTOR Traveller; Ops (optional)		PRE-CONDITIONS Document in <i>Needs verification</i> state.
#	Actor	Action / System Response
1	User	Clicks Verify on a document row.
2	System	Opens PRF-04 with stored image and values.
3	User	Either corrects fields and clicks Confirm & Save , or clicks Re-upload to replace the image (restarts UF-02.2).
4	System	For documents flagged by risk rules (name mismatch with bookings, tampered image), routes to Ops verification queue.
5	Ops	Ops reviews image and data; clicks Approve or Reject with reason .
6	System	Updates status to <i>Verified by FlightOne</i> or <i>Rejected</i> and notifies the user.
POST-CONDITIONS Document verified or user asked to re-upload.		

UF-02.4 Add CNIC, Visa or Residence Permit		
PRIMARY ACTOR Traveller		PRE-CONDITIONS Signed in.
#	Actor	Action / System Response
1	User	Clicks + Add Document and selects CNIC / National ID, Visa, or Residence Permit.
2	User	Uploads front (and back for CNIC) or scans with camera.
3	System	OCR extracts: CNIC number (13-digit format), name, father/husband name, DOB, expiry — or visa type, country, entries, validity, duration of stay.
4	User	Verifies fields and links a visa/permit to the passport it is stamped in; clicks Save .
5	System	Saves document; Visa Engine (Module 8) uses it for eligibility checks; schedules expiry reminders.
POST-CONDITIONS Document stored and linked.		

app.flightone.pk/profile/companions

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Loyalty

Programme	Number	Tier
Emirates Skywards	EK●●●3321	Silver
Marriott Bonvoy	●●●8870	Member
+ Add programme		

Preferences

Seat

Aisle

Meal

MOML

Special assistance

None

Companions & Family

Name	Relation	Passport
Ali Ahmed	Spouse	VerifiedEdit
Zara Ali (8y)	Child	ExpiringEdit
Omar Khan	Colleague	MissingRequest
+ Add family member + Add companion		

Emergency Contacts

Name	Relation	Phone
Ali Ahmed	Spouse	+92 321 ●●●0099
+ Add contact		

UF-02.5 Add Loyalty Programme, Preferences, Family, Companions & Emergency Contacts		
PRIMARY ACTOR Traveller		PRE-CONDITIONS Signed in.
#	Actor	Action / System Response
1	User	Opens Loyalty tab → clicks + Add programme → selects airline or hotel programme from list → enters membership number and tier → clicks Save .
2	System	Validates number format per programme; loyalty number is auto-added to future PNRs/hotel bookings for that brand/alliance.
3	User	Opens Preferences tab → sets seat, meal (IATA SSR code), cabin, preferred/avoided airlines, hotel star rating, bed type, special assistance (WCHR etc.) → clicks Save .
4	User	Opens Companions & Family → clicks + Add family member → enters name as per passport, relation, DOB, gender, nationality → optionally uploads their passport (OCR flow) → clicks Save .
5	System	Minors (<18) are stored under the guardian; child/infant passenger type is derived from DOB at travel date.
6	User	Clicks + Add companion (non-family) → enters email/mobile → clicks Send request .
7	System	Sends consent request; companion accepts and chooses which documents to share; until then only name is stored.
8	User	Opens Emergency Contacts → clicks + Add contact → name, relation, phone, email → Save (max 3).
9	System	Emergency contacts are used in PNR (PCTC SSR) and for Group/Live Journey emergency broadcasts.
POST-CONDITIONS Profile enriched for faster, accurate bookings.		

UF-02.6 Document Expiry Notifications		
PRIMARY ACTOR System		PRE-CONDITIONS Document with expiry date saved.
#	Actor	Action / System Response
1	System	Nightly job evaluates all documents against reminder schedule: 180, 90, 60, 30, 7 days before expiry (configurable).
2	System	Sends notification via App, Email and WhatsApp: "Your passport expires on 19 Dec 2026. Renew now to avoid travel issues."

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#	Actor	Action / System Response
3	System	If an upcoming booking's travel date falls within 6 months of expiry, sends a high-priority alert and flags the booking for Ops.
4	User	Clicks the notification link.
5	System	Opens PRF-02 with the document highlighted and a Renew / Update button (starts UF-02.2 with 'replace' mode).
6	System	After expiry, document status becomes <i>Expired</i> and cannot be used for new bookings.
POST-CONDITIONS		
User informed in time; expired documents blocked.		

UF-02.7 View Travel History

PRIMARY ACTOR		PRE-CONDITIONS
Traveller		At least one completed trip.
#	Actor	Action / System Response
1	User	Opens Travel History tab.
2	System	Shows past trips with dates, route, airline/hotel, amount, profile (Personal/Corporate), and documents.
3	User	Filters by year, profile, or country; clicks a trip to view details or Rebook .
4	System	Rebook opens AI Consultant with a pre-filled Trip Brief.
POST-CONDITIONS		
History available; rebooking accelerated.		

BUSINESS RULES

1. Name as per passport is the booking name of record; mismatches block ticketing.
2. OCR data is never saved without explicit user confirmation (manual verification step is mandatory).
3. Document images are stored encrypted (AES-256) and only viewable after re-authentication.
4. Travellers can hold only one active passport per nationality; replaced passports are archived.
5. Companion data sharing requires the companion's consent; family members under a guardian do not.
6. Expiry reminder schedule and channels are configurable by Super Admin; users can mute non-critical channels but not the 30/7-day alerts.

DATA — IDENTITY DOCUMENT

Field	Type	Req.	Validation / Notes
document_id	UUID	Y	—
owner_id	UUID	Y	Traveller or family member
type	Enum	Y	passport, cnic, national_id, visa, residence_permit
number	String (enc.)	Y	Masked in UI
issuing_country	ISO-3166	Y	—
issue_date / expiry_date	Date	Y/Y	Expiry > issue
ocr_confidence	JSON	N	Per-field score
verification_status	Enum	Y	unverified, user_verified, ops_verified, rejected, expired
file_ref	String	Y	Vault object key
linked_passport_id	UUID	N	For visas/permits

AI Booking Engine

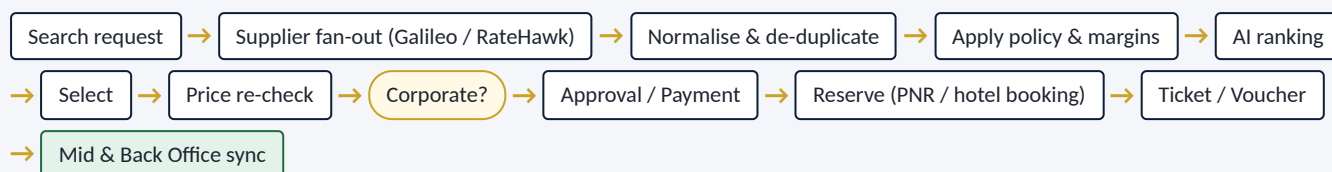
Release: Phase 1 (MVP)

Searches Galileo and RateHawk, compares and de-duplicates supplier results, ranks with AI, applies margins, reserves inventory, handles payment or approval, issues tickets and vouchers, and updates mid/back office automatically.

3.1 Functional Overview

The Booking Engine orchestrates search, comparison, ranking, pricing, reservation, payment/approval, ticketing, voucher generation and mid/back-office updates. Phase 1 suppliers: **Galileo GDS** (flights) and **RateHawk** (hotels). It is used both by the AI Consultant (conversational) and by the manual Search & Book screens.

END-TO-END BOOKING PIPELINE



SCREEN INVENTORY

ID	Screen	Purpose	Access
BK-01	Search (Flights / Hotels / Flight+Hotel)	Manual search form	Traveller
BK-02	Flight Results	AI top picks + full sortable/filterable list	Traveller
BK-03	Flight Details & Fare Rules	Segments, baggage, fare family, change/refund rules	Traveller
BK-04	Hotel Results	Map/list, AI top picks, filters	Traveller
BK-05	Hotel Details & Room Selection	Photos, amenities, rooms, meal plans, cancellation	Traveller
BK-06	Traveller Details	Select profile/companions, contact, SSRs, loyalty	Traveller
BK-07	Review & Price Check	Final itinerary, price breakdown, policy status	Traveller
BK-08	Payment	Card / wallet / bank transfer / credits / corporate billing	Traveller
BK-09	Confirmation	PNR, ticket numbers, vouchers, add-ons	Traveller
BK-10	My Trips — Booking Detail	Status, documents, manage booking	Traveller

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app.flightone.pk/search

FlightOneTraveller Portal

FlightsHotelsFlight + Hotel

ReturnOne-wayMulti-city

From

Lahore (LHE)

To

Dubai (DXB)

Depart

Fri 19 Sep

Return

Mon 22 Sep

Travellers

2 Adults, 0 Child

Cabin

Economy

Preferred airline

Any

Profile

Personal

☒ Direct only☐ Flexible dates ±3 days☒ Include nearby airports (ISB, SKT, MUX)☐ Refundable only

Search Flights

Ask AI instead

UF-03.1 Search Flights (Galileo GDS)		
PRIMARY ACTOR		PRE-CONDITIONS
Traveller (or AI on their behalf)		Signed in (guest search allowed; booking needs sign-in).
#	Actor	Action / System Response
1	User	Clicks Search & Book → Flights tab.
2	User	Selects trip type (Return / One-way / Multi-city).
3	User	Types origin; system autocompletes cities/airports (IATA); user selects. Repeats for destination.
4	User	Picks dates from calendar (calendar shows lowest cached fare per day), selects travellers (adults/children/infants), cabin and options.
5	User	Clicks Search Flights .
6	System	Validates: origin ≠ destination, return ≥ depart, infants ≤ adults, max 9 passengers.
7	System	Booking Orchestrator calls Galileo (Air Low Fare Search) — in parallel for alternate airports/dates if selected (UF-03.2).
8	Supplier	Galileo returns itineraries with fares, fare basis, brand/fare family, baggage and availability.
9	System	Normalises results, applies corporate policy flags (Module 6) and FlightOne margins (Module 5).
10	AI	Ranks results (Module 4) and selects Top 3 picks.
11	System	Displays BK-02 within target ≤ 6 s (skeleton loader and progressive results while waiting).
POST-CONDITIONS		
Search session stored (15-min validity) with result set.		
ALTERNATE & EXCEPTION FLOWS		
No results — System suggests flexible dates, nearby airports, or 1-stop options; offers AI or consultant help. Supplier timeout (>15 s) — Shows cached/partial results with notice and retry; logs supplier performance (Module 17). Session expired — On any action after 15 min, system re-runs the search and highlights price changes.		

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Filters

☒ Direct

☐ 1 stop

Airlines

☒ Emirates

☒ flydubai

☒ PIA

Departure time

Morning

Afternoon

Evening

Price

Max /pp

PKR 200k

☐ Refundable only

☐ Within policy

LHE → DXB · 19–22 Sep · 2 Adults · Economy | 64 results

Fly from Islamabad and save PKR 18,600 · Arrive Sharjah (30 min to Dubai) and save PKR 24,300 → [View](#)

BEST OVERALL

Emirates 09:40–12:05 · Direct

PKR 93,200 /pp

Select

CHEAPEST

flydubai 04:15–06:30 · Direct

PKR 71,450 /pp

Select

FASTEST

PIA 10:30–12:40 · Direct

PKR 85,600 /pp

Select

Recommended

Cheapest

Fastest

Earliest

Airline	Depart	Arrive	Stops	Bag	Fare	Price /pp	
Emirates EK623	09:40	12:05	Direct	30kg	Flex	93,200	Select
Qatar QR629 · via DOH	03:00	08:40	1	30kg	Saver	88,900	Select
Airblue PA410	13:15	15:30	Direct	20kg	Value	74,300	Out of policy
Emirates EK625 · Also on 2 fare brands	21:55	00:15+1	Direct	25kg	Saver	84,100	Compare

UF-03.2 Sales Strategy — Alternate Departure / Arrival Cities		
PRIMARY ACTOR System / AI		PRE-CONDITIONS Search executed.
#	Actor	Action / System Response
1	System	Looks up the <i>Alternate Airports Map</i> (configurable by Ops), e.g. LHE ↔ ISB, SKT, MUX, LYP; DXB ↔ SHJ, DWC, AUH; JED ↔ MED, TIF.
2	System	Runs parallel searches for alternates within configured distance/ground-time limits (e.g. ≤ 300 km or ≤ 4 h).
3	System	Computes effective saving = base price – alternate price – estimated ground transfer cost.
4	AI	If saving ≥ threshold (e.g. PKR 10,000 or 8 %), shows a suggestion banner on BK-02 and a one-line pitch in chat, including transfer time.
5	User	Clicks View on the banner.
6	System	Shows alternate-city results with a ‘Ground transfer’ row (and optional transfer booking in future phases).
7	User	Selects an alternate option and proceeds normally.
POST-CONDITIONS Conversion uplift via cheaper or better-timed alternatives; banner impressions/clicks tracked.		
ALTERNATE & EXCEPTION FLOWS		
• Corporate policy restricts airports — Alternates are hidden for corporate profile.		

UF-03.3 Search Hotels (RateHawk)		
PRIMARY ACTOR Traveller		PRE-CONDITIONS —
#	Actor	Action / System Response
1	User	Selects Hotels tab; enters destination (city, area, landmark or hotel name) with autocomplete.

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#	Actor	Action / System Response
2	User	Selects check-in/check-out dates, rooms and guests per room (adults, children with ages), nationality/residency, and optional star rating.
3	User	Clicks Search Hotels .
4	System	Calls RateHawk search by region/geo (SERP) and hotel static content (cached daily).
5	System	Normalises rates (room name, meal plan, cancellation policy, taxes/fees payable at hotel) and applies margins.
6	System	De-duplicates properties (UF-03.4) and applies policy caps (corporate nightly limit).
7	AI	Ranks and picks Top 3 (Module 4) considering location to trip purpose (e.g. near conference venue).
8	System	Displays BK-04 with list/map toggle.
POST-CONDITIONS		
Hotel search session stored.		

WIREFRAME — BK-04 Hotel Results / BK-05 Room Selection

app.flightone.pk/search/hotels/r/77c2

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Map with price pins

Dubai • Downtown • 19–22 Sep • 1 room, 2 adults • 214 properties

Photo

Rove Downtown ★★★

8.6 • 0.4 km to Dubai Mall

Room only • Free cancel until 17 Sep

PKR 28,900/night

BEST VALUE

Photo

Address Sky View ★★★★★

9.1 • Connected to mall

Breakfast • Non-refundable

PKR 71,400/night

PREMIUM

Rove Downtown — Select room

Room	Meal	Cancellation	Total 3 nts	
Rove Room, 1 King	Room only	Free until 17 Sep	86,700	Select
Rove Room, 1 King	Breakfast	Free until 17 Sep	97,800	Select
Rove Room, 2 Twin	Room only	Non-refundable	79,200	Select

Taxes included • Tourism Dirham AED 15/night payable at hotel

UF-03.4 Compare Suppliers, De-duplicate & Differentiate Similar Options

PRIMARY ACTOR
System / AI

PRE-CONDITIONS
Results from ≥1 supplier.

#	Actor	Action / System Response
1	System	Property/flight matching: hotels are matched across providers using a mapping key (supplier IDs → FlightOne master property ID), then fuzzy match on name, geo-coordinates (≤ 150 m), address and star rating; flights by carrier + flight number + date + fare brand.
2	System	Same name, different product: when the name matches but attributes differ, results are NOT merged. They are shown as variants and differentiated by attributes: room type/bed, meal plan, cancellation policy, payment timing (pay now/at hotel), baggage, fare brand, change fee, supplier reliability.

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#	Actor	Action / System Response
3	System	For identical products from different providers, keeps the best net price (plus reliability score) and stores others as fallbacks for booking failure.
4	AI	Similar-options pitch algorithm: for a selected or sold-out option, finds alternatives with similarity score = weighted match of location (distance), star/class, guest rating, amenities, price band ($\pm 20\%$), airline alliance/timing, and pitches the top 2 ("Similar to Rove Downtown, 300 m away, PKR 3,100 less").
5	User	Clicks Compare on a grouped result to see variants side by side, then selects one.
POST-CONDITIONS		
Clean result list without duplicates; transparent variant comparison; fallback suppliers ready.		

WIREFRAME — BK-06 Traveller Details → BK-07 Review → BK-08 Payment

app.flightone.pk/checkout/9f31

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3 Review

4 Payment

5 Confirmation

Payment method

Credit / Debit card (Visa, Mastercard, UnionPay)

Card number

4111 •••• •••• 1111

Expiry

08/28

CVV

•••

Wallet — JazzCash / Easypaisa

Bank transfer (IBFT) — hold required

Use FlightOne credits (PKR 4,250)

3-D Secure verification will open after you click Pay

I accept fare rules, hotel policies and Terms

Pay PKR 273,100

Summary

LHE-DXB-LHE · Emirates · 2 adults

Rove Downtown · 3 nights

	PKR
Air fare (base)	142,000
Taxes & surcharges	44,400
Hotel	86,700
Service fee	3,000
Promo SAVE3K	-3,000
Total	273,100

Price held until 14:52 (17 min)

UF-03.5 Select, Add Traveller Details & Reserve Inventory

PRIMARY ACTOR		PRE-CONDITIONS
Traveller		Option selected from results.
#	Actor	Action / System Response
1	User	Clicks Select on a flight (and room if Flight+Hotel).
2	System	Opens BK-03/BK-05 details; user reviews fare rules and clicks Continue .
3	System	Opens BK-06 with passenger slots matching the search (e.g. 2 adults).
4	User	For each slot selects self / companion / family member from profile, or clicks Add new traveller and enters details (name as per passport, DOB, gender, nationality, passport no. & expiry).
5	System	Auto-fills loyalty numbers, meal/seat SSRs, and contact details (email, mobile).
6	User	Optionally selects extras: seat map, extra baggage, meal, travel insurance (future).
7	User	Clicks Review .
8	System	Re-prices with Galileo (Air Price) and re-checks hotel availability/rate (RateHawk prebook).
9	System	Shows BK-07 with any price difference highlighted; user must accept changes to continue.
10	System	Creates reservation: Galileo PNR (with ticketing time limit) and/or hotel hold where supported; starts countdown timer.

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#	Actor	Action / System Response
11	System	Routes: Corporate profile → Approval (Module 6); Personal → Payment (UF-03.6).
POST-CONDITIONS Reservation created in <i>On Hold</i> status with time limit.		
ALTERNATE & EXCEPTION FLOWS Name/document mismatch — Blocks review; prompts user to correct passport data. Price increased — Shows old vs new price; options: Accept, See alternatives (similar options), Cancel. Seats no longer available — System auto-fails over to the next best identical fare or shows similar options. Fare not holdable — Proceeds directly to payment; reservation and ticketing happen together.		

UF-03.6 Process Payment		
PRIMARY ACTOR Traveller		PRE-CONDITIONS Reservation on hold; price accepted.
#	Actor	Action / System Response
1	User	On BK-08 selects payment method: card, wallet, bank transfer, FlightOne credits (Module 10), or combination (credits + card).
2	User	Enters card details (hosted, PCI-DSS compliant fields) and accepts terms.
3	User	Clicks Pay PKR
4	Supplier	Payment gateway initiates 3-D Secure; user completes OTP with bank.
5	System	Receives authorisation; captures amount; records transaction with gateway reference.
6	System	Triggers fulfilment: ticketing (UF-03.7) and hotel booking confirmation (UF-03.8).
7	System	If fulfilment fails after capture, auto-retries with fallback supplier; if still failing, refunds or routes to Ops (Module 13) and informs the user within 1 minute.
POST-CONDITIONS Payment captured; booking moves to <i>Paid – Fulfilment in progress</i> .		
ALTERNATE & EXCEPTION FLOWS Payment declined — Shows bank reason; allows another method while hold is valid. Bank transfer — Shows FlightOne account details and reference; hold remains until Finance confirms receipt (Ops confirms in Finance screen). Hold expired during payment — System re-prices before capture; if changed, asks for re-confirmation.		

UF-03.7 Issue Airline Tickets		
PRIMARY ACTOR System		PRE-CONDITIONS Payment captured or corporate approval + billing account valid.
#	Actor	Action / System Response
1	System	Sends ticketing command to Galileo (Air Ticketing) for the PNR with form of payment (agency/BSP or card).
2	Supplier	Galileo returns e-ticket numbers per passenger.
3	System	Updates booking to <i>Ticketed</i> ; generates branded e-ticket itinerary PDF (passenger, PNR, ticket numbers, segments, baggage, fare rules summary).
4	System	Stores documents in Traveller Vault; sends via email and WhatsApp; posts in chat; schedules Live Journey monitoring (Module 9).
5	System	Queues PNR for Ops review if ticketing returns warnings (e.g. schedule change, duplicate booking).
POST-CONDITIONS		

Ticket issued and delivered.

ALTERNATE & EXCEPTION FLOWS

- **Ticketing error** — Retry ×2; then escalate to Ops queue with error payload; customer informed of ETA.

UF-03.8 Confirm Hotel & Generate Voucher

PRIMARY ACTOR
System

PRE-CONDITIONS
Hotel rate prebooked; payment captured.

#	Actor	Action / System Response
1	System	Calls RateHawk booking finish with guest names and partner order ID; polls status until confirmed.
2	Supplier	RateHawk returns confirmation / supplier reference.
3	System	Generates branded hotel voucher PDF (hotel, address, map, check-in/out, room, meal plan, guest names, confirmation no., cancellation policy, payable-at-hotel items).
4	System	Delivers voucher via Vault, email, WhatsApp and chat.

POST-CONDITIONS
Hotel confirmed; voucher delivered.

WIREFRAME — BK-09 Confirmation

app.flightone.pk/trips/FO-240919-0831

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Home

AI Consultant

Search & Book

My Trips

Traveller Vault

Visa Centre

Groups

Rewards

Profile & Settings

Help

✓ Your trip is confirmed. Documents sent to email & WhatsApp.

Flight

PNR K7X2QP · Emirates

Tickets: 176-2419873321, 176-2419873322

E-ticket PDF

Add to calendar

Hotel

Rove Downtown · Conf. RH-9918273

19-22 Sep · Room only

Voucher PDF

Directions

Visa check: Pakistani passport → UAE visa required. **You have a valid UAE visa.** ✓

Earned **1,865** FlightOne credits

Manage booking

Add airport transfer

Invite companion to trip

UF-03.9 Update Mid-Office & Back-Office Automatically

PRIMARY ACTOR
System

PRE-CONDITIONS
Booking status changes (created, ticketed, changed, cancelled, refunded).

#	Actor	Action / System Response
1	System	Booking Orchestrator publishes a <i>booking event</i> to the event bus.
2	System	Mid Office Connector creates/updates the file: customer, PNR, ticket numbers, supplier, net cost, markup, selling price, taxes, service fee, agent/AI channel.
3	System	Back Office Connector posts accounting entries: sales invoice, supplier payable, receipt (payment), commission receivable, and GST/withholding tax lines.
4	System	CRM (Module 15) timeline is updated; commission tracking updated.
5	System	On failure, event is retried with exponential backoff and appears in <i>Ops Sync Exceptions</i> queue.

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POST-CONDITIONS

Operational and financial records in sync without manual entry.

UF-03.10 View & Manage Booking (My Trips)

PRIMARY ACTOR

Traveller

PRE-CONDITIONS

Booking exists.

#	Actor	Action / System Response
1	User	Clicks My Trips ; sees tabs Upcoming / Past / Cancelled / On hold.
2	User	Clicks a trip to open BK-10.
3	System	Shows live status, documents, payment, fare rules, visa status, journey alerts.
4	User	Clicks Manage booking → Change dates / Cancel / Add baggage / Select seat / Add special request / Name correction.
5	System	Routes to Refund & Reissue Engine (Module 14) or to Ops for manual services.

POST-CONDITIONS

Self-service request initiated.

BUSINESS RULES

1. Search results cached for 15 minutes per search key; price always re-verified before reservation and before payment.
2. Holds: Galileo PNR created with airline ticketing time limit (TTL); system auto-cancels unpaid PNRs 30 min before TTL to avoid ADMs.
3. Duplicate booking check: same passenger + same flight/date blocks a second booking unless confirmed by user.
4. Passenger name max length and special characters follow airline rules; titles appended automatically (MR/MRS/MS/MSTR/MISS).
5. Hotel 'payable at property' taxes/fees are displayed separately and never included in charge totals.
6. Every booking stores channel (AI / Manual / Agent), supplier, net, margin, discount and profile context for reporting.

DATA — BOOKING

Field	Type	Req.	Validation / Notes
booking_ref	String	Y	FO-YYMMDD-NNNN
type	Enum	Y	flight, hotel, package
status	Enum	Y	draft, on_hold, pending_approval, pending_payment, paid, ticketed, confirmed, changed, cancelled, refunded, failed
supplier / supplier_ref	Enum / String	Y	galileo PNR, ratehawk order ID
passengers[]	Array	Y	profile refs, name, type, doc, SSRs
segments[] / rooms[]	Array	Y	—
pricing	JSON	Y	net, markup, discount, fees, taxes, total, currency
payment_ids[]	Array	C	—
time_limit	Datetime	C	Hold expiry
channel	Enum	Y	ai, web, agent

Intelligent Recommendation Engine

Release: Phase 1 (MVP)

Scores every option on price, duration, layovers, quality, timing, preferences, loyalty, refundability and supplier reliability, then presents three curated, explainable recommendations.

4.1 Functional Overview

Instead of hundreds of options, the engine scores every result and presents **three curated recommendations**: typically Best Overall, Cheapest (acceptable), and Fastest / Best Experience. Users can always expand to the full list.

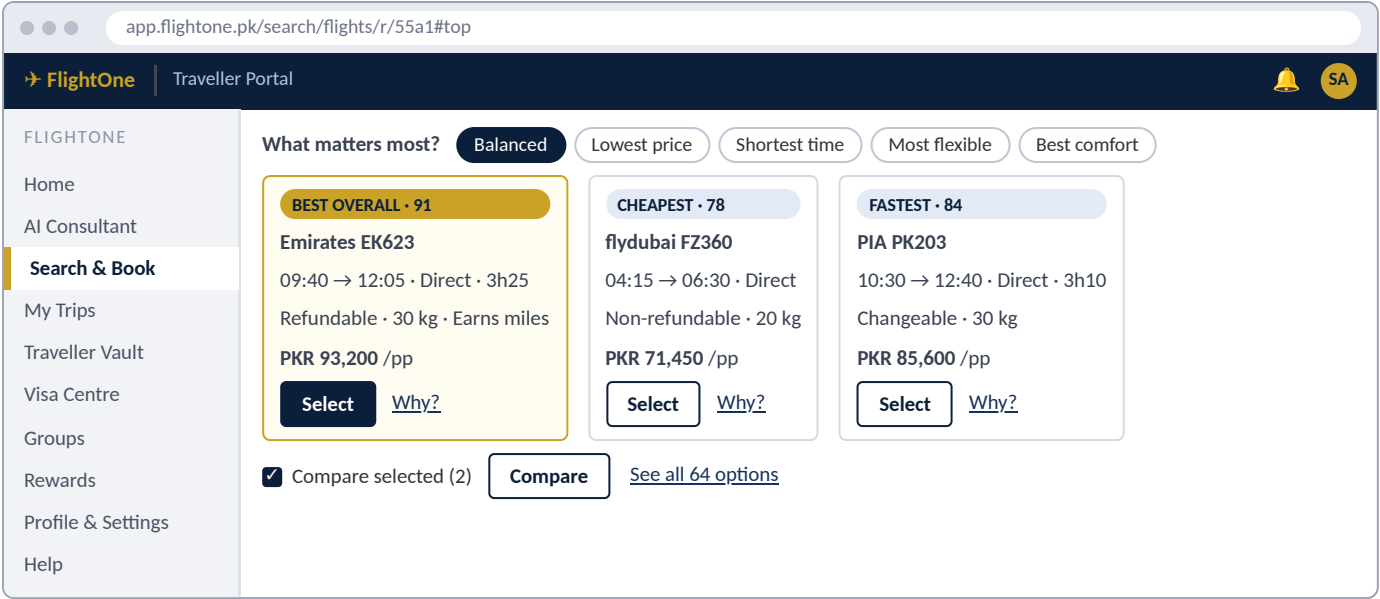
SCORING FACTORS (DEFAULT WEIGHTS — CONFIGURABLE PER PROFILE / COMPANY)

Factor	Flights	Hotels	Default Weight	Source
Price	Total fare incl. taxes & fees	Total stay cost	25 %	Supplier + Pricing Engine
Journey duration	Door-to-door elapsed time	— (distance to target)	15 %	Supplier
Layovers	Count, length (<1h risk, >6h penalty), airport change, overnight	—	10 %	Supplier
Airline / hotel quality	Airline rating, on-time %, aircraft, cabin product	Star rating, review score	10 %	Internal quality table
Arrival / check-in times	Penalise 00:00–05:00 arrivals unless preferred	Check-in time fit with flight arrival	10 %	Trip context
Customer preferences	Seat/airline/time/direct preferences	Brand, area, amenities, bed	15 %	Profile (Module 2), AI learning
Loyalty benefits	Miles earned, status benefits, alliance	Points & member rates	5 %	Profile loyalty
Refundability	Refund/change fee, fare brand	Free cancellation window	5 %	Fare rules / policy
Supplier reliability	Ticketing success, schedule change rate	Confirmation success, complaint rate	5 %	Module 17 supplier KPIs

Score = $\Sigma(\text{weight} \times \text{normalised factor score } 0\text{--}100)$ – policy penalty (out-of-policy options are down-ranked or hidden per company setting) + business boost (preferred supplier/margin, capped at 5 points so the customer's interest remains primary).

SCREEN INVENTORY

ID	Screen	Purpose	Access
REC-01	Top-3 Recommendation Panel	Shown in chat and at top of results	Traveller
REC-02	Side-by-Side Compare	Compare 2–4 options on all factors	Traveller
REC-03	Priority Adjuster	User re-weights (Cheapest ↔ Comfort ↔ Speed)	Traveller
REC-04	Recommendation Rules Admin	Weights, quality tables, boosts, thresholds	Ops / Pricing



UF-04.1 Generate Three Curated Recommendations		
PRIMARY ACTOR		PRE-CONDITIONS
AI / System		Normalised results available.
#	Actor	Action / System Response
1	System	Receives normalised result set and context: Trip Brief, profile preferences, loyalty, corporate policy, purpose (business/leisure).
2	System	Removes invalid options (insufficient connection time, visa-incompatible transits flagged by Module 8, expired documents).
3	AI	Computes factor scores and total score for every option.
4	AI	Selects slots: Best Overall = highest score; Cheapest = lowest price with score ≥ 60; Fastest (or Most Flexible/Best Comfort if fastest duplicates slot 1).
5	AI	Ensures diversity: the three picks must differ by carrier or departure window where possible.
6	AI	Generates a one-line label and reason for each pick.
7	System	Returns picks to the chat (AI-01) and results page (BK-02); logs scores for analytics and model training.
POST-CONDITIONS		Three explainable recommendations shown.

UF-04.2 Adjust Priorities & Re-rank		
PRIMARY ACTOR		PRE-CONDITIONS
Traveller		Recommendations displayed.
#	Actor	Action / System Response
1	User	Clicks a priority chip (e.g. Lowest price) or says “cheapest please” in chat.
2	System	Applies preset weight profile (e.g. Price 60 %, Duration 10 %, others scaled).
3	AI	Re-ranks and refreshes the three cards with animation; labels update.
4	System	Stores the choice as a preference signal (Module 1 UF-01.5).
POST-CONDITIONS		Re-ranked recommendations.

app.flightone.pk/search/flights/r/55a1/compare

→ FlightOne

Traveller Portal



SA

FLIGHTONE	Factor	Emirates EK623	flydubai FZ360	PIA PK203
Home	Price /pp	PKR 93,200	PKR 71,450	PKR 85,600
AI Consultant	Depart / Arrive	09:40 / 12:05	04:15 / 06:30	10:30 / 12:40
Search & Book	Duration	3h25	3h15	3h10
My Trips	Baggage	30 kg	20 kg	30 kg
Traveller Vault	Refund	Yes (PKR 15k fee)	No	Fee PKR 12k + fare diff
Visa Centre	Airline quality	4.6	3.9	3.4
Groups	Loyalty	~1,240 Skywards miles	—	—
Rewards	Policy	In policy	In policy	In policy
Profile & Settings		Select	Select	Select
Help				

UF-04.3

Compare Options

PRIMARY ACTOR		PRE-CONDITIONS
Traveller		≥2 options ticked.
#	Actor	Action / System Response
1	User	Ticks Compare on 2–4 cards/rows and clicks Compare .
2	System	Opens REC-02 with each factor as a row; best value per row in bold.
3	User	Clicks Select under the chosen option → continues to UF-03.5.
POST-CONDITIONS		Informed selection.

UF-04.4

Show All Options

PRIMARY ACTOR		PRE-CONDITIONS
Traveller		—
#	Actor	Action / System Response
1	User	Clicks See all N options (or asks AI “show everything”).
2	System	Opens the full list sorted by score, with sort (Recommended, Price, Duration, Departure) and filters (stops, airline, time, price, baggage, refundable, policy).
3	System	Each row shows its score badge so users understand the ranking.
POST-CONDITIONS		Full transparency available.

UF-04.5

Learn from Outcomes (Feedback Loop)

PRIMARY ACTOR		PRE-CONDITIONS
System		Recommendation shown.
#	Actor	Action / System Response
1	System	Captures impressions, clicks, selections, bookings, abandonments and post-trip ratings per pick slot.
2	System	Nightly training job updates personal preference vectors and global weight suggestions.

#	Actor	Action / System Response
3	Ops	Pricing/Product manager reviews suggested weight changes in REC-04 and clicks Approve (A/B test first) or Reject .
4	System	Publishes new weights version; all rankings tagged with model version for audit.
POST-CONDITIONS		
Continuously improving recommendations under human control.		

WIREFRAME — REC-04 Recommendation Rules Admin

ops.flightone.pk/recommendation/weights

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Weight profile: Default (v12)

Price

Duration

Layovers

Quality

Arrival fit

Preferences

Loyalty

Refundability

Reliability

25%

15%

10%

10%

10%

15%

5%

5%

5%

Settings

Cheapest slot minimum score

60

Business boost cap (points)

5

Out-of-policy handling

Down-rank & label

A/B test traffic

10 %

Save draft

Start A/B test

Publish

- BUSINESS RULES
1. The customer typically sees exactly three curated picks; fewer only if fewer valid options exist.

2. Out-of-policy options can never be labelled 'Best Overall' for corporate profiles.

3. Business boosts (margin/preferred supplier) are capped and disclosed in the 'Why?' explanation as 'FlightOne preferred partner'.

4. Weight changes require approval and are versioned; every recommendation log stores the version used.

Pricing & Margin Engine

Release: Phase 1 (MVP)

Rule-driven pricing that applies markups, fees, corporate agreements, promotions and controlled discounts (including AI discount limits), with full profitability reporting.

5.1 Functional Overview

Converts supplier net fares/rates into selling prices using configurable rules, and governs every discount — promotional, corporate, agent-given or AI-negotiated — while reporting profitability per booking.

PRICE BUILD-UP (EVALUATION ORDER)

#	Component	Description	Example (PKR)
1	Supplier net	Net fare/rate from Galileo (net/commissionable) or RateHawk (net)	71,000
2	Dynamic markup	Rule-based: % or fixed; by supplier, airline, route, cabin, hotel star, season, days-to-departure, channel, customer segment, demand	+ 4,260 (6 %)
3	Service fee	Per booking / per passenger / per segment	+ 1,500
4	Negotiation buffer	Hidden headroom within markup that AI/agents may give away	(of which 1,800)
5	Corporate pricing	Contracted markup/fees override standard rules	n/a (Personal)
6	Promotional discount	Promo codes / automatic campaigns	- 1,000
7	AI / agent discount	Within permission limits; cannot exceed buffer	- 800
8	Taxes on fees	Sales tax on services per province rules	+ 240
	Selling price	Floor rule: selling price $\geq$ net + minimum margin	75,200

SCREEN INVENTORY

ID	Screen	Purpose	Access
PR-01	Pricing Rules List	All markup/fee rules with priority and status	Pricing Manager
PR-02	Rule Editor	Conditions + actions + validity + simulation	Pricing Manager
PR-03	Promotions & Promo Codes	Campaigns, codes, usage limits	Marketing / Pricing
PR-04	Corporate Price Agreements	Company-specific fees/markups	Pricing / Sales
PR-05	Discount Permissions	Per role / per agent / AI discount limits	Pricing Manager
PR-06	Discount Approval Queue	Requests above limits	Pricing Manager
PR-07	Profitability Report	Margin by product, supplier, channel, agent, customer	Management / Finance

ops.flightone.pk/pricing/rules/new

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New markup rule

Rule name

GCC Economy – Summer

Priority

20

Status

Draft

Conditions (all must match)

Product

Flights

Supplier

Galileo

Airline(s)

EK, QR, FZ

Origin

PK (country)

Destination

AE, SA, QA

Cabin

Economy

Travel dates

01 Jun – 31 Aug

Days to departure

0 – 7

Segment

Personal

Actions

Markup type

Percentage

Value

6 %

Min / Max (PKR)

2,000 / 12,000

Service fee /pax

1,500

Negotiation buffer

40 % of markup

Min margin floor

1,500

Cancel

Save draft

Publish

Simulate

Sample net fare

71,000

Markup 6 % = 4,260

Service fee = 1,500

Buffer available = 1,704

Sell = 76,760

Conflicts: overrides rule #34 (GCC all cabins)

UF-05.1 Create & Publish a Markup Rule		
PRIMARY ACTOR Pricing Manager		PRE-CONDITIONS Role has pricing.rules.write.
#	Actor	Action / System Response
1	Ops	Opens Pricing Rules and clicks + New Rule .
2	Ops	Enters rule name, priority and product (Flights / Hotels / Package).
3	Ops	Adds conditions: supplier, airline/hotel chain, origin/destination (airport/city/country/region), cabin/star rating, travel & booking date ranges, days-to-departure, channel (AI/Web/Agent), customer segment (Personal/Corporate/VIP), demand level.
4	Ops	Defines actions: markup type (% / fixed / per pax / per night), value, min/max caps, service fee, negotiation buffer, minimum margin floor.
5	Ops	Enters sample net value in Simulate panel.
6	System	Calculates selling price and lists conflicting/overlapping rules by priority.
7	Ops	Clicks Publish (or Save draft).
8	System	If rule impact exceeds threshold (e.g. >2 % average price change), requires second approver (maker-checker).
9	System	Versions and activates the rule; cache invalidated; change written to audit log.
POST-CONDITIONS Rule live for new searches.		
ALTERNATE & EXCEPTION FLOWS		
Invalid ranges / negative margin — Validation errors prevent publishing. Schedule — Rule can be scheduled to activate/deactivate automatically.		

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UF-05.2 Runtime Price Calculation

PRIMARY ACTOR
System

PRE-CONDITIONS
Supplier results normalised.

#	Actor	Action / System Response
1	System	For each option, loads applicable rules (corporate agreement first, then standard rules by priority; first match per rule group wins).
2	System	Computes markup, fees, buffer, taxes on fees; applies automatic promotions.
3	System	Enforces floor: if selling < net + minimum margin, adjusts upward.
4	System	Stores a price breakdown snapshot (rule IDs and versions) with the search result and later with the booking.
5	System	Returns customer-facing price (total and per person); breakdown hidden from customer except base/taxes/fees.

POST-CONDITIONS
Consistent, auditable prices.

WIREFRAME — PR-03 Promotions & PR-05 Discount Permissions

ops.flightone.pk/pricing/discounts

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Promo codes

Code	Type	Value	Used	Valid to	
SAVE3K	Fixed	PKR 3,000	412 / 1,000	30 Sep	Active
UMRAH10	%	10 % fees	88 / ∞	31 Dec	Active
EID25	%	2.5 %	1,000/1,000	15 Jun	Exhausted

+ New promotion

Discount limits

Role / Actor	Per booking	Monthly cap
AI Consultant	≤ 50 % of buffer · max 2,000	PKR 1.5M
Junior Consultant	≤ 2,500	PKR 150k
Senior Consultant	≤ 7,500	PKR 500k
Team Lead	≤ buffer	—
Above limit	Approval queue	—

Edit limits

UF-05.3 Create a Promotion / Promo Code

PRIMARY ACTOR
Marketing or Pricing Manager

PRE-CONDITIONS
—

#	Actor	Action / System Response
1	Ops	Opens Promotions → + New promotion.
2	Ops	Selects type: promo code or automatic campaign; discount type (fixed / % of fare / % of fees / free service fee).
3	Ops	Sets eligibility: product, routes/destinations, dates, min spend, new customers only, segment, channel, first booking, referral-linked.
4	Ops	Sets usage limits: total uses, per customer, budget cap; validity dates; stackable yes/no.
5	Ops	Clicks Activate .
6	System	Validates code uniqueness; activates; promotion becomes visible in eligible results (automatic) or applicable at checkout (code).

POST-CONDITIONS
Promotion live and tracked.

UF-05.4 Customer Applies Promo Code

PRIMARY ACTOR
Traveller

PRE-CONDITIONS
On Review/Payment screen.

#	Actor	Action / System Response
1	User	Clicks Have a promo code? , enters code, clicks Apply .
2	System	Checks validity, eligibility, usage limits and stacking rules.
3	System	Recalculates total (floor rule still enforced) and shows discount line.
4	System	Reserves one usage; releases it if booking not completed within hold time.

POST-CONDITIONS
Discount applied.

ALTERNATE & EXCEPTION FLOWS

- **Invalid / expired / not eligible** — Shows specific message, e.g. "This code is valid for Umrah packages only".

UF-05.5 AI Negotiation within Discount Limits

PRIMARY ACTOR
Traveller / AI

PRE-CONDITIONS
Options shown; buffer exists.

#	Actor	Action / System Response
1	User	Types in chat: "Can you do any better on the price?"
2	AI	Checks conversion signals (price sensitivity, competing option viewed, loyalty tier, basket size) and available buffer.
3	AI	Offers a discount up to the AI limit in a single step (no haggling loops), e.g. "I can take PKR 1,200 off if you book in the next 30 minutes."
4	User	Accepts.
5	System	Applies discount as <i>AI discount</i> with expiry; logs reason and amount against AI monthly cap.
6	AI	If the customer asks for more than the limit, offers alternatives (cheaper option) or escalation to a consultant.

POST-CONDITIONS
Discount applied within guardrails; conversion tracked.

UF-05.6 Agent Discount & Approval Above Limit

PRIMARY ACTOR
Consultant; Pricing Manager

PRE-CONDITIONS
Agent working a booking.

#	Actor	Action / System Response
1	Agent	In booking detail clicks Apply discount , enters amount and reason.
2	System	If within the agent's limit, applies immediately.
3	System	If above, creates request in Discount Approval Queue (PR-06) and notifies Pricing Manager.
4	Ops	Pricing Manager reviews margin impact and clicks Approve / Reject (with comment).
5	System	Applies or rejects; notifies agent; logs decision.

POST-CONDITIONS
Controlled discounting.

UF-05.7 Set Corporate Pricing Agreement

PRIMARY ACTOR

PRE-CONDITIONS

Pricing / Corporate Sales		Company exists (Module 6).
#	Actor	Action / System Response
1	Ops	Opens Corporate Price Agreements → selects company → + Agreement .
2	Ops	Sets transaction fees (domestic/international/hotel/changes), markup overrides, negotiated airline/hotel deal codes, validity and billing currency.
3	Ops	Clicks Save & Activate .
4	System	Agreement takes precedence for all bookings made under that company's corporate profile.
POST-CONDITIONS Contracted pricing applied automatically.		

WIREFRAME
— PR-07 Profitability Report

ops.flightone.pk/reports/profitability

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Period

01-15 Sep 2026

Group by

Channel

Product

All

Export XLSX

GROSS SALES

PKR 184.2M

+12 % MoM

NET MARGIN

PKR 11.6M

6.3 %

DISCOUNTS GIVEN

PKR 1.4M

AI 38 % · Agents 62 %

AVG MARGIN / BOOKING

PKR 4,120

Channel	Bookings	Sales	Net cost	Margin	Margin %	Discounts
AI Consultant	1,284	72.4M	67.6M	4.8M	6.6 %	0.53M
Web (manual)	902	51.0M	48.1M	2.9M	5.7 %	0.21M
Consultants	610	60.8M	56.9M	3.9M	6.4 %	0.66M

UF-05.8 View Profitability Report		
PRIMARY ACTOR		PRE-CONDITIONS
Management / Finance		—
#	Actor	Action / System Response
1	Ops	Opens Reports → Profitability ; selects period, grouping (product, supplier, airline, route, channel, agent, company, customer) and filters.
2	System	Displays KPIs and table from booking price snapshots (net, markup, fees, discounts, refunds, commissions).
3	Ops	Clicks a row to drill down to bookings; clicks Export for XLSX/CSV.
POST-CONDITIONS		
Profitability visibility at any level.		

BUSINESS RULES 1. No selling price may fall below net + minimum margin floor, except with Pricing Manager override (logged). 2. Discounts cannot exceed the negotiation buffer unless approved. 3. AI discounts: one offer per conversation, time-bound, and counted against a monthly AI cap. 4. Rule precedence: Corporate agreement → Specific rule (higher priority) → Generic rule → Default markup. 5. All price calculations are reproducible from the stored snapshot (rule IDs + versions).
--

Corporate Travel

Release: Phase 1 (approvals) · Full scope Phase 1–2

Enterprise travel management — companies, departments, cost centres, policies, approvals, budgets, credit and billing — with clean separation of corporate and personal travel.

6.1 Functional Overview

- Multi-company (multi-tenant) corporate accounts with departments, cost centres and project codes.
- Travel policy builder, pre-trip approval workflows, budget enforcement, credit limits and billing cycles with consolidated invoicing.
- Corporate travellers can switch between Corporate and Personal profiles, each with separate payment methods and booking histories.
- Travel arrangers can book on behalf of colleagues.

SCREEN INVENTORY

ID	Screen	Purpose	Access
CT-01	Company Onboarding (Ops)	Create company, agreement, credit, admins	Ops / Sales
CT-02	Corporate Dashboard	Spend, pending approvals, travellers abroad, budget use	Corporate Admin
CT-03	Employees	Invite, bulk import, roles, approver mapping	Corporate Admin
CT-04	Departments, Cost Centres & Project Codes	Organisation structure and coding	Corporate Admin
CT-05	Travel Policy Builder	Rules by grade/department	Corporate Admin
CT-06	Approval Workflow Builder	Approval chains and conditions	Corporate Admin
CT-07	Approval Request	Approver decision screen (web/email/WhatsApp)	Approver
CT-08	Budgets	Budgets by department/cost centre/project	Corporate Admin
CT-09	Credit & Billing	Credit limit, utilisation, billing cycle	Corporate Admin / Finance
CT-10	Invoices & Statements	Consolidated invoices, payments, downloads	Corporate Admin / Finance
CT-11	Corporate Reports	Spend, savings, policy compliance	Corporate Admin

ops.flightone.pk/corporate/new

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1 Company

2 Agreement & pricing

3 Credit & billing

4 Admins

5 Review

Company

NTN / STRN

Industry

Credit limit (PKR)

Billing cycle

Payment terms

Invoice format

Billing contact

Security deposit / guarantee

Allow personal cards as fallback

Block bookings when credit exhausted

Back

Continue

UF-06.1 Onboard a Corporate Company		
PRIMARY ACTOR		PRE-CONDITIONS
FlightOne Corporate Sales / Ops		Signed contract.
#	Actor	Action / System Response
1	Ops	Opens Corporate → + New Company .
2	Ops	Step 1: enters legal name, trade name, NTN/STRN, address, industry, logo, company email domains (for auto-linking).
3	Ops	Step 2: selects/creates Corporate Price Agreement (UF-05.7).
4	Ops	Step 3: sets credit limit, billing cycle (weekly/fortnightly/monthly), payment terms, invoice format, currency, security instrument.
5	Ops	Step 4: adds Corporate Admin(s) and Finance contact.
6	Ops	Step 5: reviews and clicks Create Company .
7	System	Creates tenant, default department & policy template; sends admin invitations (UF-00.6).
POST-CONDITIONS		Company active; admins invited.

corporate.flightone.pk/employees

FlightOneCorporate Admin Portal

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Dashboard

Employees

Travel Policies

Approvals

Cost Centres

Budgets

Invoices

Reports

Company Settings

Search employees...

Department: All

Bulk import CSV

+ Invite employee

Name	Emp. ID	Department	Cost centre	Grade	Approver	Role	Status
Sara Ahmed	AP-1043	Sales	CC-104	M2	Imran Qureshi	Traveller	Active
Imran Qureshi	AP-0210	Sales	CC-104	D1	Nadia Khan	Approver	Active
Omar Khan	AP-1188	Marketing	CC-201	M1	Hina Raza	Arranger	Invited
Bilal Sheikh	AP-0988	Operations	CC-310	E3	Hina Raza	Traveller	Deactivated

UF-06.2 Invite Employees (Single & Bulk)

PRIMARY ACTOR
Corporate Admin

PRE-CONDITIONS
Company active.

#	Actor	Action / System Response
1	User	Opens Employees → clicks + Invite employee .
2	User	Enters name, work email, mobile, employee ID, department, cost centre, grade, approver, role(s) (Traveller / Arranger / Approver / Admin / Finance); clicks Send invite .
3	System	Validates email domain against company domains; sends invitation (UF-00.6).
4	User	For bulk: clicks Bulk import CSV → downloads template → uploads filled file.
5	System	Validates rows, shows preview with errors per row (duplicate email, unknown cost centre).
6	User	Fixes or excludes error rows and clicks Import .
7	System	Creates pending employees and sends invites; shows summary (imported / skipped).

POST-CONDITIONS
Employees invited.

ALTERNATE & EXCEPTION FLOWS

- **Employee leaves** — Admin clicks **Deactivate**; corporate profile disabled; personal profile unaffected; future corporate bookings flagged to admin.

UF-06.3 Manage Departments, Cost Centres & Project Codes

PRIMARY ACTOR
Corporate Admin

PRE-CONDITIONS
—

#	Actor	Action / System Response
1	User	Opens Cost Centres → tabs Departments / Cost Centres / Project Codes.
2	User	Clicks + Add , enters code, name, owner, parent (hierarchy), validity; for project codes sets whether mandatory at booking and allowed departments.
3	User	Clicks Save .
4	System	Codes become selectable in booking checkout for eligible employees; appear on invoices and reports.

POST-CONDITIONS
Organisation structure configured.

FLIGHTONE

Dashboard

Employees

Travel Policies

Approvals

Cost Centres

Budgets

Invoices

Reports

Company Settings

corporate.flightone.pk/policies/grade-m

FlightOne Corporate Admin Portal

Policy: Managers (M1-M3)

Applies to: Grades M1, M2, M3 · All departments

Flights

Hotels

Booking rules

Exceptions

Domestic cabin

Economy

International cabin

Economy; Business if > 6h

Max fare – domestic

PKR 45,000

Max fare – international

Lowest logical fare + 15 %

Advance booking

≥ 7 days (international)

Preferred airlines

PK, EK, QR

☒ Refundable fares only if +≤10 %

☐ Block basic/lite fares

Out-of-policy action

Allow with reason + approval

Save policy

Hotels

Max nightly: Dubai PKR 35k · London GBP 180 · Default PKR 25k

Star rating ≤ 4★

Reason codes

Client meeting timing · No policy fare · Safety · Other

UF-06.4 Configure Travel Policy		
PRIMARY ACTOR		PRE-CONDITIONS
Corporate Admin		—
#	Actor	Action / System Response
1	User	Opens Travel Policies → + New policy ; names it and assigns grades/departments/employees.
2	User	Flights tab: cabin by duration/region, max fare (absolute or lowest-logical-fare + %), advance purchase, preferred/blocked airlines, fare types, baggage.
3	User	Hotels tab: city-wise nightly caps, star rating, preferred chains, meal plan.
4	User	Booking rules tab: who may book for whom, mandatory project code, max trip length, weekend travel.
5	User	Exceptions tab: out-of-policy action (Block / Allow with reason / Allow with reason + approval), reason code list.
6	User	Clicks Save policy .
7	System	Versions the policy; the booking engine and AI apply it immediately to affected employees.
POST-CONDITIONS		
Policy active.		

UF-06.5 Configure Approval Workflow		
PRIMARY ACTOR		PRE-CONDITIONS
Corporate Admin		Policies exist.
#	Actor	Action / System Response
1	User	Opens Approvals → Workflows → + New workflow .
2	User	Defines trigger conditions: all trips / international only / out-of-policy / above amount / specific grades.
3	User	Adds levels: L1 line manager (from employee record), L2 department head if > PKR 500k, L3 finance if over budget; sets SLA (e.g. 4 h) and escalation/auto-action on timeout (remind, escalate, auto-reject).
4	User	Sets delegate rules (approver on leave) and clicks Save .

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#	Actor	Action / System Response
5	System	Activates workflow for new booking requests.
POST-CONDITIONS Approval chain configured.		

WIREFRAME — CT-07 Approval Request (Approver view)

UF-06.7 Approver Approves / Rejects

PRIMARY ACTOR
Approver

PRE-CONDITIONS
Approval request exists.

#	Actor	Action / System Response
1	Approver	Receives notification and clicks Review (or uses Approve/Reject buttons in WhatsApp/email — secured by signed one-time links).
2	System	Opens CT-07 with trip details, costs, policy status, reason, budget impact and cheaper alternative.
3	Approver	Clicks Approve , Approve in-policy alternative , or Reject (comment required).
4	System	If more levels exist, routes to next approver; otherwise marks approved.
5	System	On final approval: re-prices, tickets on company credit, sends documents, posts to invoice cycle.
6	System	On rejection: releases hold, notifies traveller with comment; AI offers in-policy options.

POST-CONDITIONS
Decision executed and audited.

ALTERNATE & EXCEPTION FLOWS

- **SLA timeout** — Reminder at 50 % SLA; escalation to delegate/next level at 100 %.
- **Price changed on approval** — If increase $\leq$ tolerance (e.g. 5 %) auto-proceeds; else returns to approver.

UF-06.8 Budget Enforcement

PRIMARY ACTOR
System; Corporate Admin

PRE-CONDITIONS
Budgets configured.

#	Actor	Action / System Response
1	User	Admin opens Budgets → + Budget : scope (department / cost centre / project), period, amount, thresholds (80 %, 100 %), action at 100 % (warn / require finance approval / block).
2	System	At checkout computes committed spend (ticketed + pending) + current trip.
3	System	Shows warnings; applies configured action (e.g. adds Finance approval level).
4	System	Sends threshold alerts to budget owner.

POST-CONDITIONS
Spend stays within budgets.

WIREFRAME — CT-09 Credit & Billing / CT-10 Invoices

corporate.flightone.pk/billing

FlightOne

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CREDIT LIMIT

PKR 15.0M

UTILISED

PKR 9.8M

65 %

AVAILABLE

PKR 5.2M

NEXT INVOICE

30 Sep

Fortnightly

Invoice	Period	Bookings	Amount	Due	Status	
INV-AP-2609-02	01–15 Sep	46	4,812,400	30 Sep	Unpaid	PDF · XLSX
INV-AP-2608-04	16–31 Aug	39	3,990,100	15 Sep	Overdue	Pay now
INV-AP-2608-03	01–15 Aug	52	5,120,800	30 Aug	Paid	PDF · XLSX

UF-06.9 Credit Limit Check, Billing Cycle & Corporate Invoicing

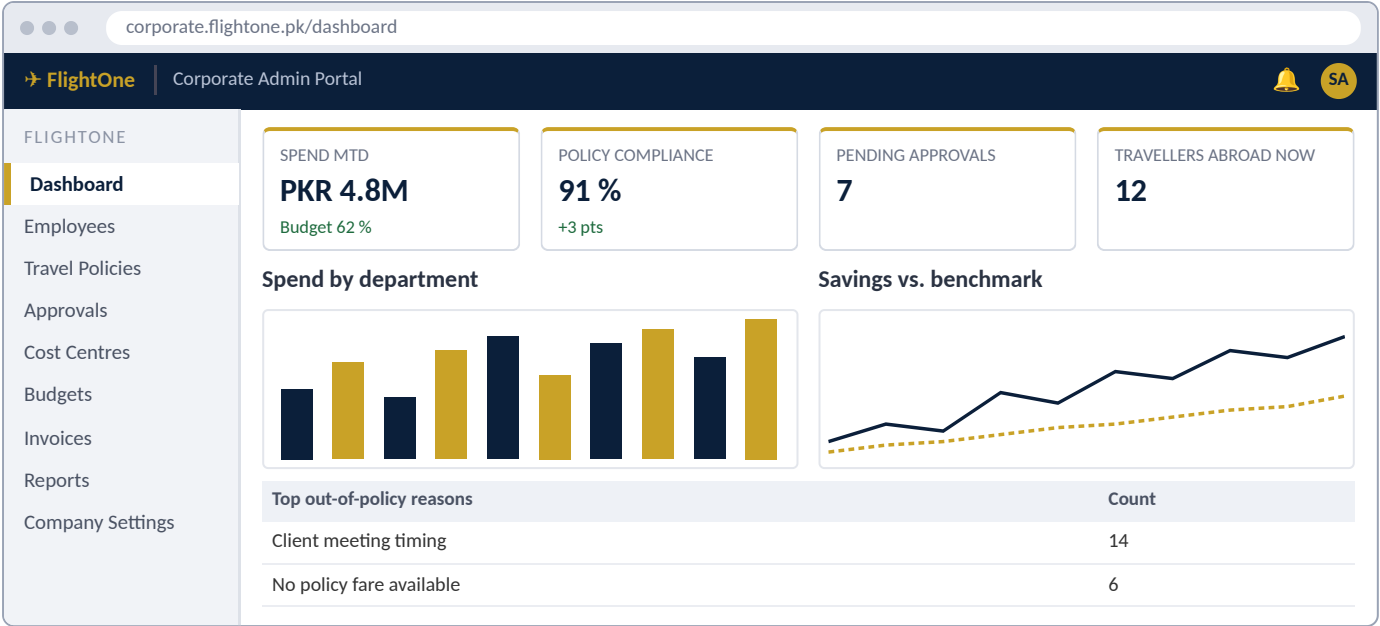
PRIMARY ACTOR		PRE-CONDITIONS
System; Corporate Finance		Company on credit.
#	Actor	Action / System Response
1	System	Before ticketing, checks available credit = limit – (unpaid invoices + uninvoiced tickets + pending approvals).
2	System	If insufficient: blocks or allows fallback personal/company card as configured; notifies Corporate Admin and FlightOne Finance.
3	System	At cycle end, generates consolidated invoice (with per-traveller breakdown by cost centre/project, taxes, service fees, credit notes for refunds).
4	System	Emails invoice (PDF + XLSX) to billing contact; posts to Back Office.
5	User	Corporate Finance opens Invoices , downloads, or clicks Pay now (bank transfer reference / online payment).
6	Ops	FlightOne Finance matches receipt and marks invoice <i>Paid</i> ; credit is released.
7	System	Overdue invoices trigger reminders at +1, +7 days and optional booking suspension.
POST-CONDITIONS		
Accurate billing and credit control.		

UF-06.10 Switch Between Corporate & Personal Profiles

PRIMARY ACTOR		PRE-CONDITIONS
Corporate Traveller		User linked to a company.
#	Actor	Action / System Response
1	User	Clicks the profile switcher in the top bar and selects Personal .
2	System	Reloads context: personal payment methods, personal booking history, personal rewards; corporate policy no longer applied; banner shows 'Personal profile'.
3	User	Books a personal trip — paid by personal card; not visible to the employer.
4	User	Switches back to Corporate .
5	System	Restores corporate context; AI conversation also switches context (asks to confirm if a trip was in progress).
POST-CONDITIONS		
Strict separation of data, payment and history.		

UF-06.11 Book on Behalf of a Colleague (Arranger)

PRIMARY ACTOR		PRE-CONDITIONS
Travel Arranger		Arranger role for the traveller's department.
#	Actor	Action / System Response
1	User	Clicks Book for selector and searches colleague name.
2	System	Loads colleague's corporate profile, documents (consented fields only) and policy.
3	User	Completes search and checkout as normal; traveller receives copies of all notifications.
4	System	Approval routes to the traveller's approver, not the arranger's.
POST-CONDITIONS		
Booking created on behalf of traveller.		



UF-06.12 Corporate Dashboard & Reports		
PRIMARY ACTOR Corporate Admin		PRE-CONDITIONS —
#	Actor	Action / System Response
1	User	Opens Dashboard : spend, compliance, pending approvals, travellers abroad (duty of care map link).
2	User	Opens Reports : selects report (spend by department/cost centre/project, policy exceptions, unused tickets, savings, carbon — Phase 3), period, filters.
3	User	Clicks Export or Schedule (weekly email).
POST-CONDITIONS Corporate visibility and control.		

- BUSINESS RULES
1. Corporate and personal profiles never share payment methods or booking visibility.

2. Policy evaluation result and version are stored on every corporate booking.

3. Approvals via email/WhatsApp use signed, single-use links that expire with the request.

4. A traveller cannot approve their own request; arrangers cannot approve bookings they made.

5. Credit exhausted → behaviour per company setting; FlightOne Finance can grant temporary limit (logged).

Traveller Vault

Release: Phase 1 (booking documents) · Phase 2 (full Vault)

An encrypted digital wallet for passports, visas, tickets, vouchers, insurance, loyalty and travel certificates — auto-filed, organised by trip, and securely shareable.

VAULT CONTENTS

Category	Source	Examples
Passports	Profile (Module 2) — OCR upload	Bio-page scans
Visas	Profile / Visa Engine (Module 8)	e-Visas, stamped visa pages
Tickets	Auto — Booking Engine	E-ticket itineraries, EMDs, boarding passes (uploaded)
Hotel vouchers	Auto — Booking Engine	RateHawk vouchers
Insurance	Upload / future insurance integration	Policy certificates
Frequent Flyer numbers	Profile loyalty	Digital card view
Loyalty cards	Profile loyalty	Hotel/retail loyalty cards
Travel certificates	Upload	Vaccination, NOC, invitation letters, polio certificate, bank statements

SCREEN INVENTORY

ID	Screen	Purpose	Access
VLT-01	Vault Home	Browse by Trip / by Category / Expiring	Traveller
VLT-02	Document Viewer	Preview, download, share, details	Traveller
VLT-03	Add Document	Upload with category, trip link, expiry	Traveller
VLT-04	Share Document	Secure expiring link / share with companion	Traveller
VLT-05	Trip Pack	All documents for a trip as one PDF/ZIP	Traveller

WIREFRAME — VLT-01 Vault Home

app.flightone.pk/vault

→ FlightOne

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By trip

By category

Expiring soon

Shared

+ Add document

Passports

2 documents

Visas

3 · 1 expiring

Tickets

14

Vouchers

9

Insurance

1

Frequent flyer

2 cards

Loyalty cards

3

Certificates

4

Recent	Category	Trip	Added	
E-ticket K7X2QP	Ticket	Dubai · Sep 2026	Today	View · Share
Rove Downtown voucher	Voucher	Dubai · Sep 2026	Today	View · Share
Polio vaccination cert.	Certificate	—	02 Aug	View · Share

UF-07.1 Automatic Filing of Booking Documents

PRIMARY ACTOR
System

PRE-CONDITIONS
Ticket/voucher generated.

#	Actor	Action / System Response
1	System	On ticketing/voucher generation, stores the PDF in the Vault under the traveller(s) and the trip folder.
2	System	Tags document with booking ref, PNR, dates, and passengers; companions receive their own copies if they have accounts.
3	System	On reissue/cancellation, marks old document <i>Superseded</i> and files the new one.

POST-CONDITIONS
Documents always current in the Vault.

UF-07.2 Open the Vault & View a Document

PRIMARY ACTOR
Traveller

PRE-CONDITIONS
Signed in.

#	Actor	Action / System Response
1	User	Clicks Traveller Vault .
2	System	Requests re-authentication (password/OTP/2FA) if last auth > 10 minutes (sensitive area).
3	User	Chooses a tab (By trip / By category / Expiring soon) and clicks a document.
4	System	Opens VLT-02 viewer with watermark "FlightOne - {user} - {date}"; shows metadata (number masked, expiry, linked trip).
5	User	Clicks Download (original) or Print .
6	System	Logs access event.

POST-CONDITIONS
Document viewed/downloaded; access audited.

WIREFRAME — VLT-03 Add Document / VLT-04 Share

app.flightone.pk/vault/share/doc-2231

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Add document

Category

Insurance

File

travel-insurance.pdf

Link to trip

Dubai · Sep 2026

Document no.

TI-88213

Expiry

22 Sep 2026

Belongs to

Sara Ahmed

Save

Share "E-ticket K7X2QP"

Secure link (view only)

Expires in

24 hours

Passcode

Auto (sent separately)

Share with FlightOne user / companion

Send to email / WhatsApp

Recipient

ali.ahmed@gmail.com

Allow download

Create link

UF-07.3 Add a Document Manually

PRIMARY ACTOR		PRE-CONDITIONS
Traveller		—
#	Actor	Action / System Response
1	User	Clicks + Add document .
2	User	Selects category (Insurance, Certificate, Loyalty card, Boarding pass, Other), uploads file (PDF/JPG/PNG ≤ 10 MB) or scans.
3	System	Runs malware scan and OCR to suggest document number and expiry (user verifies).
4	User	Links to trip (optional), sets owner (self/family), expiry, and clicks Save .
5	System	Encrypts and stores; schedules expiry reminders if applicable.
POST-CONDITIONS		Document stored.

UF-07.4 Share a Document Securely

PRIMARY ACTOR		PRE-CONDITIONS
Traveller		Document exists.
#	Actor	Action / System Response
1	User	Clicks Share on a document.
2	User	Chooses: secure link (with expiry and optional passcode), share with FlightOne user/companion, or send via email/WhatsApp.
3	User	Sets permissions (view only / allow download) and clicks Create link / Send .
4	System	Generates tokenised URL; sends passcode through a separate channel; lists share under <i>Shared</i> tab.
5	User	Clicks Revoke at any time to disable access.
POST-CONDITIONS		Controlled sharing with audit trail.

UF-07.5 Download Trip Pack

PRIMARY ACTOR		PRE-CONDITIONS
Traveller		Trip with documents.
#	Actor	Action / System Response
1	User	Opens a trip folder and clicks Download trip pack .
2	System	Compiles itinerary summary, tickets, vouchers, visa copies, insurance and emergency contacts into one PDF (or ZIP).
3	System	Offers Send to WhatsApp for offline access while travelling.
POST-CONDITIONS		Single document pack for travel.

UF-07.6 View Frequent Flyer & Loyalty Cards

PRIMARY ACTOR		PRE-CONDITIONS
Traveller		Loyalty programmes saved.
#	Actor	Action / System Response
1	User	Opens By category → Frequent flyer or Loyalty cards .

#	Actor	Action / System Response
2	System	Displays digital cards (programme, number, tier, barcode/QR if provided).
3	User	Clicks Copy number or Edit (opens Profile loyalty tab).
POST-CONDITIONS Quick access to membership numbers.		

UF-07.7 Delete / Archive a Document		
PRIMARY ACTOR Traveller		PRE-CONDITIONS User-uploaded document.
#	Actor	Action / System Response
1	User	Clicks ... → Delete .
2	System	Warns if document is linked to an upcoming trip; asks for confirmation.
3	User	Confirms.
4	System	Soft-deletes (recoverable 30 days) then permanently purges; booking-generated documents can be archived but not deleted while the booking is active.
POST-CONDITIONS Document removed per retention policy.		

BUSINESS RULES

1. Encryption at rest (AES-256, per-tenant keys) and in transit (TLS 1.2+).
2. Re-authentication required to open the Vault after 10 minutes of inactivity.
3. Shared links default to 24-hour expiry and view-only; maximum 30 days.
4. Corporate admins cannot view personal Vault documents; only documents shared to the corporate profile.
5. Retention: documents kept while account active; purged 90 days after account closure unless legally required.

Visa Intelligence

Release: Phase 2 (basic visa check in Phase 1 booking flow)

Destination and transit visa requirements, document checklists, embassy details, processing timelines, appointment tracking and visa expiry reminders — linked to the traveller’s profile and bookings.

SCREEN INVENTORY

ID	Screen	Purpose	Access
VIS-01	Visa Checker	Nationality + residence + destination + transit + purpose	All users
VIS-02	Requirement Result	Visa type, eligibility, validity rules, fees, timeline	All users
VIS-03	Document Checklist	Required documents with upload/status	Traveller
VIS-04	Embassy / Consulate Info	Addresses, hours, VFS/Gerry's centres, contacts	All users
VIS-05	My Visa Applications & Appointments	Tracker with status and dates	Traveller
VIS-06	Visa Rules Admin	Maintain rules, sources, review dates	Ops / Knowledge Manager

WIREFRAME — VIS-01 / VIS-02 Visa Checker & Result

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Check visa requirements

Passport nationality

Pakistan

Country of residence

Pakistan

Destination

United Kingdom

Transit via

Doha (DOH)

Purpose

Tourism

Travel dates

10-24 Dec

17

Stay

14 days

Check

Visa required — UK Standard Visitor Visa

PROCESSING

15 working days

Priority: 5 days

FEE

Live tariff

+ VFS service fee

VALIDITY

6 months

Multiple entry

Transit Doha: No transit visa needed if airside < 24 h

Key requirements

☐ Passport valid for the whole stay

☐ Bank statements (6 months)

☐ Employment letter / NOC

☐ Hotel & flight reservation (FlightOne can provide)

☐ TB test certificate (stays > 6 months only)

Source: UK Government (gov.uk) · Last verified 02 Sep 2026

Start checklist

Embassy & VFS info

Get visa assistance

UF-08.1 Check Destination Requirements

PRIMARY ACTOR		PRE-CONDITIONS
Traveller (or AI in chat)		—
#	Actor	Action / System Response
1	User	Opens Visa Centre (or asks AI “Do I need a visa for the UK?”).
2	System	Pre-fills nationality and residence from profile and any existing visas/permits (e.g. valid US/UK/Schengen visa that grants visa-on-arrival elsewhere).

#	Actor	Action / System Response
3	User	Enters destination, transit points, purpose, dates and length of stay; clicks Check .
4	System	Evaluates rules engine (nationality × residence × destination × purpose × held visas) using the Visa Knowledge Base (Module 16) and third-party data feed (e.g. Timatic-equivalent) where licensed.
5	System	Displays VIS-02: status (Visa-free / e-Visa / Visa on arrival / Visa required / Not allowed), visa type, fees, processing time, validity, passport validity rule, entry conditions, and source with last-verified date.
6	AI	In chat, summarises in two lines and offers “Start checklist” or “Get visa assistance”.
POST-CONDITIONS		
Requirement result shown; saved to trip if linked.		

UF-08.2 Transit Rules Check During Booking

PRIMARY ACTOR		PRE-CONDITIONS
System		Itinerary with connections.
#	Actor	Action / System Response
1	System	For each connection, evaluates transit rules (airside/landside, terminal change, self-transfer, duration, overnight).
2	System	If a transit visa is required and user does not hold one, marks the option ‘Transit visa required’ and down-ranks it (Module 4).
3	System	At checkout, requires user acknowledgement; AI proposes an alternative routing without transit visa.
POST-CONDITIONS		
Travellers avoid being denied boarding.		

WIREFRAME — VIS-03 Checklist & VIS-05 Application Tracker

app.flightone.pk/visa/applications/UK-2291

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UK Visitor Visa — Checklist

Document	Status	
Passport (Vault)	Ready	View
Photograph 45×35	Uploaded	View
Bank statements	Missing	Upload
Employment letter	Missing	Upload
Flight reservation	Ready	Generate

Checklist

60%

Application tracker

1 Documents

2 Form submitted

3 Appointment

4 Biometrics

5 Decision

6 Passport returned

Appointment

VFS Lahore · 25 Sep · 10:30

17

17

GWF reference

GWF0771...

Status

Appointment booked

☒ Remind me 1 day before

Save

Add to calendar

UF-08.3 Complete Document Checklist

PRIMARY ACTOR		PRE-CONDITIONS
Traveller		Requirement result displayed.
#	Actor	Action / System Response
1	User	Clicks Start checklist .
2	System	Creates a visa application record with the checklist for that visa type; auto-marks documents already in the Vault.
3	User	Clicks Upload next to each missing item and uploads the file.

#	Actor	Action / System Response
4	System	Validates format/size; for photos checks dimensions and background; updates progress bar.
5	User	Clicks Generate for flight/hotel reservation letter (FlightOne service, may be chargeable).
6	System	When 100 % complete, offers Download application pack and Get visa assistance .
POST-CONDITIONS Checklist completed.		

UF-08.4 View Embassy / Consulate Information

PRIMARY ACTOR Any user		PRE-CONDITIONS —
#	Actor	Action / System Response
1	User	Clicks Embassy & VFS info .
2	System	Shows VIS-04: embassy/consulate and visa application centres for the user's city, addresses, map, working hours, holidays, contacts, submission/collection timings, fee payment modes.
3	User	Clicks Directions or Official website .
POST-CONDITIONS User has correct submission details.		

UF-08.5 Track Appointment & Processing Timeline

PRIMARY ACTOR Traveller		PRE-CONDITIONS Application record exists.
#	Actor	Action / System Response
1	User	Opens My Visa Applications and selects an application.
2	User	Enters appointment date/time, centre, and reference number; clicks Save .
3	System	Schedules reminders (1 day and 2 hours before); shows expected decision date = submission date + processing time.
4	User	Updates status as it progresses (or Ops updates for assisted applications).
5	System	If a flight is booked before the expected decision date, warns and recommends refundable fares.
POST-CONDITIONS Application tracked end-to-end.		

UF-08.6 Visa Expiry Reminders

PRIMARY ACTOR System		PRE-CONDITIONS Visa saved with validity and duration of stay.
#	Actor	Action / System Response
1	System	Schedules reminders at 60, 30 and 7 days before visa expiry.
2	System	During a trip, tracks <i>duration of stay</i> from arrival date and alerts at 7 and 2 days before the allowed stay ends.
3	System	Notifies via App, Email and WhatsApp with actions: Renew, Start new application.
POST-CONDITIONS No overstays or expired-visa travel.		

UF-08.7 Request Visa Assistance (Service)

PRIMARY ACTOR

Traveller; Visa Ops

PRE-CONDITIONS

—

#	Actor	Action / System Response
1	User	Clicks Get visa assistance , selects service package, and pays service fee.
2	System	Creates Ops case in Visa queue with checklist and documents.
3	Ops	Visa officer reviews documents, requests corrections via the application thread, fills forms, books appointment and updates status.
4	System	Traveller sees every status change in VIS-05 and receives notifications.

POST-CONDITIONS

Assisted visa processing.

UF-08.8 Maintain Visa Rules

PRIMARY ACTOR

Knowledge Manager

PRE-CONDITIONS

Role permission.

#	Actor	Action / System Response
1	Ops	Opens Visa Rules Admin → selects destination.
2	Ops	Edits rule rows per nationality: status, visa types, fees, processing times, document checklist, transit rules, source URL, next review date.
3	Ops	Clicks Submit for review ; second reviewer approves.
4	System	Publishes; content feeds AI Knowledge Base; flags rules past review date on the admin dashboard.

POST-CONDITIONS

Rules accurate and current.

BUSINESS RULES

1. Every visa answer shows its source and last-verified date, plus the disclaimer that final decisions rest with the embassy/immigration authority.
2. Rules past their review date are flagged 'Verify with embassy' in the UI.
3. Passport validity rules (e.g. 6 months) are checked against the traveller's passport automatically.
4. Visa assistance cases follow SLA per service package.

Live Journey Management

Release: Phase 2

Real-time monitoring of flights and stays after booking — delays, gate and terminal changes, weather, boarding, hotel check-in, transfers and immigration advisories — with proactive disruption assistance.

MONITORED EVENTS

Event	Source	Default timing / threshold	Channels
Flight delay	Flight status API + Galileo schedule change queue	Delay ≥ 15 min; updates on every change	App, WhatsApp, Email
Gate change	Flight status API	Immediately	App, WhatsApp
Terminal change	Flight status API / Galileo	Immediately; plus 24 h before	App, WhatsApp, Email
Cancellation / schedule change	Galileo queues, airline	Immediately	All + Ops alert
Weather disruption	Weather API (origin, destination, connection)	Severe alerts 48 h → departure	App, Email
Online check-in open	Airline rules	When check-in opens (24–48 h)	App, WhatsApp
Boarding reminder	Schedule	3 h to airport · 45 min boarding	App, WhatsApp
Hotel check-in	Booking	Day before + on arrival	App, WhatsApp
Airport transfer	Transfer booking (future) / advice	On landing	App, WhatsApp
Immigration advisory	Knowledge Base (Module 16)	48 h before international departure	App, Email

SCREEN INVENTORY

ID	Screen	Purpose	Access
LJ-01	Live Trip Timeline	Chronological trip view with live status	Traveller
LJ-02	Flight Status Card	Times, gate, terminal, baggage belt, aircraft	Traveller
LJ-03	Disruption Assistant	Options to rebook/refund when disrupted	Traveller
LJ-04	Notification Preferences	Channels, quiet hours, followers	Traveller
LJ-05	Disruption Control Board	All affected trips for Ops	Ops

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app.flightone.pk/trips/FO-240919-0831/live

→ FlightOne Traveller Portal

SA

Dubai • 19–22 Sep

EK623 LHE → DXB

DELAYED 45 min

DEPARTS

10:25

was 09:40

TERMINAL / GATE

T1 / A7

Gate changed

ARRIVES

12:50

T3 • Belt 7

Check-in open • Seat 23C • Boarding 09:55

Boarding pass

Share live status

Airport transfer • Suggested pickup moved to 13:30

Notify driver

Rove Downtown • Check-in 15:00 • Late-arrival note sent ✓

Immigration advisory • UAE: keep return ticket & hotel voucher handy (sample Knowledge Base advisory)

→ EK624 DXB → LHE • 22 Sep 21:55 • On time

Alerts

09:02 Gate changed A3 → A7

08:40 Delay +45 min (late inbound aircraft)

Weather: Dubai 41°C, clear

UF-09.1 Start Journey Monitoring		
PRIMARY ACTOR System		PRE-CONDITIONS Booking ticketed/confirmed.
#	Actor	Action / System Response
1	System	Creates monitoring subscriptions for each flight segment (carrier, flight number, date) and hotel stay.
2	System	Subscribes to flight status webhooks; polls Galileo schedule-change queue every 15 minutes.
3	System	Schedules time-based reminders (check-in, leave for airport, boarding, hotel check-in, advisories).
4	System	Adds trip to LJ-01 with status 'Upcoming'; becomes 'Live' 48 h before first departure.
POST-CONDITIONS Trip monitored.		

UF-09.2 Flight Delay / Gate / Terminal Change Alert		
PRIMARY ACTOR System → Traveller		PRE-CONDITIONS Monitoring active.
#	Actor	Action / System Response
1	Supplier	Flight status provider pushes an update (new ETD, gate, terminal).
2	System	Compares with last known state; ignores noise below threshold (<15 min).
3	System	Updates LJ-02 and trip timeline; recalculates impacts: connection time, hotel check-in, transfer pickup.
4	System	Sends message: "EK623 delayed 45 min — now departs 10:25 from T1 Gate A7." via preferred channels (respecting quiet hours except critical).
5	System	If connection time becomes below minimum connecting time → escalates to Disruption flow (UF-09.4).

#	Actor	Action / System Response
6	System	Notifies hotel of late arrival (if arrival after 22:00 or next day) and updates transfer pickup time.
7	User	Clicks alert → opens LJ-01; may click Share live status to send a tracking link to family/driver.
POST-CONDITIONS Traveller and dependants informed.		

UF-09.3 Weather Disruption Advisory

PRIMARY ACTOR System		PRE-CONDITIONS Trip within 48 h.
#	Actor	Action / System Response
1	System	Checks severe weather feeds for origin, connection and destination airports.
2	System	If severe warning (storm, fog, heat, snow), sends advisory with possible impact and airline waiver info if published (Knowledge Base).
3	AI	Offers proactive options: move to earlier/later flight if waiver permits, or monitor.
POST-CONDITIONS Early awareness.		

WIREFRAME — LJ-03 Disruption Assistant

app.flightone.pk/trips/FO-240919-0831/disruption

FlightOne
Traveller Portal
SA

FLIGHTONE
Home
AI Consultant
Search & Book
My Trips
Traveller Vault
Visa Centre
Groups
Rewards
Profile & Settings
Help

QR629 cancelled — you will miss your connection to London. We have protected options for you.

RECOMMENDED

QR631 dep 08:10 · via DOH · arrive LHR 15:40
+2h35 later · No cost (airline waiver)
Accept

EK623 + EK001 · via DXB · arrive LHR 14:55
Fare difference PKR 0 (involuntary)
Accept

Full refund to original payment
Involuntary — no penalty
Request refund

Talk to a consultant
Hotel for tonight?

Options held for 30 minutes · Consultant alerted automatically

UF-09.4 Cancellation / Missed Connection — Disruption Assistance

PRIMARY ACTOR System; AI; Traveller; Agent		PRE-CONDITIONS Cancellation or broken connection detected.
#	Actor	Action / System Response
1	System	Marks trip 'Disrupted'; creates high-priority case on LJ-05 Disruption Board.
2	AI	Searches re-accommodation options within airline involuntary rules (same carrier/alliance first) and refund option; ranks them.
3	System	Sends urgent WhatsApp + app push: "Your flight is cancelled. Tap to choose a new option."
4	User	Opens LJ-03, reviews options, clicks Accept on one (or Request refund).
5	System	Executes reissue/revalidation via Module 14 or queues to Ops if manual; issues updated e-ticket; updates hotel/transfer.

#	Actor	Action / System Response
6	Agent	If user clicks Talk to a consultant or does not respond in 30 min for a departure < 12 h, a consultant calls the traveller (Module 13).
POST-CONDITIONS Traveller re-accommodated or refunded quickly.		
ALTERNATE & EXCEPTION FLOWS VIP / corporate traveller — Consultant is assigned immediately; travel manager is notified. Overnight delay — AI offers hotel booking near the airport and informs about airline care entitlements.		

UF-09.5 Check-in & Boarding Reminders

PRIMARY ACTOR System		PRE-CONDITIONS Trip live.
#	Actor	Action / System Response
1	System	When online check-in opens, sends link to airline check-in with PNR and surname (copy buttons).
2	System	Sends 'Leave for the airport' reminder based on departure time, terminal and typical traffic (3 h international / 2 h domestic default).
3	System	Sends boarding reminder with gate 45 minutes before departure (after gate is known).
4	User	Uploads boarding pass (optional) → filed in Vault.
POST-CONDITIONS Traveller on time.		

UF-09.6 Hotel Check-in, Airport Transfer & Immigration Advisories

PRIMARY ACTOR System		PRE-CONDITIONS Trip live.
#	Actor	Action / System Response
1	System	Day before arrival: sends hotel details, check-in time, address/map and voucher.
2	System	On landing (flight status 'landed'): sends welcome message with immigration tips, baggage belt, transfer pickup point, and hotel directions.
3	System	If a transfer is booked (future integration), shares driver details and live location; otherwise suggests options.
4	System	48 h before international departure: immigration advisory (entry forms, customs limits, required documents, e-gates eligibility) from Knowledge Base.
POST-CONDITIONS Smooth arrival experience.		

UF-09.7 Manage Notification Preferences & Trip Followers

PRIMARY ACTOR Traveller		PRE-CONDITIONS —
#	Actor	Action / System Response
1	User	Opens Profile & Settings → Notifications (LJ-04).
2	User	Toggles channels per event type, sets quiet hours and language.
3	User	Adds Trip followers (family/assistant) by mobile/email for selected trips.
4	System	Followers receive status alerts only (no documents); critical alerts always delivered.

POST-CONDITIONS
Personalised notifications.

WIREFRAME — LJ-05 Disruption Control Board (Ops)

ops.flightone.pk/disruptions

FlightOne | Operations Console

SA

FLIGHTONE
Work Queue
Bookings
Escalations
Refunds & Reissues
Suppliers
Pricing Rules
Knowledge Base
Finance
Audit Logs
Users & Roles

DISRUPTED TRIPS
23

UNRESOLVED
9
3 departing < 6h

AUTO-RESOLVED
61 %

Priority	Traveller	PNR	Issue	Departs	AI options sent	Owner	
P1	Omar Khan (VIP)	Q2LK9A	QR629 cancelled	4h 10m	Yes · no reply	Ayesha	Open
P2	Sara Ahmed	K7X2QP	Delay 45m, MCT OK	1h 50m	n/a	Auto	Open

UF-09.8 Ops Disruption Control Board

PRIMARY ACTOR
Ops / Consultant

PRE-CONDITIONS
—

#	Actor	Action / System Response
1	Ops	Opens Disruption Board ; list sorted by priority (VIP, time to departure, severity).
2	Ops	Clicks Open on a case; sees timeline, AI options sent, traveller replies.
3	Ops	Takes ownership, contacts traveller, executes change, and closes case with resolution code.

POST-CONDITIONS
No disrupted traveller left unattended.

BUSINESS RULES

1. Critical alerts (cancellation, missed connection, gate change within 2 h) bypass quiet hours.
2. WhatsApp messages use pre-approved templates (Meta) with deep links.
3. Duplicate alerts for the same event are suppressed within 5 minutes.
4. All alerts are stored on the trip timeline and in the Notification Centre.

Rewards & Referrals

Release: Phase 2

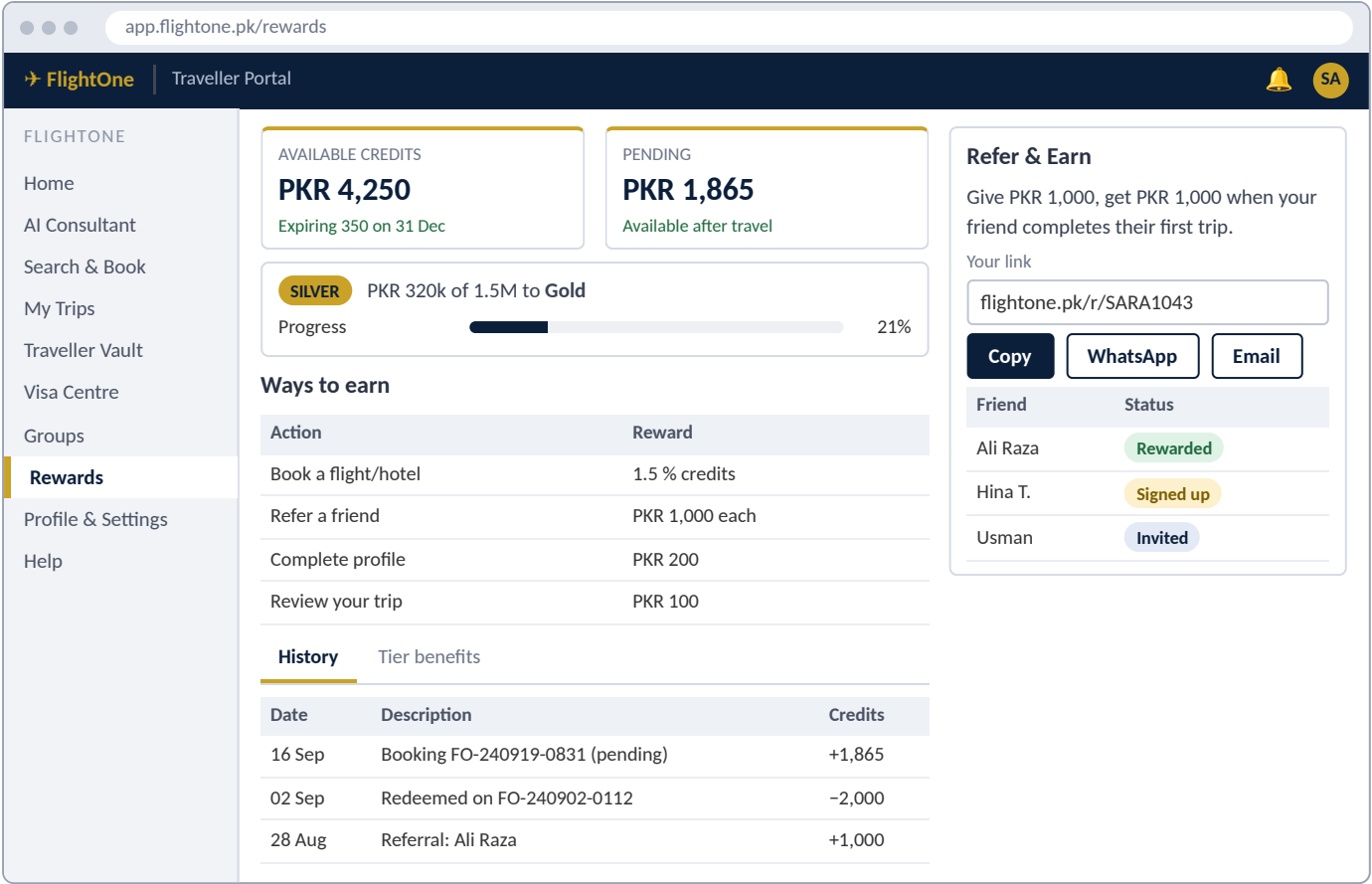
Booking rewards, redeemable credits, referral rewards, tiered loyalty and corporate reward programmes.

PROGRAMME STRUCTURE (CONFIGURABLE)

Tier	Qualification (12 months)	Earn rate	Benefits
Blue	Sign-up	1 % of fare value (excl. taxes)	Member prices
Silver	PKR 500k spend or 5 trips	1.5 %	Priority chat queue, free date change once/yr
Gold	PKR 1.5M or 12 trips	2 %	Dedicated consultant, waived service fees
Platinum	PKR 4M or 25 trips	3 %	VIP handling, lounge offers, upgrades on request

SCREEN INVENTORY

ID	Screen	Purpose	Access
RW-01	Rewards Home	Balance, tier, progress, ways to earn	Traveller
RW-02	Credits History	Earned, pending, redeemed, expired	Traveller
RW-03	Redeem at Checkout	Apply credits to payment	Traveller
RW-04	Refer & Earn	Referral link, invites, status	Traveller
RW-05	Tier Benefits	Tier comparison	Traveller
RW-06	Corporate Rewards Programme	Company-level rewards setup and statements	Corporate Admin / Ops
RW-07	Rewards Rules Admin	Earn/burn rules, tiers, expiry, campaigns	Ops / Marketing



UF-10.1 Earn Credits on a Booking		
PRIMARY ACTOR System		PRE-CONDITIONS Booking paid.
#	Actor	Action / System Response
1	System	Calculates credits = eligible amount × tier earn rate (+ campaign bonus).
2	System	Adds credits as <i>Pending</i> with expected release date (after travel completion / hotel check-out).
3	System	Shows 'You'll earn X credits' on confirmation and in RW-02.
4	System	On completion, changes status to <i>Available</i> and notifies the user; sets expiry (e.g. 12 months).
5	System	If booking is cancelled/refunded before release, pending credits are cancelled; partial refunds adjust proportionally.
POST-CONDITIONS Credits accurately accrued.		

UF-10.2 Redeem Credits at Checkout		
PRIMARY ACTOR Traveller		PRE-CONDITIONS Available credits > 0.
#	Actor	Action / System Response
1	User	On Payment screen ticks Use FlightOne credits .
2	System	Shows maximum redeemable (e.g. up to 20 % of total or configured cap) and pre-fills amount.
3	User	Adjusts amount and clicks Apply .
4	System	Reduces payable; locks credits for the hold duration; remaining balance paid by other method.

#	Actor	Action / System Response
5	System	On payment success, credits are deducted (oldest-expiring first); on failure, lock released.
POST-CONDITIONS Credits redeemed.		
ALTERNATE & EXCEPTION FLOWS Refund of a booking paid partly with credits — Credits portion returned as credits (re-validated with new expiry), cash portion to original method.		

UF-10.3 Refer a Friend		
PRIMARY ACTOR Traveller (referrer); Friend (referee)		PRE-CONDITIONS Referrer has an account.
#	Actor	Action / System Response
1	User	Opens Rewards → Refer & Earn ; clicks Copy , WhatsApp or Email to share personal link.
2	User	Friend opens link, signs up (UF-00.2); referral code is attached.
3	System	Gives friend welcome credits usable on first booking.
4	System	When friend completes first trip, awards referrer credits; updates status to <i>Rewarded</i> .
5	System	Fraud checks: same device/payment card/household address, self-referral, abnormal volume → held for review.
POST-CONDITIONS Referral rewarded.		

UF-10.4 Tier Progression		
PRIMARY ACTOR System		PRE-CONDITIONS Rolling 12-month activity.
#	Actor	Action / System Response
1	System	Recomputes qualifying spend/trips nightly.
2	System	On upgrade: notifies user, applies new benefits immediately.
3	System	Annual review: downgrade by maximum one tier if requirements not met; 30-day advance notice.
POST-CONDITIONS Tier accurate.		

UF-10.5 Corporate Reward Programme		
PRIMARY ACTOR FlightOne Ops; Corporate Admin		PRE-CONDITIONS Corporate company active.
#	Actor	Action / System Response
1	Ops	Opens company → Rewards ; selects model: company-level rebate (credit note), traveller-level credits (personal use allowed or not), or both.
2	Ops	Sets rebate slabs (e.g. 0.5 % above PKR 10M quarterly), payout (invoice credit note) and rules; clicks Save .
3	System	Accrues rewards on corporate bookings; shows quarterly statement in RW-06.
4	User	Corporate Admin views statement and clicks Apply to next invoice .
5	System	Creates credit note on next invoice (Module 6).
POST-CONDITIONS		

Corporate loyalty incentive in place.

UF-10.6 Credit Expiry

PRIMARY ACTOR

System

PRE-CONDITIONS

Credits with expiry.

#	Actor	Action / System Response
---	-------	--------------------------

1	System	Sends reminders 30 and 7 days before expiry.
---	--------	--

2	System	On expiry date, moves balance to <i>Expired</i> and logs ledger entry for Finance (liability release).
---	--------	--

POST-CONDITIONS

Liability managed.

UF-10.7 Configure Rewards Rules

PRIMARY ACTOR

Marketing / Ops

PRE-CONDITIONS

Permission.

#	Actor	Action / System Response
---	-------	--------------------------

1	Ops	Opens Rewards Rules Admin : tiers, earn rates by product, exclusions (taxes, corporate), redemption cap, expiry, referral amounts, bonus campaigns.
---	-----	--

2	Ops	Clicks Publish (maker-checker).
---	-----	--

3	System	Applies to new bookings from effective date.
---	--------	--

POST-CONDITIONS

Programme configured.

BUSINESS RULES

1. Credits are non-transferable and have no cash value; 1 credit = PKR 1.
2. Credits earned on corporate bookings follow company setting.
3. Rewards ledger is double-entry and reconciled with Back Office monthly (liability account).

MODULE 11

Group Travel

Release: Phase 2

Collaborative group trip management with shared itinerary, documents, live flight status, announcements, polls, attendance, emergency broadcasts, shared gallery and AI trip memories.

GROUP ROLES

Role	Permissions
Organiser (Owner)	All settings, invite/remove, itinerary, bookings, announcements, polls, attendance, emergency broadcast, delete group
Co-organiser / Group Leader	Same as organiser except delete group and billing
Member (Traveller)	View itinerary & documents, vote, mark attendance (self check-in), upload photos, comment
Guardian (student/minor groups)	View-only for their dependant; receives all announcements and emergency alerts
FlightOne Consultant	Manage bookings, update itinerary, post announcements (if enabled)

SCREEN INVENTORY

ID	Screen	Purpose	Access
GR-01	My Groups	List of groups with status	Traveller
GR-02	Create Group Wizard	Type, details, template	Organiser
GR-03	Group Feed	Announcements, updates, comments	Members
GR-04	Members & Invitations	Invite, roles, document readiness	Organiser
GR-05	Shared Itinerary	Day-by-day plan with live updates	Members
GR-06	Shared Documents	Group repository	Members
GR-07	Flight Status	All group flights live	Members
GR-08	Polls	Create and vote	Members
GR-09	Attendance	Roll call / QR check-in	Organiser / Members
GR-10	Emergency Broadcast	Urgent message to all + guardians	Organiser
GR-11	Gallery	Shared photos	Members
GR-12	AI Trip Memories	Auto-generated story/album	Members

app.flightone.pk/groups/new

FlightOneTraveller Portal

SA

FLIGHTONE

Home

AI Consultant

Search & Book

My Trips

Traveller Vault

Visa Centre

Groups

Rewards

Profile & Settings

Help

1 Group type2 Details3 Travel plan4 Invite members5 Review

What kind of group trip?

Corporate tour

Student group

Umrah / Hajj

Leisure tour

Sports team

Family

Umrah template adds: Mahram mapping, Nusuk/visa document checklist, Makkah-Madinah itinerary blocks, Ziyarat schedule, group leader (Muallim) role, prayer-time aware reminders.

Group name

Dates

Expected size

Al-Noor Umrah — Dec 2026

05 – 19 Dec 2026

45

Cancel

Continue

UF-11.1 Create a Group		
PRIMARY ACTOR		PRE-CONDITIONS
Organiser		Signed in.
#	Actor	Action / System Response
1	User	Clicks Groups → + Create group .
2	User	Step 1: selects group type (Corporate tour, Student, Umrah/Hajj, Leisure, Sports, Family).
3	System	Loads type template (roles, checklists, itinerary blocks, privacy defaults).
4	User	Step 2: enters name, dates, destination(s), expected size, cover photo, privacy (invite-only / link join with approval).
5	User	Step 3: chooses Request group booking (UF-11.4) or Members book individually or Add existing bookings .
6	User	Step 4: invites members (UF-11.2) — can skip.
7	User	Step 5: reviews and clicks Create group .
8	System	Creates group workspace with feed, itinerary, documents, polls, attendance and gallery.
POST-CONDITIONS		Group created.

UF-11.2 Invite Members & Join		
PRIMARY ACTOR		PRE-CONDITIONS
Organiser; Member		Group exists.
#	Actor	Action / System Response
1	User	Organiser opens Members → Invite ; adds by mobile/email, uploads CSV (name, mobile, email, role, room-sharing preference), or copies invite link/QR.
2	System	Sends WhatsApp/email invitations with group name and join link.
3	User	Member clicks link → signs in or registers → clicks Join group .
4	System	Adds member (or queues for organiser approval); prompts member to complete profile & passport (for document readiness).
5	System	Members list shows readiness: profile, passport validity, visa, ticket, payment.
POST-CONDITIONS		

Members onboarded.

ALTERNATE & EXCEPTION FLOWS

- **Minor in student group** — Invitation goes to guardian, who adds the child as a family member and joins as Guardian.

WIREFRAME — GR-03 Group Feed with Shared Itinerary

app.flightone.pk/groups/alnoor-dec26

FlightOne

Group Workspace

SA

FLIGHTONE

Feed

Itinerary

Members

Documents

Flight Status

Polls

Attendance

Gallery

Memories

Settings

Al-Noor Umrah — Dec 2026

45 members · Leader: Maulana Tariq · Consultant: Ayesha (FlightOne)

ANNOUNCEMENT

Pinned · Organiser · 2h

Bus to Madinah departs **07:30** from hotel lobby. Bring passports.

41 read · 4 unread · 6

POLL

Ziyarat day preference

☒ Thursday 10 Dec (27 votes)

☐ Friday 11 Dec (12 votes)

Closes in 5 h

FLIGHT

SV739 LHE→JED · On time · Gate 4 · 44/45 checked in

Write an announcement...

Post

Emergency

Itinerary

Day 1 · 05 Dec

SV739 LHE→JED 03:15

Bus to Makkah · Hilton Suites

Day 2 · Umrah with group 05:00

Day 6 · Makkah → Madinah bus

07:30 (updated)

UF-11.3 Build & Update Shared Itinerary (Live)

PRIMARY ACTOR

Organiser / Consultant

PRE-CONDITIONS

Group exists.

#	Actor	Action / System Response
1	User	Opens Itinerary → + Add item : flight, hotel, transfer, activity, meal, meeting point, free time.
2	System	Booked items (flights/hotels) are inserted automatically and stay in sync with live status (Module 9).
3	User	Enters time, location (map), description, applicable members (all or sub-group) and attachments; clicks Save .
4	User	Edits an item later (e.g. time change) and ticks Notify members .
5	System	Updates itinerary in real time for all members; posts change note to Feed; sends push/WhatsApp to affected members.

POST-CONDITIONS

Everyone sees the same, current plan.

UF-11.4 Request a Group Booking

PRIMARY ACTOR

Organiser; Consultant

PRE-CONDITIONS

Group has size/dates.

#	Actor	Action / System Response
1	User	Clicks Request group booking ; enters route, dates, number of passengers, hotel needs (rooms by type), budget, special requirements.
2	System	Creates a Group Request case for FlightOne consultants (groups of 10+ require airline group desk quotes).
3	Agent	Consultant obtains group fares/room blocks, builds quote in Ops Console and sends it to the group.

#	Actor	Action / System Response
4	User	Organiser reviews quote and clicks Accept ; pays deposit (organiser pays all / each member pays own share).
5	System	Tracks name-list deadline and payment milestones; members submit names/passports via their profile.
6	Agent	Consultant issues tickets/rooms; documents appear per member in Vault and in the group's Documents.
POST-CONDITIONS Group booked with milestones tracked.		

UF-11.5 Shared Document Repository		
PRIMARY ACTOR Organiser; Members		PRE-CONDITIONS —
#	Actor	Action / System Response
1	User	Opens Documents ; organiser uploads group documents (visa copies list, rooming list, guidelines, insurance, tickets group PNR).
2	User	Sets visibility per document: all members, leaders only, or individual.
3	System	Individual documents (e-tickets, visas) are shown only to the owner and organisers.
4	User	Members view/download documents; organiser sees who has viewed mandatory documents.
POST-CONDITIONS Documents centralised.		

UF-11.6 Group Flight Status Updates		
PRIMARY ACTOR System		PRE-CONDITIONS Group flights booked.
#	Actor	Action / System Response
1	System	Aggregates all group segments on Flight Status tab with per-member check-in status.
2	System	Delay/gate changes are posted automatically to the Feed and pushed to affected members (UF-09.2).
3	User	Organiser sees who has not checked in or boarded (if airline data available) and taps Remind .
POST-CONDITIONS Group moves together.		

UF-11.7 Post Announcements		
PRIMARY ACTOR Organiser / Leader		PRE-CONDITIONS —
#	Actor	Action / System Response
1	User	Types an announcement in Feed; optionally adds attachment, location, or schedules for later; chooses audience (all / sub-group / leaders); ticks Pin and Require acknowledgement .
2	User	Clicks Post .
3	System	Delivers via App + WhatsApp (and Email if chosen); tracks read receipts and acknowledgements.
4	User	Organiser clicks Read by 41 to see list and Nudge unread .
POST-CONDITIONS Information delivered and tracked.		

UF-11.8 Create a Poll & Vote

PRIMARY ACTOR

Organiser; Members

PRE-CONDITIONS

—

#	Actor	Action / System Response
1	User	Organiser clicks Polls → + New poll ; enters question, options, single/multiple choice, anonymous yes/no, closing time; clicks Publish .
2	System	Posts poll to Feed and notifies members.
3	User	Member selects option and clicks Vote (can change until closing).
4	System	Shows live results (or after close if hidden); on close, announces result; organiser can convert result into itinerary item.

POST-CONDITIONS

Group decisions made quickly.

WIREFRAME — GR-09 Attendance & GR-10 Emergency Broadcast

app.flightone.pk/groups/alnoor-dec26/attendance

FlightOne
Group Workspace
SA

FLIGHTONE
Feed
Itinerary
Members
Documents
Flight Status
Polls
Attendance
Gallery
Memories
Settings

Roll call • Bus to Madinah 07:30

PRESENT
41

MISSING
3

EXCUSED
1

Member	Method	Time	Status
Ahmed Raza	QR scan	07:12	Present
Fatima Noor	Self check-in (GPS)	07:15	Present
Khalid Mehmood	—	—	Missing
Rabia Ali	Leader	06:50	Excused

Scan QR
Call missing
Close roll call

Emergency broadcast

Type
Lost member / Medical / Security / Weather / Other

Message
Khalid Mehmood missing at Gate 3. Call 0300-... if seen.

☒ Include guardians & emergency contacts
☒ Request 'I am safe' check-in
☒ Alert FlightOne 24/7 desk

Send to all (SMS + WhatsApp + Push)

UF-11.9 Attendance Tracking

PRIMARY ACTOR

Organiser; Members

PRE-CONDITIONS

Group itinerary item exists.

#	Actor	Action / System Response
1	User	Organiser opens Attendance → Start roll call for an itinerary item (bus, flight gate, activity).
2	System	Notifies members to check in; generates group QR.
3	User	Members check in by scanning QR, tapping I'm here (with location check within radius), or leader marks manually.
4	System	Shows live present/missing counts; highlights missing members with phone buttons.
5	User	Organiser clicks Close roll call ; system saves attendance log and notifies guardians of students if configured.

POST-CONDITIONS

Attendance recorded.

UF-11.10 Emergency Broadcast

PRIMARY ACTOR

Organiser / Leader / FlightOne

PRE-CONDITIONS

Emergency situation.

#	Actor	Action / System Response
1	User	Clicks 🚨 Emergency .
2	User	Selects type, writes message, chooses to include guardians/emergency contacts, requests 'I am safe' responses, alerts FlightOne 24/7 desk.
3	User	Clicks Send and confirms (double confirmation).
4	System	Sends via SMS + WhatsApp + push with highest priority (bypasses quiet hours); creates P1 case for Ops (Module 13).
5	System	Tracks safety responses live; re-sends to non-responders every 10 min (configurable).
6	User	Organiser clicks Resolve with note; all recipients receive the all-clear message.

POST-CONDITIONS

Rapid, tracked emergency communication.

UF-11.11 Shared Photo Gallery

PRIMARY ACTOR

Members

PRE-CONDITIONS

—

#	Actor	Action / System Response
1	User	Opens Gallery → Upload ; selects photos/videos; tags day/location.
2	System	Compresses, strips precise GPS metadata (keeps city), applies content moderation, and adds to albums by day.
3	User	Members like/comment and download; organiser can hide/remove items; members can request removal of photos they appear in.

POST-CONDITIONS

Shared memories collected.

UF-11.12 AI-Generated Trip Memories

PRIMARY ACTOR

AI; Members

PRE-CONDITIONS

Trip completed (or daily during trip).

#	Actor	Action / System Response
1	AI	After the trip, compiles itinerary, check-ins, top-liked photos and announcements into a story (timeline highlights, map route, stats — distance flown, cities, group photo).
2	System	Notifies members: "Your AI-Noor Umrah memories are ready."
3	User	Member opens Memories , edits title/cover, removes items, and clicks Share (link/WhatsApp) or Download PDF album .
4	AI	Suggests rebooking or next group trip ("Plan Ramadan Umrah 2027?").

POST-CONDITIONS

Engaging post-trip content and re-engagement.

BUSINESS RULES

1. Only organisers/leaders can post announcements and broadcasts (members can comment unless disabled).
2. Personal documents are never visible to other members.
3. Emergency broadcasts are logged and cannot be deleted.
4. Groups are archived 90 days after end date (gallery downloadable for 1 year).

MICE Platform

Release: Phase 2

Management of Meetings, Incentives, Conferences and Exhibitions — registration, travel, stay, transfers, agenda, badges, QR check-in, attendance, budget, sponsors and reporting.

SCREEN INVENTORY

ID	Screen	Purpose	Access
MC-01	Events List	All events and status	Event Manager
MC-02	Create Event Wizard	Type, dates, venue, settings	Event Manager
MC-03	Registration Form Builder	Fields, ticket types, approval	Event Manager
MC-04	Public Registration Page	Delegate self-registration	Delegates
MC-05	Delegates	List, import, status, communications	Event Manager
MC-06	Travel & Stay Allocation	Flights, rooms, transfers per delegate	Event Manager / Consultant
MC-07	Agenda Builder	Sessions, speakers, rooms, tracks	Event Manager
MC-08	Badge Designer & Printing	Templates, QR, print queue	Event Manager
MC-09	Check-in Console	QR scan, manual lookup, walk-ins	Event Staff
MC-10	Attendance	Session-level attendance	Event Manager
MC-11	Budget	Planned vs actual	Event Manager / Finance
MC-12	Sponsors	Packages, deliverables, leads	Event Manager
MC-13	Event Reports	Registration, travel, attendance, finance	Event Manager / Client

WIREFRAME — MC-02 Create Event Wizard

mice.flightone.pk/events/new

FlightOne | MICE Console

FLIGHTONE

Events

Delegates

Travel & Stay

Agenda

Check-in

Badges

Sponsors

Budget

Reports

1 Basics2 Venue & dates3 Registration4 Travel & stay5 Budget6 Publish

Event name

Pakistan Pharma Summit 2026

Type

Conference

Client

Acme Pharma (Pvt) Ltd

Start

12 Nov 2026

End

14 Nov 2026

Venue

Expo Centre, Karachi

Expected delegates

600

Currency

PKR

Event manager

Hina Raza

☐ Include flights

☐ Include hotels

☐ Include transfers

☐ Sponsors module

Save draft

Continue

UF-12.1 Create an Event

PRIMARY ACTOR
Event Manager

PRE-CONDITIONS
MICE role.

#	Actor	Action / System Response
1	User	Opens Events → + New event .
2	User	Basics: name, type (Meeting / Incentive / Conference / Exhibition), client (corporate account), logo, description.
3	User	Venue & dates: venue (map), halls/rooms, start/end, time zone, travel window (arrival/departure dates).
4	User	Enables modules: flights, hotels, transfers, sponsors, badges, paid tickets.
5	User	Sets budget headers and approvers; clicks Create event .
6	System	Creates event workspace with default registration form, badge template, and communication templates.

POST-CONDITIONS
Event in Draft.

WIREFRAME — MC-03 Registration Form Builder & MC-04 Public Registration

The wireframe shows the FlightOne MICE Console interface. On the left is a sidebar with a menu: FLIGHTONE, Events, Delegates (highlighted), Travel & Stay, Agenda, Check-in, Badges, Sponsors, Budget, and Reports. The main content area is titled 'Form builder' and contains a table of fields for the registration form:

Field	Type	Req.
Full name (as per CNIC/passport)	Text	✓
Organisation / Designation	Text	✓
Mobile (WhatsApp)	Phone	✓
Delegate category	Select	✓
Need flight? From city	Select	
Need hotel? Room sharing	Select	
Dietary requirements	Select	
Photo (for badge)	Upload	

Below the table are two sections: 'Ticket types' with a dropdown showing 'Delegate · Speaker · VIP · Exhibitor', and 'Approval' with a dropdown showing 'Manual for VIP'. At the bottom are buttons for '+ Add field' and 'Publish form'.

To the right of the form builder is a preview of the public registration form titled 'Register — Pakistan Pharma Summit'. It includes an 'Event banner' placeholder, a 'Full name' input field with 'Dr. Kamran Ali', a 'Delegate category' dropdown with 'Speaker' selected, 'Flight from' and 'Hotel' dropdowns with 'Lahore' and 'Yes — single' selected respectively, and a 'Register' button.

UF-12.2 Delegate Registration

PRIMARY ACTOR
Event Manager; Delegate

PRE-CONDITIONS
Event created.

#	Actor	Action / System Response
1	User	Event Manager builds the form in MC-03: adds fields, ticket/delegate categories (with price if paid), capacity, approval rule, confirmation message; clicks Publish form .
2	System	Generates public registration URL/QR and invitation templates.
3	User	Delegate opens link, fills the form, uploads photo, selects travel/hotel needs, pays (if paid), clicks Register .
4	System	Validates; creates delegate record; if approval required → <i>Pending</i> ; else <i>Confirmed</i> .
5	System	Sends confirmation (email + WhatsApp) with QR code and a delegate portal link (agenda, travel details, badge).

#	Actor	Action / System Response
6	User	Event Manager can also Import delegates via CSV or Invite a closed list (pre-filled links).
POST-CONDITIONS Delegates registered.		
ALTERNATE & EXCEPTION FLOWS Capacity reached — Waitlist enabled; delegates auto-promoted when seats free up. Duplicate registration — Matched by email/mobile; user asked to update existing registration.		

WIREFRAME — MC-06 Travel & Stay Allocation

mice.flightone.pk/events/pps26/travel

FlightOne

MICE Console

SA

FLIGHTONE

Events

Delegates

Travel & Stay

Agenda

Check-in

Badges

Sponsors

Budget

Reports

NEED FLIGHTS

212

168 ticketed

NEED ROOMS

340

Block 320 · +20 overflow

TRANSFERS

298

14 vehicles

Filter: From Lahore · Flight not booked

Bulk search flights

Assign rooms

Build transfer manifest

Delegate	From	Arrive / Depart	Flight	Hotel · Room	Transfer
Dr. Kamran Ali (Speaker)	LHE	11 / 14 Nov	PK302 Ticketed	Mövenpick · Single	T-04 11 Nov 10:30 <a>Edit
Sana Iqbal	ISB	11 / 14 Nov	Pending approval	Mövenpick · Twin w/ A. Khan	T-06 <a>Edit
Faisal Butt	Local	—	Not required	—	— <a>Edit

UF-12.3 Manage Delegate Flights

PRIMARY ACTOR

Event Manager / Consultant

PRE-CONDITIONS

Delegates need flights.

#	Actor	Action / System Response
1	User	Opens Travel & Stay ; filters delegates needing flights by origin city and dates.
2	User	Clicks Bulk search flights ; system groups delegates by origin/date.
3	System	Searches Galileo per group; AI recommends best options within event travel policy/budget; group fares requested for large blocks.
4	User	Assigns options to delegates (bulk) and clicks Book (or Send for client approval).
5	System	Creates PNRs, tickets on event billing account, sends e-tickets to delegates, and updates delegate portal.
6	System	Delegate change requests (date/time) arrive in the event queue and follow Module 14.
POST-CONDITIONS Delegate flights booked.		

UF-12.4 Manage Hotels (Room Blocks & Rooming List)

PRIMARY ACTOR

Event Manager

PRE-CONDITIONS

Hotel needed.

#	Actor	Action / System Response
1	User	Creates room block: hotel (RateHawk or direct contract), room types, quantity, rates, cut-off date, attrition terms.
2	User	Clicks Assign rooms : auto-assign by category/gender/sharing preferences or drag-and-drop.

#	Actor	Action / System Response
3	System	Generates rooming list (XLSX/PDF) and sends to hotel; overflow booked via RateHawk.
4	System	Warns before cut-off date on unused rooms; hotel vouchers sent to delegates.
POST-CONDITIONS Accommodation allocated.		

UF-12.5 Manage Transfers

PRIMARY ACTOR Event Manager		PRE-CONDITIONS Flights/hotels allocated.
#	Actor	Action / System Response
1	User	Clicks Build transfer manifest .
2	System	Groups arrivals/departures by time window (e.g. 60 min) and destination; proposes vehicles by capacity.
3	User	Adjusts groupings, assigns vendor/vehicle/driver, and clicks Confirm .
4	System	Sends pickup details to delegates (WhatsApp) and manifests to drivers; syncs with live flight status (Module 9) to adjust pickups.
POST-CONDITIONS Transfers organised.		

UF-12.6 Build Event Agenda

PRIMARY ACTOR Event Manager		PRE-CONDITIONS Event created.
#	Actor	Action / System Response
1	User	Opens Agenda → adds days, tracks, rooms.
2	User	Clicks + Session : title, type (keynote, panel, workshop, break), time, room, speakers (from delegates with Speaker category), capacity, description, requires registration yes/no.
3	System	Checks conflicts (room double-booking, speaker overlap) and warns.
4	User	Clicks Publish agenda .
5	System	Agenda appears on delegate portal; delegates can add sessions to 'My agenda'; changes notify affected delegates.
POST-CONDITIONS Agenda live.		

FlightOne

MICE Console

SA

FLIGHTONE

Events

Delegates

Travel & Stay

Agenda

Check-in

Badges

Sponsors

Budget

Reports

Badge template • Speaker

Event logo

Photo

DR. KAMRAN ALI

Chief Medical Officer • Getz Pharma

SPEAKER

QR

Edit layout

Print batch (58)

Check-in • Hall A entrance

Camera — scan delegate QR

✓ Dr. Kamran Ali • Speaker • Checked in 08:42 • Badge printed

Search name / mobile

Walk-in registration

CHECKED IN

412 / 588

WALK-INS

17

UF-12.7 Badge Generation		
PRIMARY ACTOR Event Manager		PRE-CONDITIONS Delegates confirmed.
#	Actor	Action / System Response
1	User	Opens Badges ; selects template per category; places fields (name, organisation, category colour, photo, QR).
2	User	Clicks Preview , then Print batch (pre-print) or enables Print on check-in .
3	System	Generates print-ready PDF (CR80 / A6) with unique QR per delegate; supports e-badge in delegate portal/WhatsApp.
POST-CONDITIONS Badges ready.		

UF-12.8 QR Check-in & Attendance Tracking		
PRIMARY ACTOR Event Staff; Delegate		PRE-CONDITIONS Event day.
#	Actor	Action / System Response
1	User	Staff opens Check-in Console on a laptop/tablet with camera and selects checkpoint (main entrance or session room).
2	User	Delegate shows QR (badge/e-badge/WhatsApp).
3	System	Scans and validates (event, date, category access rights, not already checked in at this checkpoint).
4	System	Shows success with delegate details; prints badge if print-on-check-in; records timestamp.
5	User	If no QR, staff searches by name/mobile and checks in manually; walk-ins registered via quick form.
6	System	Session-level scans build attendance per session; dashboard updates live (MC-10).
7	System	Offline mode caches delegate list and syncs when connection returns.
POST-CONDITIONS Attendance captured.		
ALTERNATE & EXCEPTION FLOWS		
• Access denied — Red screen with reason (e.g. VIP-only session); staff can override with supervisor PIN.		

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UF-12.9 Budget Management

PRIMARY ACTOR

Event Manager; Finance

PRE-CONDITIONS

Budget headers set.

#	Actor	Action / System Response
1	User	Opens Budget : lines for venue, flights, hotels, transfers, F&B, AV, printing, gifts, staff, contingency.
2	System	Auto-posts actuals from bookings (flights/hotels/transfers) and vendor invoices entered in Ops.
3	User	Adds manual estimates/actuals and uploads vendor quotes/invoices.
4	System	Shows planned vs committed vs actual, variance, per-delegate cost; alerts at thresholds; client approval for overruns.

POST-CONDITIONS

Financial control over the event.

WIREFRAME — MC-11 Budget & MC-12 Sponsors

mice.flightone.pk/events/pps26/budget

FlightOne
MICE Console

FLIGHTONE
Events
Delegates
Travel & Stay
Agenda
Check-in
Badges
Sponsors
Budget
Reports

Budget

Line	Planned	Committed	Actual	Var.
Flights	18.0M	16.9M	15.2M	-6 %
Hotels	22.5M	23.4M	—	+4 %
Transfers	2.1M	1.9M	—	-10 %
Venue & AV	9.0M	9.0M	4.5M	0 %

Sponsors

Sponsor	Package	Amount	Deliverables	Paid
Getz Pharma	Platinum	5.0M	8 / 10	Paid
Searle	Gold	2.5M	4 / 6	50 %
Hilton Pharma	Booth	0.8M	2 / 2	Due

+ Add sponsor

UF-12.10 Sponsor Management

PRIMARY ACTOR

Event Manager

PRE-CONDITIONS

Sponsors module enabled.

#	Actor	Action / System Response
1	User	Defines packages (Platinum/Gold/Booth) with price and deliverables (logo placements, booth, speaking slot, passes, delegate kit inserts).
2	User	Clicks + Add sponsor : company, contacts, package, contract upload, payment schedule.
3	System	Creates deliverables checklist and sponsor passes (as delegates in Exhibitor/Sponsor category).
4	User	Tracks deliverables and payments; sponsor staff can scan delegate badges for leads (with delegate consent).
5	System	Generates sponsor report (logo impressions, booth visits, leads).

POST-CONDITIONS

Sponsor commitments delivered and tracked.

UF-12.11 Event Reporting

PRIMARY ACTOR

Event Manager; Client

PRE-CONDITIONS

Event ongoing/finished.

#	Actor	Action / System Response
---	-------	--------------------------

1	User	Opens Reports ; selects report: registrations (by category/day/source), travel (flights/rooms/transfers), attendance (event & session), budget, sponsors, feedback survey.
---	------	---

2	System	Displays charts and tables; post-event sends feedback survey to delegates and includes results.
---	--------	---

3	User	Clicks Export (PDF/XLSX) or Share with client (read-only client portal link).
---	------	---

POST-CONDITIONS

Event outcomes documented.

BUSINESS RULES

1. Delegate data is visible only to the event team and the client; sponsor lead scans require delegate consent.
2. QR codes are signed and event-specific; badges can be re-issued (old QR invalidated).
3. Hotel cut-off and attrition dates generate alerts at 14, 7 and 2 days.
4. Event billing uses the client corporate account (Module 6) with a dedicated event cost centre.

Human Agent Escalation

Release: Phase 1 (MVP)

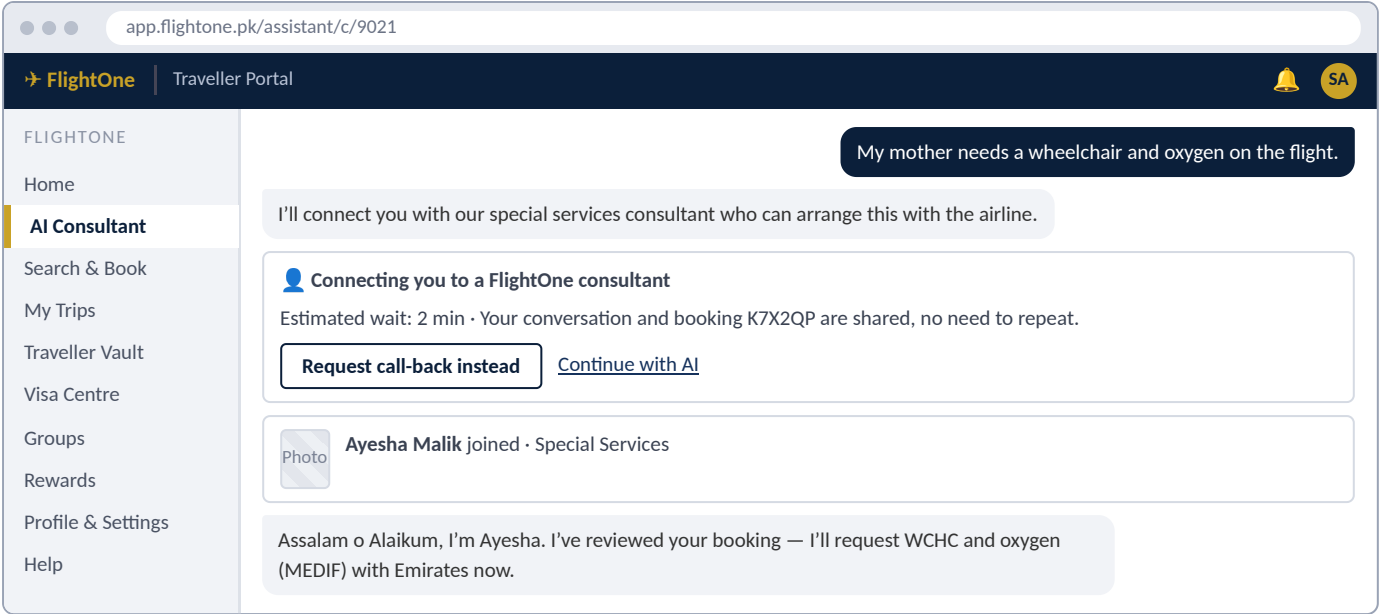
Intelligent hand-off from AI to FlightOne consultants for customer requests, VIPs, complex itineraries, supplier failures, refund disputes, medical and special service needs — with full conversation history.

ESCALATION TRIGGERS

Trigger	Detection	Priority	Routing
Requested by customer	“Talk to a human” button / intent detected (“agent”, “call me”)	P3 (P2 if booking in progress)	Next available consultant
VIP bookings	Customer tier Platinum / VIP flag / corporate VIP list / booking value > threshold	P2	VIP desk / dedicated consultant
Complex itineraries	Multi-city > 4 segments, open-jaw with mixed carriers, group ≥ 10, special fares (student/umrah/seaman), infants + special assistance	P3	Senior consultant
Supplier failures	Ticketing/booking error after payment, hotel not confirmed within 10 min, API outage	P1	Ops fulfilment desk
Refund disputes	Customer rejects refund quote, chargeback notice, negative sentiment on refund	P2	Refunds team
Medical assistance	Keywords (wheelchair, oxygen, pregnancy, medical emergency), MEDIF needed	P1 if in-trip / P2	Special services desk
Special service requests	Pets, unaccompanied minors, sports equipment, extra seat, name correction	P3	Special services desk
AI low confidence / loop	Confidence < threshold twice, 3 failed clarifications, negative sentiment	P3	Next available

SCREEN INVENTORY

ID	Screen	Purpose	Access
ESC-01	Customer Handover (in chat)	Transfer notice, wait time, call-back option	Traveller
ESC-02	Consultant Workspace	Queue, live conversation, Customer 360, AI co-pilot	Consultant
ESC-03	Escalation Rules & Routing Admin	Triggers, skills, SLAs, hours	Ops Manager
ESC-04	Supervisor Live View	Queue health, SLA, reassign	Team Lead



UF-13.1 Customer Requests a Human		
PRIMARY ACTOR		PRE-CONDITIONS
Traveller		In conversation.
#	Actor	Action / System Response
1	User	Clicks Talk to a consultant or types “I want to speak to someone”.
2	AI	Confirms in one line and asks for the topic if unclear (chips: New booking / Change / Refund / Other).
3	System	Creates escalation case with priority, topic, language, customer tier, linked booking and full transcript; checks business hours and queue wait.
4	System	Shows ESC-01: estimated wait, options Wait in chat , Request call-back (choose time), or Continue with AI .
5	System	Routes to consultant by skill (language, product, desk), availability and workload.
6	Agent	Consultant accepts; system posts “Ayesha joined” in chat.
7	Agent	Consultant replies in the same thread; the AI stays silent but offers suggestions to the consultant (co-pilot).
POST-CONDITIONS		
Customer is talking to a human with full context.		
ALTERNATE & EXCEPTION FLOWS		
No consultant available / after hours — Offers call-back slot or 24/7 emergency line for in-trip issues; AI continues for simple tasks. Customer abandons — Case remains open for follow-up via WhatsApp within 1 hour.		

UF-13.2 Automatic Escalation by Trigger		
PRIMARY ACTOR		PRE-CONDITIONS
System / AI		Trigger detected (see table).
#	Actor	Action / System Response
1	System	Evaluates escalation rules on every message and booking event.
2	System	On match, creates case with trigger reason and priority; for P1 alerts the desk via sound + WhatsApp to on-duty lead.
3	AI	Informs the customer transparently: “I’m bringing in a specialist for this.” and continues collecting useful details while waiting.

#	Actor	Action / System Response
4	System	For supplier failures after payment, customer is informed within 1 minute with ETA; refunds auto-initiated if not resolved within SLA (configurable).
POST-CONDITIONS Human involved where it adds value.		

WIREFRAME — ESC-02 Consultant Workspace

ops.flightone.pk/workspace

FlightOne

Operations Console

SA

FLIGHTONE

Work Queue

Bookings

Escalations

Refunds & Reissues

Suppliers

Pricing Rules

Knowledge Base

Finance

Audit Logs

Users & Roles

Queue

P1 Supplier failure · Hotel

Bilal S. · 03:12 waiting

P2 Medical · K7X2QP

Sara A. · Active

P3 Complex multi-city

Omar K. · 01:05

Sara Ahmed · Medical assistance

AI summary: Mother (Farzana Ahmed) on EK623 19 Sep needs wheelchair to seat (WCHC) + oxygen. MEDIF required.

My mother needs a wheelchair and oxygen on the flight.

Assalam o Alaikum, I'm Ayesha...

Co-pilot suggestions

Add SSR WCHC for pax 2 ·

Send Emirates MEDIF form link ·

Note: check airline medical-clearance lead time (KB)

Add SSR

Send MEDIF

Reply... /templates

Send

Customer 360

Silver · 14 trips · CSAT 4.8

Booking K7X2QP · Ticketed

Docs: Passport ✓ Visa ✓

Open booking

Call

Hand back to AI

Close case

UF-13.3 Consultant Handles the Case		
PRIMARY ACTOR Consultant		PRE-CONDITIONS Case assigned.
#	Actor	Action / System Response
1	Agent	Opens case from queue in ESC-02; reads AI summary, transcript, Customer 360 (profile, bookings, documents, loyalty, past cases).
2	Agent	Chats, or clicks Call (click-to-call; recording per policy).
3	Agent	Performs actions from workspace: search/book, modify PNR (SSR, seats), apply discount within limit (Module 5), trigger refund/reissue (Module 14), send documents, create Ops task.
4	AI	Co-pilot suggests next steps, knowledge articles (Module 16) and draft replies; consultant accepts/edits.
5	Agent	Adds internal notes and tags (topic, resolution).
POST-CONDITIONS Customer issue progressing with full audit trail.		

UF-13.4 Hand Back to AI or Close Case		
PRIMARY ACTOR Consultant; Traveller		PRE-CONDITIONS Issue resolved or remaining work is routine.
#	Actor	Action / System Response
1	Agent	Clicks Hand back to AI with a handover note (e.g. 'MEDIF sent; follow up for upload') — or Close case with resolution code.

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#	Actor	Action / System Response
2	System	AI resumes the conversation with the note as context and follows up on pending items.
3	System	On close, sends CSAT survey (1–5 + comment) in chat/WhatsApp.
4	System	Records handling time, first response time, resolution, CSAT for Module 17.
POST-CONDITIONS Case closed; learning captured for AI improvement.		

UF-13.5 Request & Fulfil a Call-back

PRIMARY ACTOR Traveller; Consultant		PRE-CONDITIONS Customer chose call-back.
#	Actor	Action / System Response
1	User	Selects date/time slot and phone number; clicks Confirm call-back .
2	System	Books slot in team calendar; sends confirmation and reminder 10 minutes before.
3	Agent	At slot time, the case appears at top of consultant queue; consultant clicks Call .
4	System	If unanswered, retries once after 10 minutes and sends WhatsApp message.
POST-CONDITIONS Customer called at chosen time.		

UF-13.6 Configure Escalation Rules & Monitor SLAs

PRIMARY ACTOR Ops Manager / Team Lead		PRE-CONDITIONS Permission.
#	Actor	Action / System Response
1	Ops	Opens Escalation Rules : edits triggers, thresholds (VIP value), keyword lists, priority mapping, skills/desks, business hours, SLA targets (P1 1 min, P2 5 min, P3 15 min).
2	Ops	Clicks Publish .
3	Ops	Team Lead opens Supervisor Live View: queue length, waiting times, SLA breaches, agent status; clicks Reassign or Join conversation (whisper mode).
POST-CONDITIONS Escalations routed and monitored.		

BUSINESS RULES

1. The complete AI conversation, trip brief and linked bookings always accompany the transfer.
2. Customer is told when a human joins and when the AI resumes.
3. P1 cases (supplier failure after payment, in-trip medical) must be acknowledged within 1 minute, 24/7.
4. Consultants' actions are logged under their user ID; AI suggestions accepted by a consultant are marked as such.

Refund & Reissue Engine

Release: Phase 1 (manual-assisted) · Phase 2 (automation)

Automated calculation and processing of refunds, exchanges, reissues, partial refunds, schedule changes, cancellations and travel credits for flights and hotels.

REFUND / CHANGE CALCULATION MODEL

Component	Flights	Hotels
Paid amount	Total paid per passenger (fare + taxes + fees)	Total paid per room
Supplier penalty	Per fare rules (Cat 16) — cancellation/no-show fee; fare difference for changes	Per cancellation policy — free until deadline, then % / nights
Non-refundable components	Non-refundable fare; certain carrier surcharges (YQ/YR) per rules	Non-refundable rates
Refundable taxes	Unused government taxes always refundable (even on non-refundable fares)	Taxes as per supplier
Agency fee	FlightOne cancellation/reissue service fee (per Module 5)	Cancellation service fee
Rewards / promo	Credits portion returned as credits; promo discount deducted proportionally	Same
Refund value	Paid – penalty – non-refundable – agency fee (+ refundable taxes)	Paid – penalty – agency fee
Timeline	Airline processing (BSP) 7–60 days; card reversal 5–14 business days after	3–14 days depending on supplier
Travel credit option	Credit shell / open ticket valid per airline (e.g. 12 months)	FlightOne credits (optional bonus e.g. +5 %)

SCREEN INVENTORY

ID	Screen	Purpose	Access
RR-01	Manage Booking	Change / cancel / services menu	Traveller
RR-02	Cancellation Quote	Refund breakdown & options	Traveller
RR-03	Change Flight Search	New dates/flights within rules	Traveller
RR-04	Reissue Quote	Fare difference, penalty, total	Traveller
RR-05	Refund Tracker	Status timeline	Traveller
RR-06	Schedule Change Acceptance	Accept / alternative / refund	Traveller
RR-07	Refunds & Reissues Queue	Ops processing, supplier status	Ops / Refunds team

PRIMARY ACTOR		PRE-CONDITIONS
Traveller (or AI / Consultant)		Booking ticketed/confirmed; cancellation allowed.
#	Actor	Action / System Response
1	User	Opens My Trips → trip → Manage booking → Cancel booking (or asks AI).
2	User	Selects components (flight passengers/segments, hotel rooms).
3	System	Fetches fare rules (Galileo) and hotel policy (RateHawk); checks void window (same-day void before BSP cut-off = full refund minus fee).
4	System	Calculates refund per component (model above) and displays RR-02 with timeline.
5	User	Chooses refund method: original payment or travel credit; selects reason; clicks Confirm cancellation .
6	System	Asks for final confirmation (OTP for amounts over threshold).
7	System	Flights: voids or refunds ticket in Galileo (auto-refund where supported, else queues to Ops); cancels PNR segments.
8	System	Hotels: cancels via RateHawk; receives cancellation confirmation.
9	System	Creates refund record, credit note in Back Office, reverses pending rewards; status <i>Refund in progress</i> .
10	System	Sends cancellation confirmation; RR-05 tracker available.
POST-CONDITIONS		
Booking cancelled; refund initiated.		
ALTERNATE & EXCEPTION FLOWS		
• Non-refundable fare — Quote shows taxes-only refund and suggests change instead or travel credit if airline allows. • Auto-refund not supported — Case created in RR-07 queue; customer sees ‘Being processed by our team’ with SLA. • Quote expired (>15 min) — Recalculated before confirmation.		

FLIGHTONE

Home

AI Consultant

Search & Book

My Trips

Traveller Vault

Visa Centre

Groups

Rewards

Profile & Settings

Help

1 Choose change2 Pick new flight3 Quote4 Pay & reissue

Original

EK624 DXB→LHE · 22 Sep 21:55

→

New

EK622 DXB→LHE · 23 Sep 16:20

Component	PKR
Fare difference (2 pax)	11,600
Tax difference	1,240
Airline change penalty (2 × 7,500)	15,000
FlightOne reissue fee	2,000
Total to pay	29,840

Hotel: extend Rove Downtown by 1 night for PKR 29,100? [Add](#)

BackPay & reissue

UF-14.2 Change Dates / Flights (Exchange & Reissue)		
PRIMARY ACTOR		PRE-CONDITIONS
Traveller		Ticket changeable per fare rules.
#	Actor	Action / System Response
1	User	Clicks Change dates / flights ; selects segment(s) to change and new date(s).
2	System	Searches availability on permitted carriers/fare rules (same airline, same or higher fare brand as rules dictate).
3	AI	Recommends best options with total change cost.
4	User	Selects a new flight.
5	System	Calculates reissue quote: fare difference + tax difference + penalty + agency fee (residual value handled per rules — refund, credit or forfeit).
6	User	Reviews RR-04 and clicks Pay & reissue ; pays (UF-03.6) or uses company billing/approval if corporate.
7	System	Executes exchange in Galileo; issues new ticket number(s); updates PNR, e-ticket, Vault, Live Journey, Back Office (ADM/ACM entries).
8	AI	Offers to adjust dependent services (hotel nights, transfers).
POST-CONDITIONS		Ticket reissued.
ALTERNATE & EXCEPTION FLOWS		
Change not permitted — Shows reason and offers cancel + new booking comparison. Lower fare (residual) — Shows residual handling per airline rule.		

UF-14.3 Partial Refund (One Passenger or Segment)		
PRIMARY ACTOR		PRE-CONDITIONS
Traveller		Multi-passenger or multi-segment booking.
#	Actor	Action / System Response
1	User	Clicks Cancel for one traveller (or selects unused segments).
2	System	Splits PNR for the selected passenger (Galileo divide) to keep others' bookings intact.

#	Actor	Action / System Response
3	System	Calculates refund for that passenger/segment only; child/infant pricing re-validated (e.g. infant cannot remain without adult).
4	User	Confirms.
5	System	Processes refund for the split record; others unaffected; hotel occupancy updated if needed (may require room change).
POST-CONDITIONS Partial refund processed without disrupting the rest of the group.		

WIREFRAME — RR-06 Schedule Change Acceptance

app.flightone.pk/trips/FO-240919-0831/schedule-change

FlightOneTraveller Portal

FLIGHTONE

Home

AI Consultant

Search & Book

My Trips

Traveller Vault

Visa Centre

Groups

Rewards

Profile & Settings

Help

The airline changed your flight **PK302**. New departure **14:45** (was 09:30).

Accept new time

No cost

Accept

Choose another flight

Free change within airline policy

See options

Full refund

Involuntary — no penalty

Refund

Please respond by 18 Sep 23:59. Our team will contact you if we don't hear back.

UF-14.4 Handle Airline Schedule Change		
PRIMARY ACTOR System; Traveller; Ops		PRE-CONDITIONS Schedule change received in Galileo queue.
#	Actor	Action / System Response
1	System	Reads queue, compares old vs new schedule, classifies: minor (< 60 min), major (≥ 60 min / routing change), cancelled; checks connection validity.
2	System	Minor: auto-accepts, updates PNR/documents, informs traveller.
3	System	Major: sends RR-06 with options (accept / alternative / refund) as per airline involuntary policy.
4	User	Clicks an option.
5	System	Executes involuntary reissue or refund (penalty-free), or accepts; Ops queue for manual cases.
6	System	If no response within deadline, Ops follows up; auto-accept only if still valid and within policy.
POST-CONDITIONS Schedule changes resolved proactively.		

UF-14.5 Choose Travel Credit Instead of Refund		
PRIMARY ACTOR Traveller		PRE-CONDITIONS Refund quote displayed.
#	Actor	Action / System Response
1	User	Selects FlightOne travel credit (or airline credit shell where offered).

#	Actor	Action / System Response
2	System	Shows credit value (optional bonus), validity and usage rules.
3	User	Confirms.
4	System	Credits wallet (Module 10 ledger, separate 'refund credit' type) instantly after supplier cancellation succeeds; airline credit shells recorded in the profile with PNR/ticket reference and expiry reminders.
POST-CONDITIONS		
Instant credit; cash retained.		

UF-14.6 Track Refund Status

PRIMARY ACTOR Traveller		PRE-CONDITIONS Refund initiated.
#	Actor	Action / System Response
1	User	Opens trip → Refund status (RR-05).
2	System	Shows timeline: Requested → Submitted to supplier → Supplier approved → Received by FlightOne → Refunded to you (with gateway reference).
3	System	Notifies at each stage; if delayed beyond SLA, auto-escalates to Refunds team.
POST-CONDITIONS		
Transparent refund tracking.		

WIREFRAME — RR-07 Refunds & Reissues Queue (Ops)

ops.flightone.pk/refunds

→ FlightOne | Operations Console

🔔

SA

FLIGHTONE

Work Queue

Bookings

Escalations

Refunds & Reissues

Suppliers

Pricing Rules

Knowledge Base

Finance

Audit Logs

Users & Roles

Status: Awaiting supplier

Supplier: All

Age > 7 days

Export

Case	Customer	Type	Supplier	Quote	Supplier status	Age	Owner	
RF-2291	Sara Ahmed	Full refund	Galileo / EK	154,400	Submitted	3d	Asad	Open
RI-1182	Omar Khan	Reissue	Galileo / QR	29,840	Manual needed	1h	—	Assign
RF-2275	Acme Pharma	Partial	RateHawk	43,100	Received	9d	Asad	Pay out
DS-0042	Hamza T.	Dispute	Galileo / PK	12,000	Under review	2d	Maria	Open

UF-14.7 Ops Processes Manual Refund / Reissue

PRIMARY ACTOR Refunds Officer		PRE-CONDITIONS Case in RR-07.
#	Actor	Action / System Response
1	Ops	Opens case; reviews quote, fare rules, PNR history and customer choice.
2	Ops	Performs refund/exchange in Galileo terminal or supplier portal; enters supplier reference and actual refund amount.
3	System	If actual differs from quote, flags variance; officer adds reason; customer notified if amount lower (requires customer acknowledgement).
4	Ops	When supplier funds are received (BSP/ supplier statement), clicks Pay out .

#	Actor	Action / System Response
5	System	Initiates card refund/bank transfer; posts accounting entries; closes case; notifies customer.
POST-CONDITIONS Refund completed and reconciled.		

UF-14.8 Refund Dispute		
PRIMARY ACTOR Traveller; Refunds team		PRE-CONDITIONS Customer disagrees with quote/outcome.
#	Actor	Action / System Response
1	User	Clicks Dispute this amount on RR-02/RR-05, selects reason and adds details/attachments.
2	System	Creates dispute case (P2) and escalates to Refunds team (Module 13).
3	Ops	Reviews rules and history; approves adjustment (within authority), requests airline waiver, or explains decision.
4	System	Records outcome; updates refund; customer notified with explanation.
POST-CONDITIONS Dispute resolved with audit trail.		

BUSINESS RULES

1. Quotes are valid for 15 minutes and always recalculated before execution.
2. Void is attempted automatically when within the same-day void window.
3. Involuntary changes (airline-initiated) never incur FlightOne penalties; agency fee waived by default.
4. Refunds go back to the original form of payment unless the customer selects credit.
5. Every refund/reissue creates matching Back Office entries (credit note, ADM/ACM, supplier receivable).

Operations Platform

Release: Phase 1 (core) · Phase 2 (advanced finance)

Integrated CRM, Mid Office, Back Office, Accounting & Finance, Commission tracking, Supplier reconciliation and Audit logs.

SCREEN INVENTORY

ID	Screen	Purpose	Access
OPS-01	Work Queue	Unified tasks: tickets to issue, failures, approvals, refunds, sync exceptions	Ops
OPS-02	CRM — Customer 360	Profile, bookings, conversations, cases, leads, notes	Consultant / Sales
OPS-03	CRM — Leads & Pipeline	Leads from chat/web/WhatsApp, stages, follow-ups	Sales
OPS-04	Mid Office — Booking File	Consolidated file per booking: segments, costs, sales, documents	Ops
OPS-05	Back Office — Accounting	Invoices, credit notes, journals, ledgers	Finance
OPS-06	Finance — Receipts & Supplier Payments	Collections, BSP/supplier settlements	Finance
OPS-07	Commission Tracking	Airline/hotel commissions & incentives; agent commissions	Finance
OPS-08	Supplier Reconciliation	Match supplier statements vs bookings	Finance
OPS-09	Audit Logs	Immutable record of all actions	Admin / Compliance
OPS-10	Users, Roles & Permissions	Staff accounts and RBAC	Super Admin
OPS-11	Integration Monitor & Sync Exceptions	Connector health and failed events	Ops / IT

ops.flightone.pk/queue

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SA

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Work Queue

Bookings

Escalations

Refunds & Reissues

Suppliers

Pricing Rules

Knowledge Base

Finance

Audit Logs

Users & Roles

All (48)

Ticketing (6)

Failures (3)

Refunds (14)

Schedule changes (9)

Sync exceptions (2)

Assign to me

Leads (14)

Priority	Task	Reference	Customer	SLA	Owner	
P1	Hotel not confirmed after payment	FO-240916-0114	Bilal Sheikh	00:45 left	—	Take
P2	Ticket by TTL today 18:00	PNR Q2LK9A	Omar Khan	3h	Asad	Open
P3	Back Office posting failed	INV-2609-118	Acme Pharma	1d	Finance	Retry
P3	New lead — Umrah group 30 pax	LD-5521	Masjid Committee	4h	Sales	Open

UF-15.1 Work the Operations Queue

PRIMARY ACTOR

Ops staff

PRE-CONDITIONS

Signed in with Ops role (2FA).

#	Actor	Action / System Response
1	Ops	Opens Work Queue ; filters by type/priority/owner.
2	Ops	Clicks Take on a task (assigns to self; SLA timer shown).
3	System	Opens the linked record (booking file, case, refund, lead) with required actions.
4	Ops	Completes the action and clicks Resolve with resolution code/notes.
5	System	Updates status, notifies customer where relevant, writes audit log, and measures SLA.

POST-CONDITIONS

Tasks completed within SLA.

ops.flightone.pk/crm/customers/C-10043

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Work Queue

Bookings

Escalations

Refunds & Reissues

Suppliers

Pricing Rules

Knowledge Base

Finance

Audit Logs

Users & Roles

Sara Ahmed

Silver · Personal + Corporate (Acme Pharma) · Since 2024 · LTV PKR 2.4M

Timeline

Bookings

Conversations

Cases

Documents

Rewards

Notes

When	Event
Today 14:35	Booking FO-240919-0831 ticketed (AI channel)
Today 14:20	AI conversation — Dubai trip (converted)
02 Sep	Case CS-9912 closed · CSAT 5
28 Aug	Referred Ali Raza

Tags & segments

Frequent GCC

Price-sensitive

Family

Next best action

Offer Dubai hotel+transfer bundle

New lead

Add note

UF-15.2 CRM — Customer 360 & Lead Management

PRIMARY ACTOR
Consultant / Sales

PRE-CONDITIONS
—

#	Actor	Action / System Response
1	Ops	Searches customer by name, phone, email, PNR or ticket; opens Customer 360.
2	System	Shows unified timeline: bookings, conversations (AI + human), cases, payments, documents status, rewards, notes, marketing consent.
3	System	Leads are auto-created from unconverted AI conversations with high intent, website forms, WhatsApp and group/MICE enquiries.
4	Ops	Opens Leads pipeline (New → Contacted → Quoted → Won/Lost), clicks a lead, logs a call, sends a quote (from Booking Engine), sets follow-up reminder.
5	Ops	Marks lead Won (linked booking) or Lost (reason).
6	System	Tracks conversion by source and consultant for Module 17.
POST-CONDITIONS Complete customer relationship view.		

WIREFRAME — OPS-04 Mid Office — Booking File

ops.flightone.pk/files/FO-240919-0831

→ FlightOne

Operations Console

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FLIGHTONE

Work Queue

Bookings

Escalations

Refunds & Reissues

Suppliers

Pricing Rules

Knowledge Base

Finance

Audit Logs

Users & Roles

File FO-240919-0831 · Sara Ahmed · Channel: AI · Profile: Personal Ticketed

Item	Supplier	Ref	Net cost	Markup	Fees	Discount	Sell	Tax
Air — 2 tickets	Galileo / EK	K7X2QP	178,000	5,400	3,000	-1,000	186,400	480
Hotel — 3 nights	RateHawk	RH-9918273	81,600	4,600	500	-2,000	86,700	80
Total			259,600	10,000	3,500	-3,000	273,100	560

Payments: Card · PKR 273,100 · Gateway ref 88A1...

Back Office: INV-P-2609-5512 posted ✓ · Supplier payable posted ✓

Commission: EK 0 % · RateHawk n/a

UF-15.3 Mid Office — Booking File Management

PRIMARY ACTOR
Ops

PRE-CONDITIONS
Booking exists.

#	Actor	Action / System Response
1	System	Creates/updates a booking file automatically from booking events (UF-03.9).
2	Ops	Opens file: sees all services, suppliers, references, cost/sell breakdown, payments, documents, history.
3	Ops	Adds offline services (e.g. manual visa service, transfer via local vendor) with cost and sell price; clicks Save .
4	System	Recalculates file totals and posts incremental entries to Back Office.
POST-CONDITIONS Single source of truth per booking.		

UF-15.4 Back Office — Accounting & Finance

PRIMARY ACTOR
Finance

PRE-CONDITIONS
Booking/financial events.

#	Actor	Action / System Response
1	System	Receives postings: sales invoice, credit notes, supplier bills, receipts, refunds, commissions, rewards liability; maps to chart of accounts.
2	Ops	Opens Accounting ; reviews invoices, journals and ledgers; filters by date/customer/supplier.
3	Ops	Records receipts for bank transfers (matches customer reference) and clicks Allocate .
4	Ops	Opens Supplier payments ; reviews BSP billing and RateHawk invoices; schedules payments; clicks Mark paid with bank reference.
5	System	Updates customer/supplier balances; exports to external accounting system (e.g. QuickBooks/Oracle/SAP connector) if used.
6	Ops	Runs period close: trial balance, receivables ageing, payables ageing, tax (sales tax on services, withholding) reports.

POST-CONDITIONS
Accurate books without double entry.

WIREFRAME — OPS-08 Supplier Reconciliation

ops.flightone.pk/finance/reconciliation/bsp-2609-1

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Operations Console

FLIGHTONE

Work Queue

Bookings

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Finance

Audit Logs

Users & Roles

Supplier

BSP Pakistan

Statement

BSP 01-07 Sep 2026

Upload & match

MATCHED

1,142

99.1 %

AMOUNT MISMATCH

6

PKR 38,210

IN STATEMENT ONLY

3

IN SYSTEM ONLY

1

Ticket	System	Statement	Difference	Suggested cause	
176-2419873321	93,200	93,200	0	—	Matched
157-2208812201	88,900	91,300	2,400	Tax change on reissue	Resolve
214-9981120034	—	74,300	74,300	Ticket issued outside platform	Import

UF-15.5 Supplier Reconciliation

PRIMARY ACTOR
Finance

PRE-CONDITIONS
Supplier statement available.

#	Actor	Action / System Response
1	Ops	Opens Reconciliation ; selects supplier (BSP/airline, RateHawk, other) and uploads statement (CSV/XLSX/HOT file) or fetches via API.
2	System	Auto-matches lines by ticket/confirmation number, date and amount (tolerance configurable).
3	System	Shows matched, amount mismatch, statement-only and system-only items with suggested causes.
4	Ops	Resolves exceptions: accept supplier amount (posts adjustment), raise dispute with supplier (ADM dispute), import missing booking, or mark as timing difference.
5	Ops	Clicks Close reconciliation .
6	System	Locks period for that supplier and posts adjustments.

POST-CONDITIONS

Supplier balances agree with books.

UF-15.6 Commission Tracking

PRIMARY ACTOR

Finance

PRE-CONDITIONS

Commission rules configured.

#	Actor	Action / System Response
1	Ops	Opens Commission → Rules : airline commissions/PLB (productivity-linked bonus) by carrier/route/period, hotel commissions, and staff/agent sales commissions.
2	System	Calculates expected commission per booking; accumulates towards PLB targets and shows progress (e.g. EK target 60 % achieved).
3	Ops	Records received commissions/incentives from suppliers; system reconciles expected vs received.
4	Ops	Runs staff commission report for payroll period; clicks Approve .

POST-CONDITIONS

Commission revenue and staff incentives tracked.

WIREFRAME — OPS-09 Audit Logs & OPS-10 Users & Roles

ops.flightone.pk/audit

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Work Queue

Bookings

Escalations

Refunds & Reissues

Suppliers

Pricing Rules

Knowledge Base

Finance

Audit Logs

Users & Roles

Audit log

User: any

Entity: Pricing rule

Last 7 days

Time	User	Action	Entity	Change
14:02:11	Hamid (Pricing)	UPDATE	Rule #52	markup 5 % → 6 %
13:40:55	AI Consultant	DISCOUNT	FO-... 0831	-800 (limit 2,000)
13:12:03	Asad (Ops)	REFUND	RF-2275	Pay out 43,100

Export

Roles

Role	Users	2FA
Consultant	18	Required
Refunds Officer	4	Required
Finance	6	Required
Pricing Manager	2	Required
Super Admin	2	Required

+ Invite staff

UF-15.7 Audit Logs

PRIMARY ACTOR

Admin / Compliance

PRE-CONDITIONS

Permission.

#	Actor	Action / System Response
1	System	Writes an append-only event for every create/update/delete, login, permission change, price/discount, payment, refund, document access, AI action (with model/version).
2	Ops	Opens Audit Logs ; filters by user, entity, action, date; clicks an event to see before/after values, IP, device.
3	Ops	Clicks Export (CSV) for audits.

POST-CONDITIONS

Full traceability; logs retained ≥ 7 years (tamper-evident).

UF-15.8 Manage Staff Users & Roles

PRIMARY ACTOR

PRE-CONDITIONS

Super Admin —		
#	Actor	Action / System Response
1	Ops	Opens Users & Roles → + Invite staff ; enters name, email, team, role(s), desks/skills, discount limit, working hours.
2	System	Sends invite; enforces 2FA on first login.
3	Ops	Edits a role's permission matrix (module × view/create/edit/approve/export) and clicks Save (maker-checker for sensitive permissions).
4	Ops	Deactivates leavers; system revokes sessions immediately and reassigns open tasks.
POST-CONDITIONS		
Least-privilege access maintained.		

UF-15.9 Integration Monitor & Sync Exceptions		
PRIMARY ACTOR		PRE-CONDITIONS
Ops / IT		—
#	Actor	Action / System Response
1	System	Tracks each connector (Galileo, RateHawk, payment gateway, WhatsApp, email, flight status, accounting) — uptime, latency, error rate.
2	System	Failed events (e.g. Back Office posting) go to Sync Exceptions with payload and error.
3	Ops	Opens exception, fixes data if needed, clicks Retry or Mark resolved .
4	System	Alerts IT on-call when error rate exceeds threshold.
POST-CONDITIONS		
Integrations healthy; no silent data loss.		

BUSINESS RULES

1. Financial postings are immutable; corrections are made through reversing entries.
2. Period locks prevent back-dated changes after close (Finance Manager override logged).
3. All Ops staff require 2FA and are subject to IP/device policies.
4. PII in logs is masked; access to unmasked data requires elevated permission.

MODULE 16

AI Knowledge Platform

Release: Phase 1 (MVP)

A governed, versioned knowledge base of SOPs, airline policies, supplier contracts, visa procedures, corporate agreements and travel policies that grounds the AI's answers.

KNOWLEDGE COLLECTIONS

Collection	Examples	Visibility	Owner
SOPs	Ticketing SOP, refund SOP, disruption handling, VIP handling	Staff + AI (internal)	Ops Manager
Airline policies	Baggage, fare brands, change/refund rules, special services, pets, waivers	Staff + AI (customer-facing)	Knowledge Manager
Supplier contracts	RateHawk terms, airline deals, hotel contracts, commission terms	Staff (restricted) + AI (internal reasoning only)	Finance / Contracts
Visa procedures	Country rules, checklists, embassy details	Staff + AI (customer-facing)	Visa team
Corporate agreements	Company-specific fees, SLAs, deal codes	Only that company's context + staff	Corporate Sales
Travel policies	Company travel policies (synced from Module 6) and FlightOne T&Cs	Company users + AI	Corporate Admin / Legal

SCREEN INVENTORY

ID	Screen	Purpose	Access
KB-01	Knowledge Library	Browse/search collections	Staff
KB-02	Upload & Ingest	Upload files/URLs, metadata	Knowledge Manager
KB-03	Article Editor	Create/edit structured articles	Knowledge Manager
KB-04	Review & Approval	Maker-checker publishing	Reviewer
KB-05	AI Test Console	Ask questions, see retrieved sources	Knowledge Manager
KB-06	Knowledge Gaps & Feedback	Unanswered questions, wrong answers	Knowledge Manager

ops.flightone.pk/knowledge

FlightOne | Operations Console

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Work Queue

Bookings

Escalations

Refunds & Reissues

Suppliers

Pricing Rules

Knowledge Base

Finance

Audit Logs

Users & Roles

Search knowledge... e.g. 'Emirates infant baggage'

+ Upload

All

SOPs

Airline policies

Supplier contracts

Visa

Corporate

Travel policies

Title	Collection	Version	Review by	Status	Used (30d)
Emirates — Baggage & fare brands (PK)	Airline	v7	30 Nov	Published	1,284
QR — Schedule change waiver Sep-26	Airline	v1	30 Sep	Published	212
PIA — Special assistance (MEDIF)	Airline	v3	12 Sep	Review overdue	44
flydubai — Change fees 2026	Airline	v2 draft	—	In review	—

Upload knowledge

File / URL

EK_fare_brands_Oct26.pdf

Collection

Airline policies

Airline

EK

Market

PK

Effective

01 Oct

Review by

31 Dec

Visibility

Customer-facing

☒ Supersedes v7

Upload & process

UF-16.1 Upload & Ingest a Knowledge Document		
PRIMARY ACTOR Knowledge Manager		PRE-CONDITIONS Permission.
#	Actor	Action / System Response
1	Ops	Opens Knowledge Base → + Upload .
2	Ops	Uploads file (PDF, DOCX, XLSX, HTML) or URL; sets collection, tags (airline, supplier, country, company), market, effective date, review date, visibility, and whether it supersedes an older version.
3	Ops	Clicks Upload & process .
4	System	Extracts text (OCR for scans), splits into sections, generates embeddings, extracts key facts (e.g. baggage allowances, fees) into structured fields for review.
5	System	Creates a Draft version and sends it to review.
POST-CONDITIONS Draft ready for review.		

UF-16.2 Review & Publish		
PRIMARY ACTOR Reviewer		PRE-CONDITIONS Draft exists.
#	Actor	Action / System Response
1	Ops	Opens Review queue (KB-04); compares draft with previous version (diff view).
2	Ops	Verifies extracted facts; edits where needed; clicks Approve & publish or Request changes .
3	System	Publishes new version; previous version archived (still retrievable for bookings made under it); AI index updated within minutes.
POST-CONDITIONS Only approved knowledge reaches the AI.		

UF-16.3 Create / Edit an Article (SOP)

PRIMARY ACTOR

Knowledge Manager

PRE-CONDITIONS

—

#	Actor	Action / System Response
1	Ops	Clicks + New article ; chooses template (SOP, policy summary, FAQ); writes steps, adds attachments and links to related articles.
2	Ops	Sets audience (staff / customer-facing) and clicks Submit for review .
3	System	Follows UF-16.2.

POST-CONDITIONS

Structured SOP available.

WIREFRAME — KB-05 AI Test Console

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Work Queue

Bookings

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Question

Can a passenger on this airline's Economy Saver fare carry 2 checked bags?

Context

Customer: Personal · Market PK

Ask

AI answer (sample): The Saver fare from Pakistan uses the weight concept, so two bags are allowed as long as the combined weight is within the fare's allowance.

Confidence: 0.92

Retrieved sources

Emirates — Baggage & fare brands (PK) · v7 · \$3.2

score 0.89

EK fare brands Oct-26 (draft) — *excluded: not published*

Report wrong answer

UF-16.4 AI Retrieves Knowledge to Answer (with Citations)

PRIMARY ACTOR

AI

PRE-CONDITIONS

Customer or consultant question.

#	Actor	Action / System Response
1	AI	Detects that the answer requires policy/procedure knowledge.
2	System	Searches only published, in-effect content the requester is allowed to see (e.g. corporate agreement of the user's company only; supplier contracts never quoted to customers).
3	AI	Composes a short answer grounded in retrieved sections; stores citations with the message (visible to staff; customers see 'Source: Emirates baggage policy').
4	AI	If no reliable source is found or confidence is low, says so and offers a consultant (Module 13); logs a knowledge gap.

POST-CONDITIONS

Accurate, traceable answers.

UF-16.5 Consultant Searches Knowledge

PRIMARY ACTOR
Consultant

PRE-CONDITIONS
—

#	Actor	Action / System Response
---	-------	--------------------------

1	Agent	Types a query in KB search (or in workspace co-pilot).
---	-------	--

2	System	Returns ranked sections with highlights and a generated summary with citations.
---	--------	---

3	Agent	Clicks Insert into reply to use approved customer-facing text.
---	-------	---

POST-CONDITIONS
Faster, consistent servicing.

UF-16.6 Knowledge Gaps, Feedback & Review Reminders

PRIMARY ACTOR
Knowledge Manager

PRE-CONDITIONS
—

#	Actor	Action / System Response
---	-------	--------------------------

1	System	Collects unanswered questions, low-confidence answers, 'Report wrong answer' flags and consultant corrections into KB-06.
---	--------	---

2	Ops	Opens KB-06, groups similar gaps, creates or updates content, and marks gaps resolved.
---	-----	--

3	System	Sends reminders for content approaching review date; overdue content is flagged and (optionally) de-prioritised in AI retrieval.
---	--------	--

POST-CONDITIONS
Knowledge base improves continuously.

BUSINESS RULES

1. Only Published and in-effect versions are used by the AI; drafts and archived versions are excluded (except for historical booking reference).
2. Access control is enforced at retrieval time (tenant, role, visibility).
3. Supplier contract terms are never disclosed to customers.
4. Each AI answer stores the knowledge versions used for audit (Module 15).

Management Dashboard

Release: Phase 1 (core KPIs) · Phase 3 (advanced analytics)

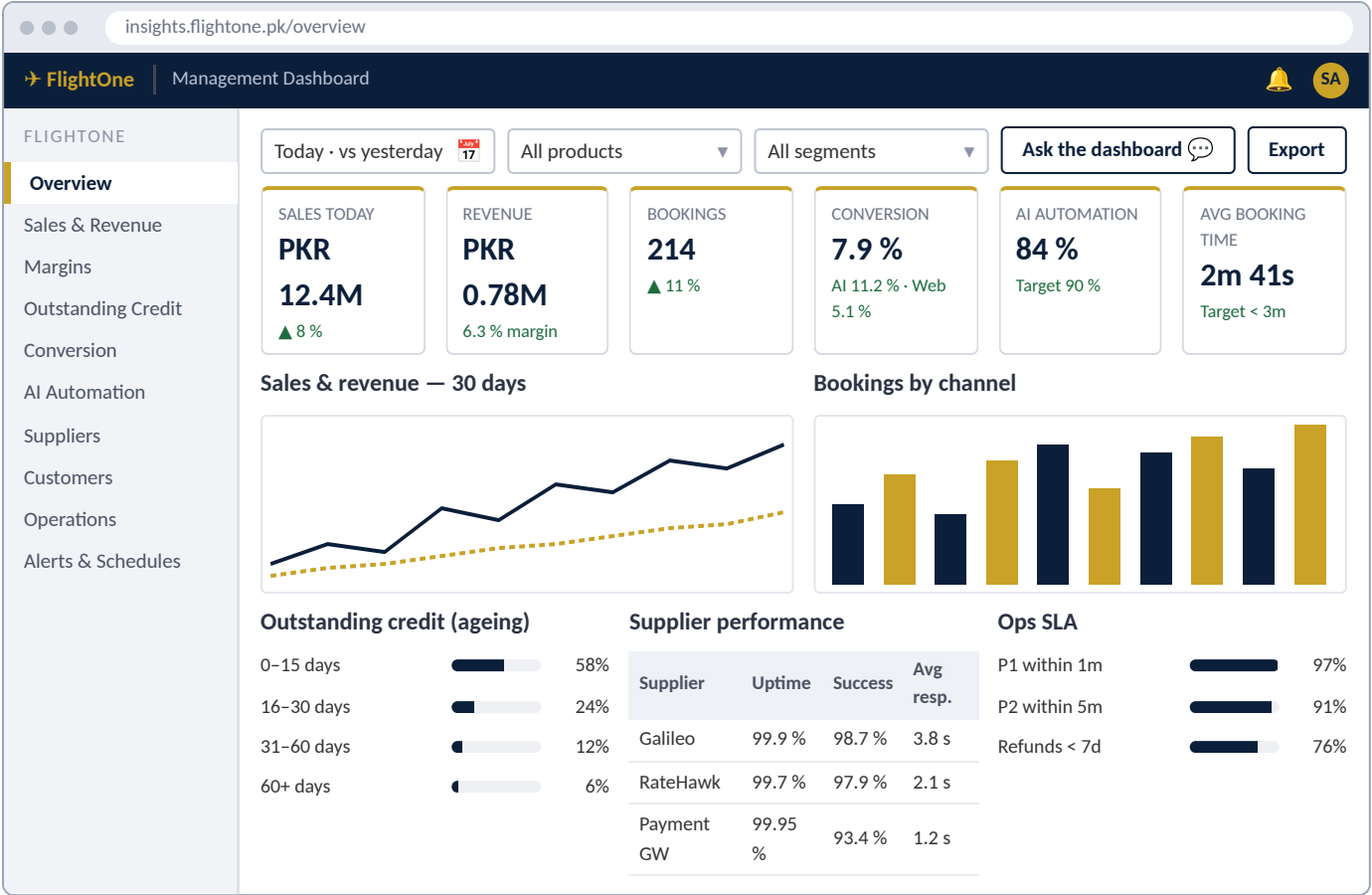
Real-time dashboards and alerts for sales, revenue, margins, outstanding credit, conversion, AI automation, supplier performance, customer analytics and operational KPIs.

KPI DEFINITIONS

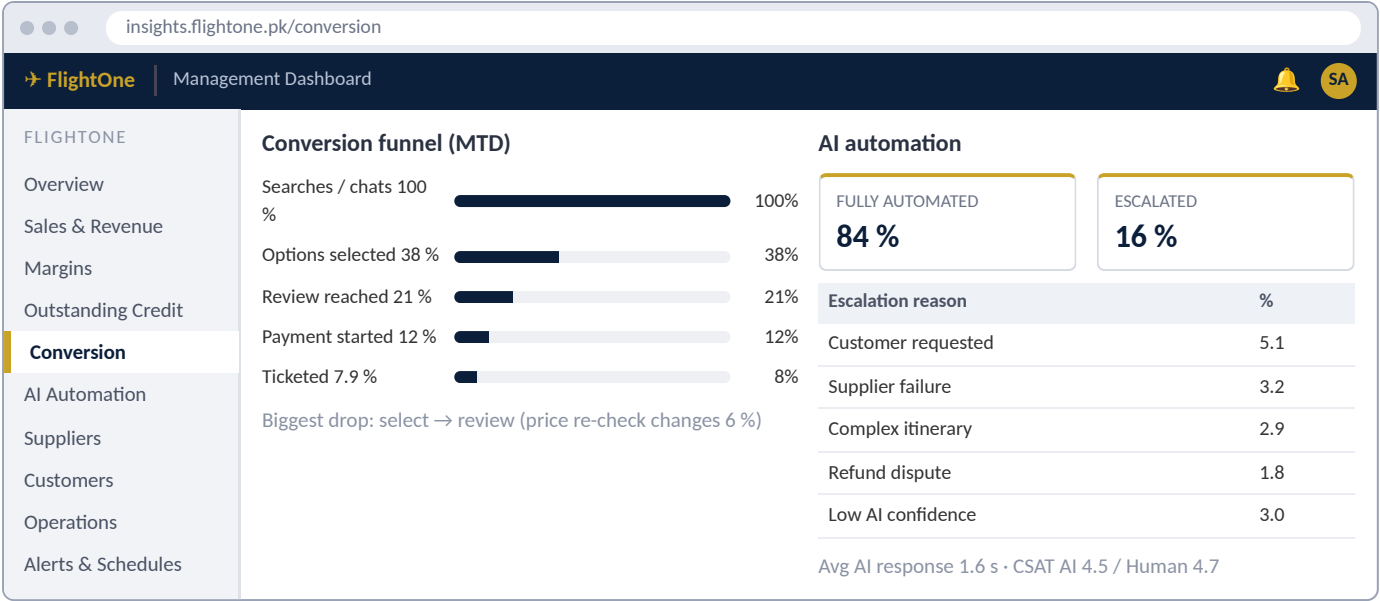
KPI	Definition	Refresh
Sales (GBV)	Gross booking value of confirmed/ticketed bookings (net of cancellations)	Live
Revenue	Markup + service fees + commissions + other income – discounts	Live
Margin %	Revenue ÷ Sales	Live
Outstanding credit	Corporate receivables: uninvoiced + invoiced unpaid; ageing buckets 0–15/16–30/31–60/60+	Hourly
Booking conversion	Bookings ÷ searches (and ÷ conversations for AI); funnel: search → select → review → pay → ticketed	Live
AI automation rate	Servicing requests fully completed by AI without human touch ÷ all requests (target 90 %)	Hourly
Supplier performance	Response time, availability, booking success %, price competitiveness, schedule-change rate, reconciliation mismatches	Hourly
Customer analytics	New vs repeat, CLV, NPS/CSAT, top routes, segments, churn risk, referral contribution	Daily
Operational KPIs	Queue size, SLA compliance, first response time, handling time, refunds ageing, disruption resolution time, booking time (target < 3 min)	Live

SCREEN INVENTORY

ID	Screen	Purpose	Access
DB-01	Executive Overview	Headline KPIs and trends	Management
DB-02 ... DB-09	Topic dashboards	Sales, Margins, Credit, Conversion, AI, Suppliers, Customers, Operations	Management / Managers
DB-10	Alerts & Scheduled Reports	Thresholds, subscriptions	Management



UF-17.1 View Executive Dashboard		
PRIMARY ACTOR Management		PRE-CONDITIONS Management role (2FA).
#	Actor	Action / System Response
1	User	Opens Management Dashboard .
2	System	Loads DB-01 with default period (Today vs Yesterday) and real-time tiles (streaming updates every 30 s).
3	User	Changes period (Today, WTD, MTD, QTD, YTD, custom), comparison (previous period / last year), and filters (product, channel, segment, company, supplier, branch).
4	System	Recalculates all widgets; saves the view as the user's default if chosen.
POST-CONDITIONS Leadership sees real-time business state.		



UF-17.2 Drill Down into a Metric		
PRIMARY ACTOR		PRE-CONDITIONS
Management / Manager		Dashboard open.
#	Actor	Action / System Response
1	User	Clicks a KPI tile or chart segment (e.g. 'Refunds < 7d 76 %').
2	System	Opens the topic dashboard filtered to that metric with breakdown tables.
3	User	Clicks a row to drill to list level (e.g. aged refund cases) and then to the record in the Operations Console (permission-checked).
POST-CONDITIONS		
Root cause visible in three clicks.		

UF-17.3 Ask the Dashboard (Natural Language Analytics)		
PRIMARY ACTOR		PRE-CONDITIONS
Management		—
#	Actor	Action / System Response
1	User	Clicks Ask the dashboard and types “Which routes lost margin this week vs last week?”.
2	AI	Translates to a governed metrics query (semantic layer; no free SQL), runs it, and returns a short answer with a chart and the filters applied.
3	User	Clicks Pin to dashboard or Export .
POST-CONDITIONS		
Fast ad-hoc insights.		

UF-17.4 Set KPI Alerts & Scheduled Reports		
PRIMARY ACTOR		PRE-CONDITIONS
Management		—
#	Actor	Action / System Response
1	User	Opens Alerts & Schedules → + Alert : metric, condition (e.g. AI automation < 80 % for 1 h; payment success < 90 %; supplier error rate > 5 %; overdue credit > PKR 5M), recipients, channels (email/WhatsApp).

#	Actor	Action / System Response
2	User	Clicks + Scheduled report : dashboard view, frequency (daily 9 am, weekly Monday), format (PDF/XLSX), recipients.
3	System	Evaluates alerts continuously; sends notifications with a deep link; delivers scheduled reports on time.
POST-CONDITIONS Management informed proactively.		

UF-17.5 Supplier, Customer & Operational Analytics		
PRIMARY ACTOR Managers		PRE-CONDITIONS —
#	Actor	Action / System Response
1	User	Opens Suppliers : compares Galileo vs RateHawk (and future suppliers) on uptime, response time, success rate, price competitiveness, reconciliation mismatches.
2	User	Opens Customers : segments (new/repeat, personal/corporate, tier), CLV, top routes/destinations, NPS/CSAT, churn-risk list (export to CRM campaign).
3	User	Opens Operations : queue volumes, SLA compliance, agent productivity, disruption resolution times, refunds ageing, average booking time.
POST-CONDITIONS Data-driven decisions across the business.		

BUSINESS RULES

1. Dashboards respect RBAC; branch/company filters are enforced for non-executive roles.
2. Metric definitions are centrally governed (semantic layer) and versioned.
3. Personally identifiable information is not shown in aggregate dashboards; drill-down to records requires Ops permissions.

Phase 1 Integrations — Galileo GDS & RateHawk

Phase 1 is deliberately limited to one flight source and one hotel source to minimise licensing cost while validating the product. Operation names below are indicative and will be finalised against the Travelport and RateHawk (Emerging Travel Group) API contracts and certification requirements.

GALILEO GDS (TRAVELPORT) — FLIGHTS

Platform Function	Galileo Capability (indicative)	Used in Flows
Flight search	Low fare search / availability (with alternate airports & flexible dates)	UF-03.1, UF-03.2, UF-12.3
Fare rules & brands	Fare rules, branded fares / ancillaries	UF-03.5, UF-04.1, UF-14.1
Price check	Air pricing (re-price)	UF-03.5, UF-14.2
Reserve	Create reservation / Universal Record (PNR) with SSRs, OSIs, FF numbers, APIS	UF-03.5, UF-02.5
Ticketing	Air ticketing / document issuance	UF-03.7, UF-06.7
Retrieve & modify	Universal Record retrieve / modify, seat map, SSR add	UF-03.10, UF-13.3
Exchange / reissue	Air exchange quote & exchange	UF-14.2, UF-14.4
Void / refund	Void ticket, refund quote & refund	UF-14.1, UF-14.3
Divide PNR	Split passenger from booking	UF-14.3
Queues	Queue read for schedule changes, TTL, waitlist	UF-09.1, UF-14.4
Cancel	Cancel segments / PNR	UF-03.5 (auto-cancel unpaid), UF-14.1

RATEHAWK API — HOTELS

Platform Function	RateHawk Capability (indicative)	Used in Flows
Static content	Hotel static data dump / hotel info (names, geo, images, amenities) — cached daily	UF-03.3, UF-03.4
Search	Search engine results by region / geo / hotel IDs	UF-03.3, UF-12.4
Rates for a hotel	Hotel page (all rates and room types)	UF-03.3
Prebook	Prebook (rate re-check and price change detection)	UF-03.5
Booking	Create booking form → finish booking → check finish status	UF-03.8
Voucher / info	Order info / documents	UF-03.8, UF-07.1
Cancel	Order cancel (with penalty calculation from policy)	UF-14.1
Reconciliation	Orders list / financial reports	UF-15.5

OTHER PHASE 1 SERVICES

Service	Purpose
Payment gateway (local acquirer + wallets: cards, JazzCash, Easypaisa, IBFT)	Payments, 3-D Secure, refunds (UF-03.6, UF-14.7)
WhatsApp Business Platform	OTP, notifications, approvals, broadcasts
Email & SMS providers	Transactional messages, OTP fallback
OCR service	Passport/CNIC/visa extraction (UF-02.2)
Flight status data provider	Delays, gates, terminals (Module 9 — Phase 2)
LLM provider & vector database	AI Conversation Engine and Knowledge retrieval
Maps / geocoding	Hotel maps, distances, transfer grouping

Future Integrations

The Supplier Integration Layer uses a common adapter contract (search, price, book, ticket/confirm, retrieve, modify, cancel, refund, reconcile) so new suppliers can be added without changing the Booking Orchestrator.

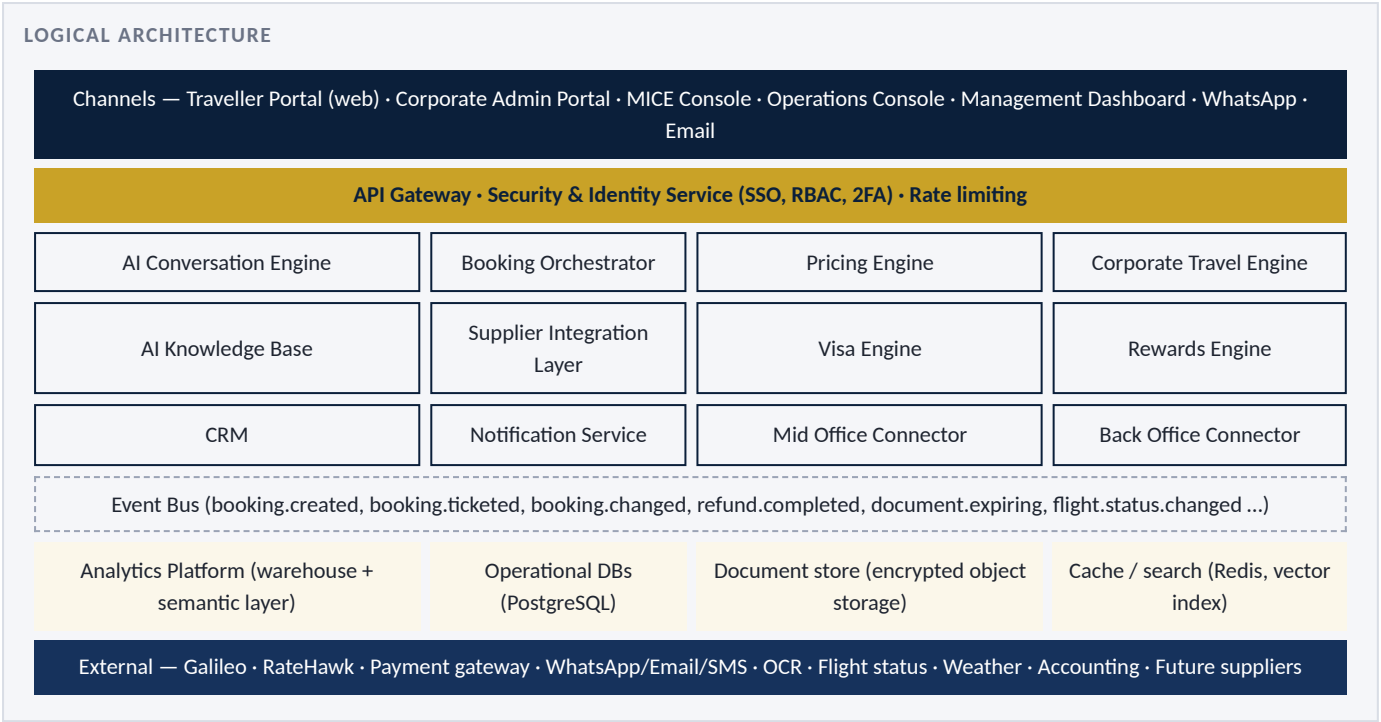
Supplier	Category	Adds	Adapter notes
Amadeus	GDS	Broader content, alternative pricing	Same flight adapter contract
Sabre	GDS	Broader content, alternative pricing	Same flight adapter contract
NDC (via aggregator or direct)	Airline retailing	Richer fare brands, ancillaries, offers/orders	Offer/Order model mapped to booking model
Airline direct APIs	Airlines (e.g. LCCs)	LCC inventory not in GDS	Payment at airline; reconciliation per airline
Hotelbeds	Hotel bedbank	More hotel inventory	Hotel adapter; property mapping required
WebBeds	Hotel bedbank	More hotel inventory (esp. Middle East)	Hotel adapter; mapping
Expedia Partner Solutions	Hotels	Large global inventory	Hotel adapter; mapping
Viator	Activities & tours	Tours, attractions for itineraries and groups	New 'activity' product type
Insurance providers	Travel insurance	Policy issuance at checkout	New 'insurance' product; Vault filing
Airport transfer providers	Ground transport	Bookable transfers, driver tracking	New 'transfer' product; Live Journey hooks

EXTENSIBILITY RULES

1. When multiple suppliers return the same product, de-duplication and best-source selection follow UF-03.4.
2. Each new supplier requires: mapping, pricing rules, reconciliation format, performance KPIs and failure fallback.

High-Level Architecture

Modular microservices behind an API gateway, communicating synchronously (REST/gRPC) for queries and asynchronously (event bus) for state changes.



SERVICE RESPONSIBILITIES

Service	Responsibilities	Key Modules
AI Conversation Engine	Intent/entity extraction, dialogue, tool calling (search, book, refund), response style guardrails, escalation detection, co-pilot	1, 4, 13
Booking Orchestrator	Search fan-out, sessions, holds, payment/approval routing, fulfilment, failover, booking state machine	3, 14
Supplier Integration Layer	Adapters for Galileo/RateHawk/future suppliers; normalisation; mapping; health metrics	3, 14, 15
Pricing Engine	Markup, fees, buffers, promotions, corporate pricing, discount limits, price snapshots	5
CRM	Customer 360, leads, cases, segments, consent	2, 13, 15
Corporate Travel Engine	Tenants, org structure, policies, approvals, budgets, credit, invoicing	6
Visa Engine	Rules evaluation, checklists, applications, reminders	8
Rewards Engine	Credits ledger, tiers, referrals, corporate rebates	10
Notification Service	Templates, channel routing (app/email/WhatsApp/SMS), quiet hours, read receipts, broadcasts	All
Mid / Back Office Connectors	Booking files, accounting postings, reconciliation feeds, exports	3, 15
Analytics Platform	Event ingestion, warehouse, KPI semantic layer, dashboards, NL analytics	17
AI Knowledge Base	Ingestion, versioning, embeddings, access-controlled retrieval, citations	16
Security & Identity Service	Authentication, SSO, 2FA, RBAC, sessions, audit events, encryption keys	0, 15

BOOKING STATE MACHINE

From	Event	To
Draft	Reserve	On Hold
On Hold	Corporate submit	Pending Approval
Pending Approval	Approved	Pending Payment / Ticketing
Pending Approval	Rejected / expired	Cancelled
On Hold	Payment captured	Paid
Paid	Ticket / confirmation success	Ticketed / Confirmed
Paid	Fulfilment failure (after retries)	Failed → Escalated (refund or manual fulfilment)
Ticketed / Confirmed	Change executed	Changed
Ticketed / Confirmed	Cancelled	Cancelled → Refund in progress → Refunded
Any hold	Time limit reached	Expired (auto-cancel)

Notification Event Catalogue

Event	Recipient	App	Email	WhatsApp	SMS	Module
OTP / verification	User		✓	✓	✓ fallback	0
New device sign-in	User	✓	✓	✓		0
Hold expiring (15 min)	Traveller	✓		✓		3
Booking confirmed / ticketed + documents	Traveller (+ arranger)	✓	✓	✓		3
Price changed at re-check	Traveller	✓				3
Approval requested / reminder / escalated	Approver	✓	✓	✓		6
Approval decision	Traveller, arranger	✓	✓	✓		6
Budget threshold reached	Budget owner	✓	✓			6
Invoice issued / overdue	Corporate finance		✓	✓		6
Document expiring (180/90/60/30/7 d)	Traveller	✓	✓	✓		2, 7
Visa appointment reminder / stay limit	Traveller	✓	✓	✓		8
Delay / gate / terminal change	Traveller, followers, group	✓		✓		9, 11
Cancellation / missed connection	Traveller, Ops	✓	✓	✓	✓	9, 14
Check-in open / leave for airport / boarding	Traveller	✓		✓		9
Credits earned / available / expiring	Traveller	✓	✓			10
Referral rewarded	Referrer	✓		✓		10
Group announcement / poll	Group members	✓	opt.	✓		11
Emergency broadcast	Group, guardians, emergency contacts, Ops	✓	✓	✓	✓	11
Event registration confirmed + QR	Delegate		✓	✓		12
Consultant joined / call-back reminder	Traveller	✓		✓		13
Refund status change	Traveller	✓	✓	✓		14
Schedule change action required	Traveller	✓	✓	✓		14
Knowledge review due	Knowledge owner	✓	✓			16
KPI alert	Management		✓	✓		17

Non-Functional Requirements

Area	Requirement
Performance	AI first response ≤ 2 s (streaming); flight search results ≤ 6 s p90; page load ≤ 2.5 s p75; end-to-end booking time < 3 minutes (PRD objective)
Availability	99.9 % for traveller portal & booking; 99.95 % for payment/fulfilment path; 24/7 P1 support
Scalability	Horizontal scaling of stateless services; 5,000 concurrent chats and 500 searches/min at launch, 10× headroom
Security	OWASP ASVS L2; TLS 1.2+; AES-256 at rest; secrets vault; WAF; 2FA for staff; least-privilege RBAC; annual penetration test
Payments	PCI-DSS SAQ-A (hosted fields / tokenisation); no card data stored by FlightOne
Privacy	Consent management; data minimisation; PII masking in logs/transcripts; data subject export/delete; compliance with Pakistan's applicable data-protection rules and GDPR for EU travellers
AI safety & quality	Grounding in live data/knowledge; confirmation before irreversible actions; discount and policy guardrails; prompt-injection defences; evaluation suite and human review of sampled conversations
Auditability	Immutable audit logs retained ≥ 7 years; price and policy snapshots per booking; AI model/knowledge versions recorded
Localisation	English and Urdu UI (RTL support for Urdu); PKR default with multi-currency display; Asia/Karachi default time zone with per-user time zone
Accessibility	WCAG 2.1 AA for web portals
Observability	Centralised logs, metrics, traces; supplier health dashboards; alerting to on-call
Backup & DR	Point-in-time recovery; RPO ≤ 15 min, RTO ≤ 1 h; multi-AZ deployment
Browser support	Latest 2 versions of Chrome, Edge, Safari, Firefox; responsive down to 360 px

Development Roadmap

Phase	Scope (from PRD)	Modules / Flows in this SDS
Phase 1 — MVP	AI Chat · Galileo · RateHawk · Customer Profiles · Pricing Engine · Payment Gateway · Corporate Approvals · Ticketing · Voucher Generation	M00; M1; M2; M3; M4; M5; M6 (policies, approvals, credit, invoicing); M7 (booking documents); M13; M14 (manual-assisted); M15 (core); M16; M17 (core KPIs)
Phase 2	Visa Vault · Rewards · Group Travel · MICE · Refund Automation · Journey Monitoring	M7 (full Vault); M8; M9; M10; M11; M12; M14 (automation); M15 (advanced finance)
Phase 3	Voice AI · Autonomous Travel Concierge · Predictive Recommendations · White-label Corporate Portals · Expense Management · Carbon Reporting · Advanced Analytics	Extensions of M1 (voice, autonomous actions), M4 (predictive), M6 (white-label portals, expense, carbon), M17 (advanced analytics)

PHASE 3 EXTENSION NOTES

Capability	Design intent
Voice AI	Voice channel for AI Consultant (web mic + phone line) using the same conversation engine and guardrails; voice confirmation of bookings with OTP.
Autonomous Travel Concierge	Pre-authorised rules (e.g. 'rebook me automatically if delayed > 2 h within PKR 20k') executed by AI without prompting, with instant notifications.
Predictive Recommendations	Anticipates trips from history/calendar, fare prediction ('book now vs wait'), proactive offers.
White-label Corporate Portals	Company-branded domain, logo and colours on the Traveller Portal; SSO.
Expense Management	Receipt capture (OCR), per-diem, expense reports linked to trips, approvals, reimbursement export.
Carbon Reporting	CO ₂ per flight/hotel night, company dashboards, policy nudges to lower-emission options.
Advanced Analytics	Forecasting, cohort analysis, pricing elasticity, supplier negotiation insights.

Requirements Traceability Matrix

Every PRD requirement mapped to the user flows that satisfy it — confirming no feature was skipped.

PRD Ref	Requirement	Satisfied by
M1	Understand natural language; ask follow-ups; short crisp decision-led answers	UF-01.1, UF-01.2; Rules M1-1
M1	Refine requirements; suggest alternatives; recommend & explain; learn preferences; continue across devices	UF-01.2 – UF-01.6
M2	Passport, CNIC, visas, residence permits	UF-02.2, UF-02.4
M2	FF numbers, hotel loyalty, travel/seat/meal preferences	UF-02.5
M2	Emergency contacts, travel history, saved companions, family members	UF-02.5, UF-02.7
M2	Expiry notifications; OCR; manual verify or re-upload	UF-02.6, UF-02.2, UF-02.3
M3	Search Galileo; search RateHawk; compare suppliers; rank with AI	UF-03.1, UF-03.3, UF-03.4, UF-04.1
M3	Sales strategy — alternate departure/arrival cities	UF-03.2
M3	Similar-option pitch; differentiate same-name products by attributes across providers	UF-03.4
M3	Apply margins; reserve; payments or approvals; issue tickets; vouchers; update mid/back office	UF-05.2, UF-03.5 – UF-03.9, UF-06.6
M4	Evaluate price, duration, layovers, quality, arrival times, preferences, loyalty, refundability, reliability; three curated picks	UF-04.1 – UF-04.5
M5	Net fares, dynamic markups, negotiation buffers, promos, corporate pricing, agent permissions, AI discount limits, profitability	UF-05.1 – UF-05.8
M6	Multiple companies, departments, cost centres, project codes	UF-06.1 – UF-06.3
M6	Travel policies, approval workflows, budget enforcement	UF-06.4 – UF-06.8
M6	Credit limits, billing cycles, corporate invoicing	UF-06.9
M6	Corporate ↔ Personal profile switching	UF-06.10
M7	Store passports, visas, tickets, vouchers, insurance, FF numbers, loyalty cards, certificates	UF-07.1 – UF-07.7
M8	Destination requirements, transit rules, documents, embassy info, timelines, expiry reminders, appointment tracking	UF-08.1 – UF-08.8
M9	Delays, gate/terminal changes, weather, boarding, hotel check-in, transfers, immigration advisories; app/email/WhatsApp	UF-09.1 – UF-09.8
M10	Booking rewards, redeemable credits, referrals, corporate programmes, tiers	UF-10.1 – UF-10.7
M11	Group creation, invitations, shared itinerary & documents, flight status, announcements, polls, attendance, emergency, live updates, gallery, AI memories	UF-11.1 – UF-11.12
M12	Delegate registration, flights, hotels, transfers, agenda, QR check-in, badges, attendance, budget, sponsors, reporting	UF-12.1 – UF-12.11
M13	Escalation on request, VIP, complex, supplier failure, refund dispute, medical, special services; full history transfer	UF-13.1 – UF-13.6
M14	Airline/hotel refund value, penalties, agency fees, timelines, credits; refunds, exchanges, reissues, partial, schedule changes, cancellations	UF-14.1 – UF-14.8
M15	CRM, mid office, back office, accounting, finance, commission, supplier reconciliation, audit logs	UF-15.1 – UF-15.9

PRD Ref	Requirement	Satisfied by
M16	SOPs, airline policies, supplier contracts, visa procedures, corporate agreements, travel policies	UF-16.1 – UF-16.6
M17	Sales, revenue, margins, outstanding credit, conversion, AI automation, supplier performance, customer analytics, operational KPIs	UF-17.1 – UF-17.5
Obj.	Booking time < 3 min; automate servicing (90 % AI); conversion; satisfaction; lower ops workload; scalable platform	M1, M3, M13, M17 KPIs; Appendix C, E

END OF DOCUMENT

FlightOne AI-TOS — Software Design Specification v1.0. Wireframes are low-fidelity and indicative; supplier operation names are subject to final API contracts and certification.